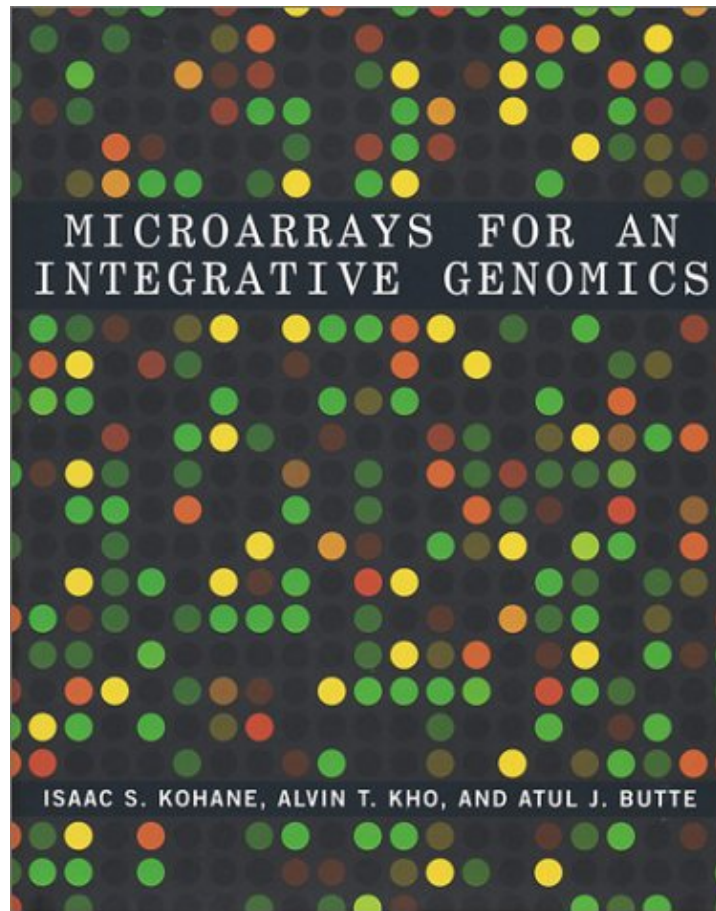




Microarrays for an Integrative Genomics

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Microarrays for an Integrative Genomics

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Foreword

The impact of microarray measurements on biology and bioinformatics has been astounding. Starting from virtually no literature a few years ago, this field has come to dominate many conferences and journals. As an example, the Intelligent Systems for Molecular Biology conference, the annual meeting of the International Society for Computational Biology held in Copenhagen in 2002, had almost 50% of its papers in the areas addressed directly or indirectly within this book. Four years ago, there were none. Bioinformatics has always been driven by the availability of data—sequential, structural, and most recently functional. The availability of sequence data brought into biology a cadre of computer scientists with special skills in string processing. The availability of structural data brought in technical experts in visualization and computational geometry. This most recent development—the availability of relatively large data sets measuring the expression of genes within cells—has helped attract yet another group of scientists—the data miners, machine learners and statisticians.

In many ways, the impact of this data on biology and informatics can be summarized in figure 1.4 of this book—the world has not quite been turned upside down, but it certainly has been turned on its side! A decade ago, if confronted with the data matrix shown on the right of this figure, a well-trained information scientist would say "This is ridiculous. Why would you ask me to analyze a data set where you clearly have a profoundly under-determined problem? There's not enough data here to distinguish between any of the zillion hypotheses that could be consistent with this data set. And who designed these experiments, anyway? How can you measure so many features of such a few examples?" Yet, these experiments are proceeding and are making major contributions to our understanding of how gene systems interact, how to distinguish different types of cancer, and how to measure the impact of the environment on a cell. Our information scientist friend is, in some sense, correct about the relative paucity of data. (Have you ever tried to convince a biologist holding a microarray with 45,000 spots that this is a relatively data-poor exercise? It's not fun.) However, the information scientist has missed the point about the design and analysis of these experiments. These data sets do indeed contain gold, but the experiments (as for all experiments) must be considered carefully in both the design and implementation phases in order to maximize value. This is where the authors of this book have made a contribution. They start from the premise that these experiments offer great potential, but must be performed and analyzed carefully. They set the context of traditional reductionist biology, and then go on to discuss the design, analysis, storage and interpretation of this first generation of functional genomics experiments. The writing is lively and candid, and the examples are taken from an array of applications. The authors' practical experience in dealing with this data comes through, and they intersperse practical advice with philosophical reverie. Sometimes, these two merge into important discussions such as on the role of ontologies in making sense of these data sets, or on the challenges of linking microarray results with phenotypic data pertinent to human disease.

The functional genomics revolution is here. We do not know how it will change our view of biology and medicine. They are both much more likely to become quantitative and systematic (as opposed to qualitative and reductionist). The informatics techniques required to address this revolution are not entirely clear, but this text gets us started in the right direction.

Russ Altman

Stanford University

March 2002

Preface

Three years ago, when a colleague would approach us with questions about functional genomics and the informatics techniques required to leverage the data obtained from measurement techniques such as DNA microarrays, we had a standard response: "Come listen to a 1-hour presentation by one of us and you'll have a foundation for further discussions." Since then, this response has become inadequate. First, the field can hardly be summarized in even eight 1-hour lectures, and second, the growth in the number of potential collaborators has far outstripped the time available to us to make the necessary presentations.

In early 2000, Ben Reis, one of the graduates of the Children's Hospital Informatics Program, had the inspired idea of formalizing our introductory presentations into a book. We immediately agreed that this was a timely suggestion and to its credit, so did The MIT Press. In our teaching duties in several courses within the Division of Health Sciences and Technology (HST) at Harvard/MIT the range of topics within functional genomics that we were covering in formal presentations grew rapidly. Subsequently, with the inception of the Bioinformatics and Integrative Genomics training program at HST and the development of a Genomic Medicine course at HST, we felt the need for this book all the more acutely.

Organization of the text

We recognize that the readership of this book will be varied due to the intrinsically multidisciplinary nature of the functional genomics enterprise (as will be emphasized in the introductory chapter). Accordingly we outline the content of the following chapters so that readers may choose for themselves the path that suits them. Nonetheless, our intent and contention is that the current ordering of the chapters provides the most efficient way of acquiring the content of this book.

Introduction. Here we establish the motivation and the scope of this book and touch upon substantial obstacles to success in the successful application of bioinformatics to an integrative genomics. The notion of an interdisciplinary functional genomics pipeline is introduced. We also review which kind of readers might find this book worthwhile. The promise and limitations of functional genomics techniques, the nature of various kinds of genomic data, and the central role played by the discipline of bioinformatics are outlined. For those who have a limited background in biological sciences, there is a subsection on the basic minimum of molecular biology concepts that will be needed to grasp the the following chapters.

Chapter 2. Experimental Design. This chapter develops a framework for approaching the design of microarray-driven functional genomics experiments. Very little here is quantitative or mathematical. Rather the emphasis is on ways of thinking about the design of experiments and how it might impact the yield of these experiments. We address challenges that are particular to computer scientists (e.g., defining a figure merit for the performance of the bioinformatics algorithms) and to biologists (e.g., discarding potentially valuable data using formal decision theory because of the scale issues in massively parallel data acquisition using noisy measurement devices), respectively. In exploring the design issues we introduce the functional genomics clustering dogma, the broad machine-learning categories of supervised and unsupervised learning, and the nature of the analyses developed using these techniques.

Chapter 3. Microarray Measurements to Analyses. We lay the foundations for performing analyses of microarray data sets. This is the first of the more quantitative and mathematical chapters. We start with a discussion of the acquisition of digital data from the two most widely employed classes of microarrays. Then we consider the two most generic problems of comparing gene expression within a single microarray, *i.e.*, *intra*-array analyses, and comparing expression across microarrays,

i.e., inter-array analyses; in so doing, we introduce the fundamental concept of (dis)similarity and similarity measures and the several kinds of such measures. These measures become the building blocks for the genomic data-mining techniques described in the following chapter.

Chapter 4. Genomic Data-Mining Techniques. When gene expression is measured in more than two samples, gene expression patterns have to be analyzed using methods that consider the coordinated interactions of genes across multiple conditions. This chapter assesses the components of biomedical experiments that can be included in a data-mining investigation. We then cover the most commonly used analytic techniques, discussing the advantages and disadvantages of each technique, as well as the postanalysis process. Where appropriate, we provide pseudocode that will allow readers with some training in computer science to understand the details of the most often used and cited data-mining algorithms. The emerging field of genetic network reverse engineering is also introduced here.

Chapter 5. Bio-Ontologies, Data Models, Nomenclature. This chapter addresses possibly the least exciting but the most pressing bioinformatics need for genomic research: creating and using comprehensive annotations of gene function, storing and organizing microarray expression data, and ensuring standardized access to these data. We review current efforts to create formalized systems of description of gene function and the various kinds of "ontologies" that support these descriptions. The challenge to design "standardized data models" for the storage of microarray data is addressed and the principal contenders claiming to be this standard are reviewed. Naming schemes—nomenclatures—most applicable to gene expression studies are described. Nomenclatures, data models, and ontologies are placed in a perspective of the general problem of analyzing the results of functional genomics experiments. Tools that leverage these standardization efforts and the on-line published literature are also described.

Chapter 6. From Functional Genomics to Clinical Relevance: Getting the Phenotype Right. Here we address the process of translating the functional genomics research agenda into one of clinical relevance. We place in this perspective the value and deficiencies of electronic medical records and standardized clinical vocabularies. Although by no means comprehensive, we provide the highlights of the privacy issues (*e.g.*, the implications of the Health Insurance Portability and Accountability Act, anonymization, cryptographic identifiers, *etc.*) that are most likely to have an impact on the clinical application of genomic technologies.

Chapter 7. The Near Future. As the techniques and goals of functional genomics are in rapid flux, we engage in some short-term forecasting to guide readers planning in this time window. Microarray technologies being developed and recently released are previewed. In this context, the problem of comparing expression measurements across generations of microarray measurement platforms is appraised. More broadly, the different kinds of software required for the successful functional genomics enterprise are described. Finally, a model to meet the training needs of this new discipline is outlined.

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Chapter 1: Introduction

The functional genomics "meltdown" is coming. At least that is what we fear is likely to occur with the confluence of the high expectations engendered by the Human Genome Project and the prevalence of highly uneven scholarship in the investigations made possible by comprehensive genomic measurement technologies, such as microarray-based expression profiling. Because the availability of these technologies has preceded the development of a substantive canon of appropriate analytic techniques, safe experimental design, and cautionary tales, the quality of these investigations and the manner in which they have been reported often results in the dissemination of highly preliminary and flimsy findings. The absence of widespread use of computational and biological validation procedures associated with these studies has led to many reports that will likely not be substantiated by follow-up studies. With the rapid decrease in the cost of genomic techniques, follow-up studies are likely to arise within the next few years. At that time, several previously well-publicized findings in basic biology, clinical diagnosis, clinical prognosis, and pharmacogenomic targeting will be found deficient. The inevitable reaction to these unfortunate developments will be attenuated if the discipline of functional genomics has matured enough by then to achieve the rigor required.

In this book, we have attempted to address some of the challenges required to begin this maturational process in the intersection between microarray expression technology, bioinformatics, and biomedical science. Despite the likelihood of an eventual disappointment and subsequent retrenchment of the ambitions of microarray applications, we are confident that the development of systematic approaches to this discipline will ultimately deliver on the exuberant predictions and promises made over the last 5 years. We hope that our experience (including our own generous share of mistakes), as communicated here, will help provide the reader with a framework to participate productively in attaining this goal.

1.1 The Future Is So Bright...

Let us be clear lest the above suggest that we are pessimistic about this field. There are very few disciplines within biomedical research with as much promise and excitement as functional genomics. Consider the example of the analysis of large B-cell lymphoma, a deadly malignancy of the lymphatic system, conducted by Alizadeh *et al.* [3]. In (a necessarily abbreviated) summary of their study, the gene expression of the lymphatic tissues of a cohort of patients with the diagnosis of large B-cell lymphoma was measured using DNA microarray technology. That is, thousands of genes expressed in these tissues were simultaneously measured in the respective lymphatic tissue sample of each patient.

When a clustering analysis was performed to see which patients resembled one another the most, based on their gene expression pattern, two distinct clusters of patients were found. When the investigators examined the patients' histories it became apparent that the two clusters corresponded to two populations of patients with dramatically different mortality rates (illustrated by the survival curves in figure 4.13, section 4.11).

The implications of these two significantly distinct mortality rates are profound. First, these investigators have discovered with genomic technologies and bioinformatics analyses a new subcategory of large B-cell lymphoma, a new *diagnosis* with clinical significance. Second, they have generated a tool that provides (pending confirmation in other studies) a new *prognosis*; patients can be given much more precise estimates of their longevity. Third, it provides a new *therapeutic opportunity*; patients with an expression pattern predicting a poor response to standard therapy may be treated with different (*e.g.*, much more aggressive) chemotherapy. Fourth, it

presents a new *biomedical research* opportunity; what is it about these two subpopulations that makes them so different in outcome and how can that be related to the differences in gene expression?

It is rare that a set of measurement and analytic techniques can so revolutionize biomedical research and clinical practice. It is precisely because the excitement and expectations surrounding this field are so high that we are compelled to inject a note of skepticism about the measurement techniques and the analytic methods. Without such skepticism, the scientific method cannot function and the field will not advance. Nevertheless, even as we have discovered for ourselves significant drawbacks in genomic methodologies, which we address in this book, we remain convinced that these problems only represent the transient growing pains of a new field of biomedical investigation.

Who is this book intended for? Answering this question has served as our constant compass throughout the book's writing. There are three audiences that we have had in mind.

1. *Experienced biologists with limited experience using expression microarrays, or who are concerned that their current approaches to this field are problematic.* For them, this book provides a systematic approach to using microarrays as a tool to investigate biology. Our particular goal for this audience is to realize the pitfalls and *caveats* relating to expression microarray technologies and their analysis. Also, we intend this book to provide these biologists with a firm foundation on which they can engage in collaborations with colleagues formally trained in bioinformatics and biostatistics. We deliberately limited the amount of mathematical treatment of this material. Where it is present, the reader can skip it without significantly reducing the comprehension of the subsequent text.
2. *Experienced informaticians with limited experience analyzing microarray data.* We believe the field of functional genomics to be one of the most fruitful and rewarding for a computer scientist or other quantitatively trained scientist to enter. It provides a source of challenges, problems, and data sets that will stimulate basic methodological development while furthering important goals in the enterprise of biological discovery and the state of the art of clinical care. For this reason, we emphasize an approach that is driven by the questions of interest to these latter goals. As a result, this book will not present the fundamental computer science underlying various techniques (*e.g.*, proofs of the soundness of various machine-learning algorithms) but will instead illustrate their application to problems that are challenging investigators in functional genomics. This is not to say that the approach we have taken is not rigorous; it just eschews details on the methodologies that are available elsewhere and that we have copiously referenced.

We are convinced that the most productive research projects will be those that involve intimate collaborations with biologists. This is in contrast to the remote and *post hoc* analysis of the outcome of experiments that the bioinformatician has little to do with, but which is frequently the norm in this discipline. We intend and hope this book can serve as the basis for collaborations in which the informatician understands the goals of biology in this scientific enterprise and in which she or he can contribute to the experimental design and analysis as a first-class member of the research team.

3. *Students entering the field of Bioinformatics.* In our own classes (*e.g.*, Medical Computing 6.872 taught at MIT), we have been gratified to note the emergence of a new generation of students who are both facile in the use of computers as experimental tools and who have a broad understanding of the biological sciences. Although this text can be used as the basis of a course, it is also designed for independent study. For those students, this text is intended to serve as a rapid introduction to the fields of microarray-driven studies of functional genomics so that they can become productive researchers even while they pursue their studies. Indeed, we have had several successful collaborations with undergraduates

who have used this material as a launching pad for graduate research or careers in industry.

1.2 Functional Genomics

Now that the human genome has been sequenced (or, more accurately, now that a handful of human genomes have been sequenced), we are said to be in a postgenomic era [200]. We find this term confusing ("genome, we hardly knew ye") because in our view, now that we know at least the draft outline for the genome of multiple organisms, we can begin for the first time to systematically deconstruct how the genetically programmed behavior of an organism's physiology is related to the constituent genes that make its individual version of its species' genome. In this deconstruction, several kinds of biological information are available: DNA sequence, physical maps, gene maps, gene polymorphisms, protein structure, gene expression, protein interaction effects,^[1] and a vast literature in MEDLINE [129].

As we use it in this book, functional genomics refers to the overall enterprise of the deconstruction of the genome to assign biological function to genes, groups of genes, and particular gene interactions. These functions may be directly or indirectly the result of a gene's transcription. Much of functional genomics has been and will continue to be the kind of hypothesis-driven biological research pursued for the past decades.

In this book, we address a computationally intensive branch of functional genomics that has emerged as a result of the practical implementation of technologies to assess gene expression thousands of genes at a time. The ability to comprehensively measure expression affords an opportunity to reduce our dependence on *a priori* knowledge (or biases) and allow the organism to point us in potentially fruitful directions of investigation. That is, we describe a hypothesis-generating effort which, if carefully crafted, can then lead to a highly productive set of investigations using more conventional hypothesis-driven research. In this we have been inspired by the work of Arkin *et al.* [12], as have others. Starting from the raw time-series measurements of the substrates of glycolysis (see figure 1.1), Arkin *et al.* were able to computationally reconstruct the glycolytic pathway (figure 1.2). This reverse engineering of the metabolic pathway without prior knowledge is in contrast to the decades of exacting hypothesis-driven elucidation of this pathway by biochemists. It turns out that, as we will discuss later, in experimental design (section 2.1.3), this metaphor is flawed but it remains an icon for one of our major goals: to use bioinformatics applied to functional genomics data to create and re-create the kind of biological pathway charts that are common in most basic biology laboratories.

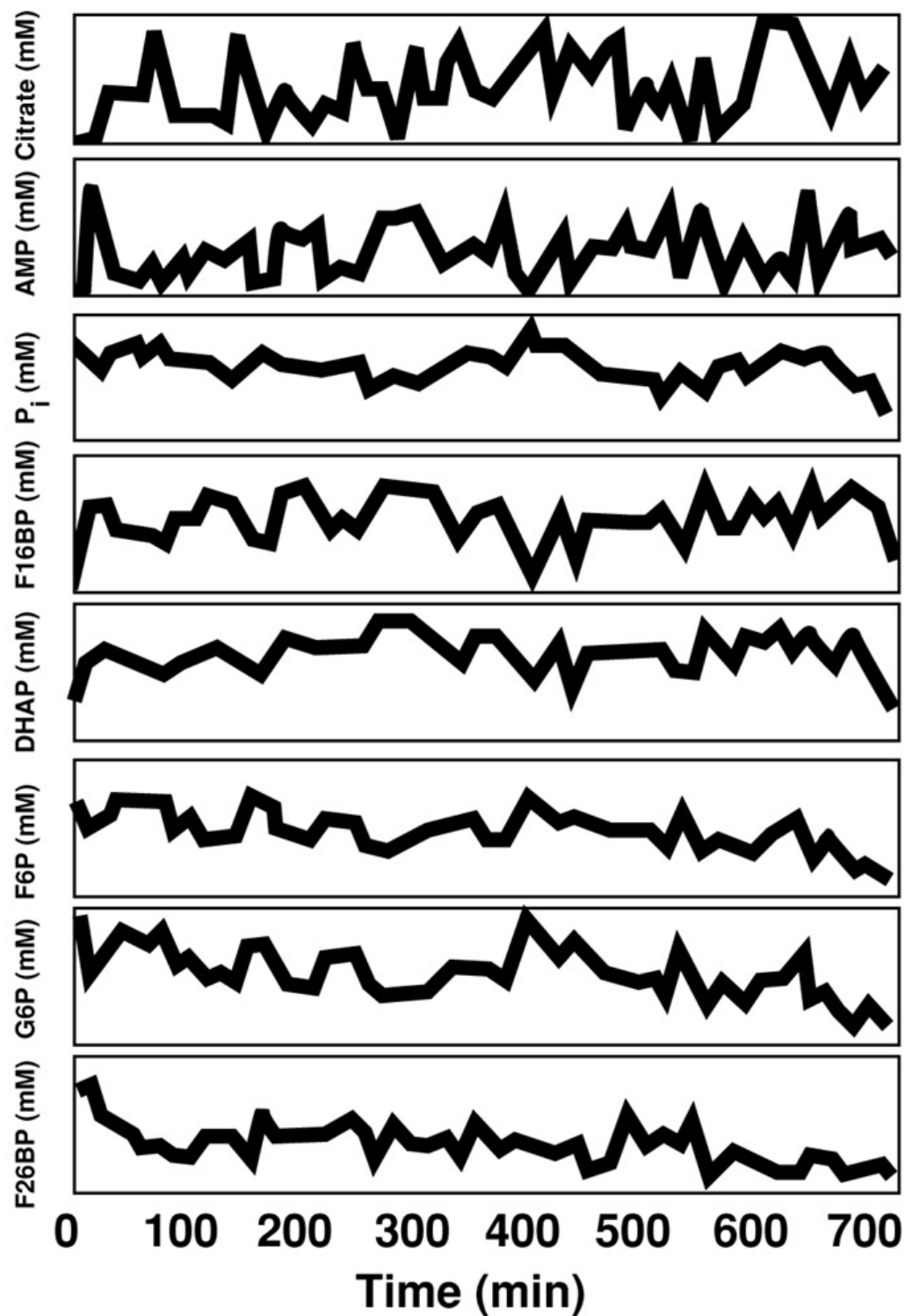


Figure 1.1: Time-series data from anaerobic metabolism. The time courses of measured concentration of the small molecule inputs, adenine monophosphate (AMP, a source of chemical energy to catalyze reactions) and citrate (a substrate), in the experiments, with the responses of the concentrations of phosphate (P_i , an inorganic ion) and of the substrates fructose-1,6-biphosphate (F16BP), dihydroxy acetone phosphate (DHAP), fructose-6-phosphate (F6P), glucose-6-phosphate (G6P), and fructose-2,6-biphosphate (F26BP). (Derived from Arkin *et al.* [12].)

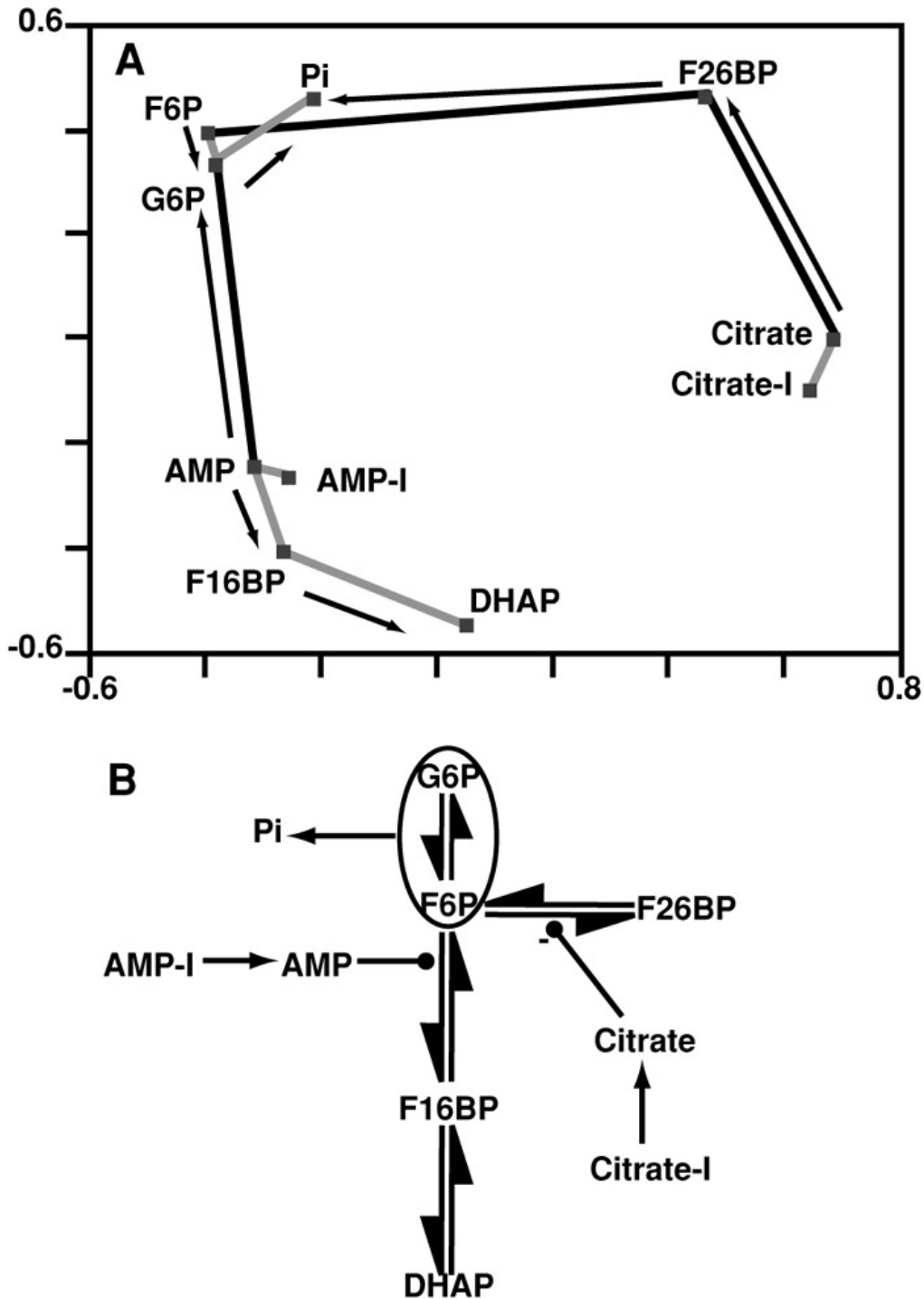


Figure 1.2: Glycolytic pathway reconstructed *ab initio* from time-series data. **A**, The two-dimensional projection of the correlation metric construction (CMC), defined by Arkin *et al.*, for the time series shown in figure 1.1. Each point represents the time series of a given species. The closer two points are, the higher the correlation between the respective time series. Black (gray) lines indicate negative (positive) correlation between the respective species. Arrows indicate temporal ordering among species based on the lagged correlations between their time series. **B**, The predicted reaction pathway-derived CMC diagram. Its correspondence to the known mechanism of glycolysis is high. (Derived from Arkin *et al.* [12].) Furthermore, we note that genomic data can be fruitfully exploited even without the assignment of function: a prognostic test for rejection of transplanted kidneys based on the expression level of three genes is useful even if the function of those three genes is poorly known or not known at all.

1.2.1 Informatics and advances in enabling technology

Gene expression detection microarrays are notable not because they can uniquely measure gene expression. There certainly have been many technologies that have allowed for the quantitative or semiquantitative measurement of expression for well over two decades. What distinguishes gene expression detection microarrays is that they are able to measure tens of thousands of genes at a time and it is this *quantitative* change in the scale of gene measurement that has led to a *qualitative* change in our ability to understand regulatory processes occurring at the cellular level. Figure 1.3 provides perhaps the best motivation for the application of information sciences to the functional genomics enterprise. Since DNA sequencing was invented 25 years ago, the number of gene sequences deposited in international repositories, such as GenBank, has grown exponentially, culminating with the entire human genome being sequenced in 2001. Distinguished from this, the amount of *knowledge* about these genes (as measured by the proxy of the number of papers published in biomedicine) has also been growing exponentially, but at a *much* slower rate. As shown, the number of GenBank entries has fast outstripped the growth of MEDLINE. As such, it serves as a proxy for the large gap that has just opened up between our knowledge of the functioning of the genome and raw genomic data. And GenBank entries are just a fraction of the various kinds of data, listed above, generated as part of our investigation of the genome. This volume of data must somehow be sifted and linked to the biological phenomena of interest. Doing so exhaustively, reliably, and reproducibly is a plausible strategy only with the application of algorithmic implementations on computers. This has led to an unprecedented demand for investigators with the knowledge of successful manipulation and analysis of large data sets. These skills may come from education and training in computational physics, chemical engineering, operations research, or financial modeling, but once they are applied to the domain of functional genomics, they can be collectively described as belonging to the domain of bioinformatics.

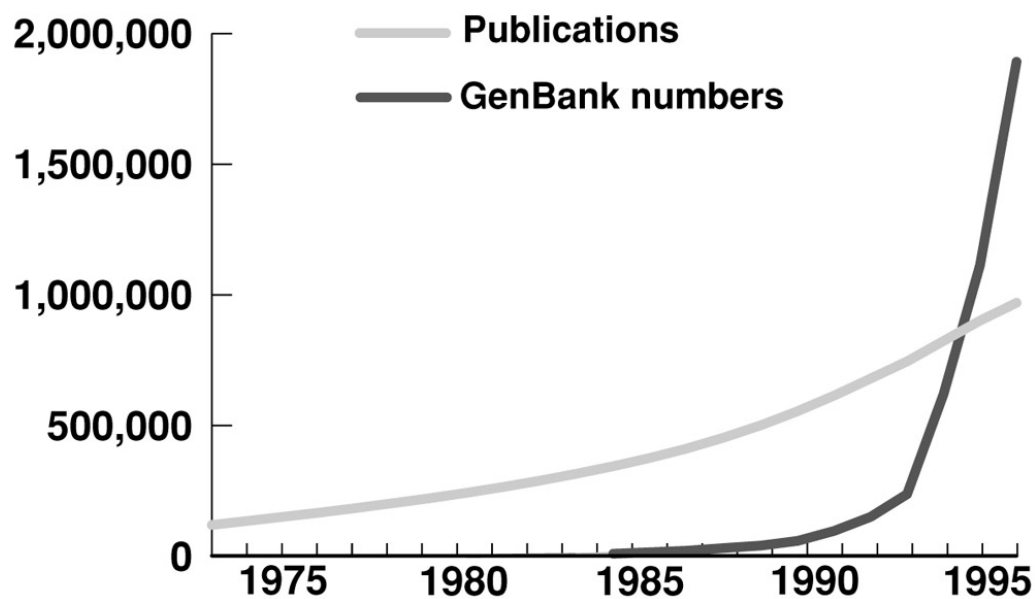


Figure 1.3: Relative growth of MEDLINE and GenBank. The industrialization of genomic data acquisition has created a growing gap between knowledge—of which MEDLINE publications are a proxy—and the information we have gathered about the genome. Information science applied to this information (*i.e.*, bioinformatics) is one of the pillars of an international strategy to overcome this knowledge gap. Cumulative growth of molecular biology and genetics literature (light gray) is compared here with DNA sequences (dark gray). Articles in the G5 (molecular biology and genetics) subset of MEDLINE are plotted alongside DNA sequence records in GenBank over the same time period. (Derived from Ermolaeva *et al.* [65].)

One reason why the number of known sequences is growing so much faster is the discovery and

use of many automated techniques, such as automated sequencers and shotgun sequencing methods. Until the recent advent of gene expression microarrays, we did not have a similar technique to automate the acquisition of knowledge about these genes' behavior in cellular physiology. The past 5 years have seen an incredible confluence of disparate technologies, such as robotics, fluorescence detection, photolithography, and the Human Genome Project,^[2] so that today, biologists can use RNA expression microarray detection technologies to obtain near-comprehensive expression data for individual cells, tissues, or organs in various states. With currently available commercial tools, a single experiment using RNA expression detection microarrays can now provide systematic quantitative information on the expression of 60,000 unique RNAs within cells in any given state. Complementary DNA (cDNA) and oligonucleotide microarray technology^[3] cannot only be used to determine the abundance of RNA transcripts. By virtue of their broad reach, these measurement platforms permit a large number of exhaustive comparisons: of transcriptional activity across different tissues in the same organism, across neighboring cells of different types in the same tissue, across groups of patients with and without a particular disease or with two different diseases. These platforms can also be used to analyze complex systems, such as traits with multigenic origins or those linked to the environment. They can be used in time series to measure how a particular intervention may start a transcriptional program, *i.e.*, change the expression of large numbers of genes in a reproducible pattern determined by inherent genetic regulatory networks. With sufficient data, they can be even used to provide insight into the underlying mechanisms of these genetic regulatory networks.

Nonetheless, the tools to extract *knowledge* from *data* collected from all of these types of experiments are still in their infancy, and novel tools are still needed to sift through the enormous databases of simultaneous RNA expression to find the true nuggets of related function. The application of techniques of information science, computer science, and biostatistics^[4] to the challenge of knowledge acquisition from genomic data is commonly known as *bioinformatics*. This appellation applies to the quantitative and computational analysis of all forms of genomic data, including gene sequence, protein interactions, protein folding, and any observable or measurable phenomenon of interest to the biomedical researcher. The breadth of this commonly used definition of bioinformatics risks relegating it to the dustbin of labels too general to be useful of which artificial intelligence, knowledge management, and systems analysis are only among the more recent. In this book our intent is to be sufficiently specific about the bioinformatics techniques employed that the matter of a sufficiently broad and yet specific definition of bioinformatics is moot.

Over the past 6 years, several approaches have been developed to analyze microarray-generated RNA expression data sets. The central hypothesis (or hope) of these methods is that, with improved techniques in bioinformatics, one can analyze larger data sets of measurements from RNA expression detection microarrays to discover the "true" biological functional pathways in gene regulation, and develop more definitive, sensitive, and specific diagnostic and prognostic characteristics of disease. However, this is only one of many important areas of bioinformatics addressed in this book. Particularly because we are still in the immediate aftermath of the Human Genome Project, many of the basic naming and data management practices of functional genomics remain in flux and are active areas of bioinformatics development. Although this activity is quite distinct from the analytic efforts touched on above, it currently consumes perhaps the largest proportion of the bioinformatics community because its resolution is urgent and a *sine qua non* for the success of any of the analytic efforts. After all, if we cannot reliably name the same gene in identical fashion across experiments, if we cannot reliably retrieve expression data from all the microarray experiments of interest, if we cannot readily access the meaning and function of genes determined by thousands of researchers, then the whole enterprise of functional genomics will be crippled if not intractable.

There is a related discipline to bioinformatics—*clinical informatics*—which refers to the application of

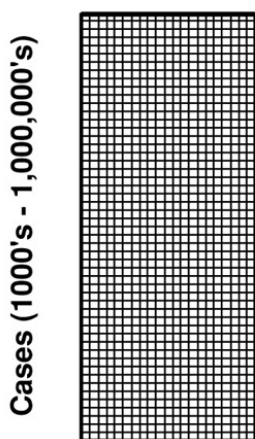
information science to various aspects of clinical care. Although clinical informatics is not addressed in this book in detail, in chapter 6 we describe many of the problems that have dogged clinical informaticians and that will confront bioinformaticians as they attempt to bring their basic science findings to clinical relevance.

1.2.2 Why do we need new techniques?

A first look at a typical genomic study might cause a quantitatively trained scientist or even a biologically trained scientist to ask the following, quite legitimate question: Why is this field not amenable to standard biostatistical techniques? After all, we are trying to understand the relationship between multiple variables and the mechanisms that the relationships reveal. And there has been a long history of the development of biostatistical techniques to analyze large studies with large numbers of cases with many variables to elucidate precisely this kind of question. Specifically, these studies ask questions such as: What risk factors are associated with heart disease? Does smoking cause disease? What is the difference in survival between a group treated with one chemotherapeutic drug versus another? On the surface these questions seem similar to many of those posed regarding genetic risk factors for acute and chronic disease. Yet a review of the bioinformatics and functional genomics literature over the past 3 years reveals that most of the analyses have been performed using techniques borrowed from the computational sciences and machine-learning communities in particular. Why is this? There are several reasons, including academic parochialism, but perhaps the most substantive one is the essentially underdetermined nature of genomic data sets as described below. If we examine figure 1.4, we see sketched out the fundamental difference between a typical clinical study and a typical genomic study. A high-quality clinical study will involve thousands to tens of thousands of cases, such as in the Nurses' Health Study [18] or the Framingham Heart Study [55] over which tens or even hundreds of variables are measured. In contrast, in a typical genomic study, there are only tens or, exceptionally, hundreds of cases, but thousands of measured variables.

A typical clinical study

Variables (10's - 100's)



A typical genomic study

Variables (10,000's - 100,000's)

Cases (10's - 100's)

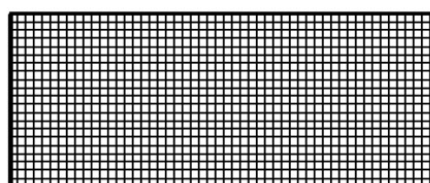


Figure 1.4: A major difference between classic clinical studies and microarray analyses. The high dimensionality of genomic data in contrast to the relatively small number of samples typically obtained results in a highly underdetermined system.

Initially, the low number of cases in functional genomic investigations may have been due to the high cost of the microarrays (on the order of several thousand dollars per microarray) but increasingly the scarcity of cases in a typical functional genomics study will relate to the scarcity of appropriate biological samples. As these experiments involve the measurement of gene expression,

a particular tissue has to be obtained under the right conditions. This is in distinction to genomic DNA samples where most blood samples will suffice. Especially in human populations, suitable tissue samples are relatively rare.^[5] Yet even though there are only tens of cases, each case involves the measurements of tens of thousands of variables corresponding to the expression of tens of thousands of genes measurable with microarray technology. The result of the large number of variables compared to the number of cases is that we have highly *underdetermined* systems. That is, we are making measurements of very high dimensionality (on the order of tens of thousands) but we are only providing a small number of cases to explore this high-dimensional space. Another way to say this is that there are many, many ways in which the variables being measured could be interrelated mechanistically, based on the relatively small number of observations. Due to this high dimensionality and the underdetermined nature of these systems, standard biostatistical techniques do not hold up well because many of the assumptions that underlie these conventional biostatistical techniques do not hold.^[6] We often provide the following analogy. To solve a linear equation of one variable (e.g., $4x = 5$) we only need one equation to find the value of the variable. To solve a linear equation of two variables (e.g., $y = 4x + b$), two equations are required. If we have tens of thousands of variables, but only hundreds of equations, then there will be thousands of potentially valid solutions. This is the essence of what constitutes an underdetermined system. In this context, we must use techniques that can maximally inform us of the relationships between variables of interest (and find out which ones are of interest) despite the underdetermined nature of the data sets. High-dimensionality data sets have been well-known to the "machine-learning" community of computer scientists in applications such as automated recognition of human faces, so it is not surprising that many of the techniques developed by that community have found their way into the functional genomics enterprise.

^[1]For those of you who are unfamiliar with what these are, we touch on defining these terms in section 1.5 and in the glossary.

^[2]Using the common shorthand title for the international public and private effort to determine the sequence of bases in the human genome.

^[3]The characteristics of the various microarray technologies are addressed in chapter 3.

^[4]The label applied often seems to be determined more by the training background of the labeler rather than any fundamental characteristic of the analytic technique.

^[5]See chapter 2 for a discussion of which tissues are appropriate for particular experiments.

^[6]Although this holds true for most of the biostatistical techniques biologists will have learned in graduate school, in fairness there has been quite a lot of research by statisticians on the analysis of underdetermined systems of high dimensionality. Their work has just not found its way into mainstream biomedical study until recently.

1.3 Missing the Forest for the Dendrograms, or One Aspect of Integrative Genomics

In the first 2 years of significant publications regarding the large scale application of microarray technologies, numerous special purpose or adapted machine-learning algorithms were described in the literature. Self-organizing maps [175], dendrograms [27,63,76], K -means clusters [101], support vector machines [33,72], neural networks [107], and several other methodologies (borrowed largely from the machine-learning community of computer science) have been employed. Most of these have worked reasonably well for the purposes described in the papers. It is one of the central contentions of this book, however, that the choice of a particular clustering or classification methodology is secondary to proper experimental design and full knowledge of the properties and limitations of massively parallel expression analysis in general and those of the specific microarray technologies employed. This contention does not constitute an overly fastidious approach to functional genomics but the insight from (often expensive) experience gleaned through our own mistakes and those of our colleagues and the reported investigations. The bioinformatics technique

centric approach has been evident in our own collaborations. Often, at the outset of a new investigation, our collaborators from both the computational community and the biological community immediately wish to address the questions of which are the appropriate clustering techniques and which one is "the best." While we certainly recognize that the choice of clustering or classification technique is important (and we devote chapter 4 to this matter), we are firmly convinced that it is only one part of a well-designed pipeline that defines a successful exploration of biology and medicine using microarray technology. This pipeline is diagrammed in figure 1.5 on page 17. We refer to this functional genomics pipeline throughout the book. We discuss at length the practicalities of assembling this pipeline in subsequent chapters, but a few characteristics of this pipeline bear mentioning here:

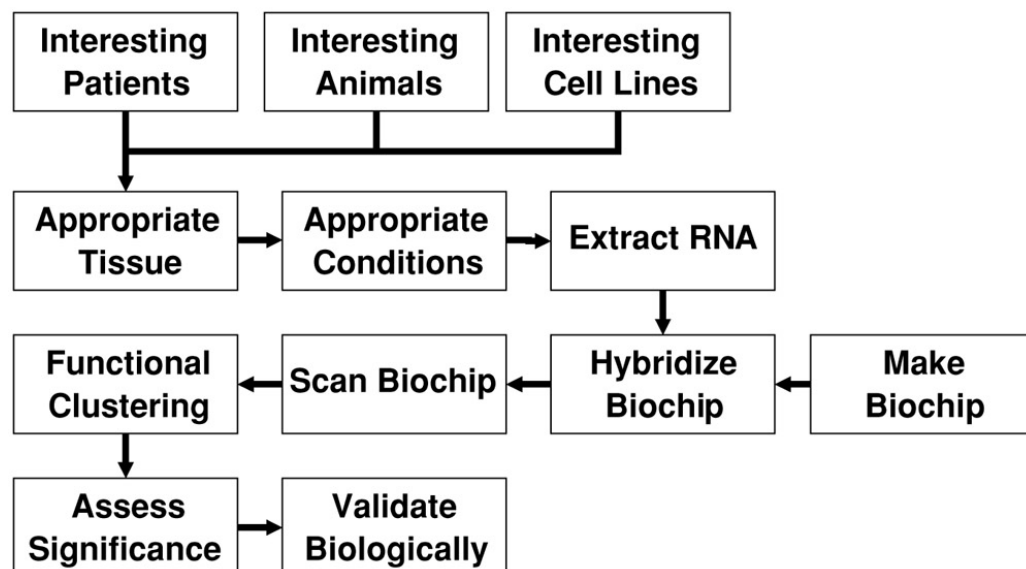


Figure 1.5: An archetypal functional genomics pipeline. Shown is a simplified view of a functional genomics pipeline solely involved in expression microarray experiments. Note the interdigitation of "wet" and "dry" components requiring close multidisciplinary collaboration and some creative consideration of the value of the individual contributions in this pipeline for a particular experiment and publication.

- *Selection of the right tissue.* Experiments in functional genomics require selection of the functionally relevant tissue or cell type. In certain experiments, like those using blood and solid cancers, the functionally relevant tissue is clear. In other analyses, the functionally relevant tissue is not so easily ascertainable or acquirable. For example, the clinical phenotype seen in type 2 diabetes mellitus, or insulin resistance, involves the coordinated physiological dysfunction of several organs and cell types, including liver, muscle, and fat cells. Schizophrenia involves a higher-order brain dysfunction, but brain cells are not easily accessible in humans. For some common diseases like hypertension, it is not clear what the functionally relevant tissue is. A successful pipeline involves collaboration with a source of tissue, such as a surgical team, a laboratory with biologically interesting animals, or a laboratory with cell lines of interest.
- *Right conditions.* Even if the appropriate tissue is selected from the organism of interest, the conditions under which the tissue is obtained (*e.g.*, number of hours post mortem) can determine whether or not the investigation is successful. An insulin-sensitive tissue such as skeletal muscle will have a different characteristic metabolic and expression profile depending on the glucose and insulin concentrations prior to the extraction of RNA. The time of day will influence the expression of genes in all tissues which have endogenous circadian rhythms or that have processes that can be entrained by physiological clocks. Awareness of these issues and cooperation from a surgeon, pathologist, or technician responsible for

obtaining the tissue is therefore an essential component to the success of the functional genomic pipeline.

- *Extracting RNA, hybridizing to microarray, and scanning.* Each of these steps in this "wet" component of a functional genomics pipeline is susceptible to operator error and is a potential source of poor or noisy measurements. The RNA extracted may be of poor quality, the hybridization conditions may vary (e.g., the room temperature), and the settings of the scanner that produces the digital image of the microarray may vary from one scan to another. Industrialization and standardization of this component has been the focus of the more successful and high-quality functional genomics efforts using expression microarrays.
- *Functional clustering.* This "dry" component of the pipeline is often thought to be what bioinformatics is about. And in fact, it may be at this stage that the algorithmic analysis of an expression profiling study to detect biologically or clinically meaningful patterns or associations is the only time a bioinformatician is involved. We will argue throughout this book that a successful functional genomics pipeline involves the bioinformatician at every step.
- *Computational validation.* As will be elaborated in this book, there are many reasons to perform bioinformatics analyses on functional genomics data sets, and many methods can be used. One unique problem with these types of data sets is that they are "short and wide," meaning that many characteristics are measured on relatively few samples. For example, current microarrays offer the quantitation of up to 60,000 expressed sequence tags (ESTs) in any given sample, but current costs may limit a single experiment to 10 to 100 samples. Because of this problem, these data sets are essentially underdetermined, as described on page 10, meaning that there are many correct ways to mathematically describe the clusters and genetic regulatory networks contained within them. Thus, some computational validation is required immediately after the bioinformatics analysis so that computationally sound but biologically spurious or improbable hypotheses are screened out.

The principal motivation for the screening out of spurious or improbable hypotheses is the efforts that follow. Each hypothesis generated that passes this step may need to be validated in a biological laboratory. Some biological laboratories may wish (and may have the resources) to pursue many hypotheses and can tolerate the eventual refutation of large numbers of false-positive hypotheses. Other biological laboratories may only be able to validate a few. Thus, a proper bioinformatics analysis includes a computational validation. An ideal computational validation does not merely provide a yes or no answer as to the potential validity of a hypothesis, but instead provides a continuum of validation, or a receiver-operating characteristic curve. With such a curve, the biologist can select the desired point of sensitivity or specificity and true and false negatives and positives (see sections 2.1.4 and 4.12.3).

- *Biological validation.* Most biological questions will not be answered using microarrays. Instead, the most likely outcome from a functional genomics analysis is the next biological question to ask. As hypotheses are generated from bioinformatics analyses, biological validation is crucial to verify these hypotheses. This verification may include, for instance, making sure a particular set of genes is truly expressed at the proper time and place as hypothesized, using conventional biological techniques such as Northern blotting and *in situ* hybridization.
- *The multidisciplinary team.* In most settings, all of these steps, from acquisition of source material, to microarray construction, to bioinformatics analysis, to biological verification, cannot be performed by a single group or laboratory. A successful functional genomics pipeline brings together resources from many disciplines and of varied backgrounds. Two anecdotes serve to illustrate the value of this multidisciplinary approach.

We were in the process of analyzing a large number of microarray expression data obtained from

skeletal muscle for some colleagues interested in muscular dystrophy—a class of genetic diseases of muscle. They were gratified when our clustering analyses found interactions between transcriptional factors and contractile proteins that they had discovered just months before using conventional molecular–biological techniques as well as several new but plausible interactions. However, because the clustering analyses were exhaustive, they also identified several hormonal interactions that were not of primary interest to these neuromuscular specialists. Using annotation tools linking the microarray data to several national databases, it became quickly apparent that these hormonal interactions were thought to be exclusive to adipocytes (cells constituting the principal component of fatty tissue) but we had just found suggestive evidence to the contrary. The multidisciplinary nature of our effort allowed the formulation of well–posed questions directly related to the interests of the biological investigators and yet kept us open to important hypotheses generated from the data.

We are participating in a study of the functional genomics of the developing brain using mouse models. We had computed, using approximately two dozen microarray data sets produced by our collaborators—developmental biology researchers—a list of approximately 100 genes that appeared to be involved in the development of a specific region of the brain. Our collaborators were in the process of selecting a subset of these for biological validation but we were worried about the outcome of the validation because the data had been derived from entire portions of the brain whereas the process the developmental biologists were interested in only occurred in a minute component of the brain. It seemed probable that many of the 100 genes were not specific to the processes we were studying. Given only the expression data from the microarrays, none of the bioinformatics techniques were able to further refine or hone the list of 100. Fortunately, the developmental biologists provided us with the following insight. They knew of one gene g that was expressed in the tiny area of the brain that they were studying and they had determined empirically that it was expressed in no other part of the brain. They suggested that we find all those genes in the list of 100 that behaved very closely to g . We found 10 such genes and our collaborators went on to successfully validate 8 of them using the techniques of conventional hypothesis–driven molecular biology. If we had not drawn on the multidisciplinary capabilities of our team for that small but crucial biological insight, then we would have been stuck with a large list of nonspecific genes of little relevance to the questions originally posed by the developmental biologists.

When the initial design of the multidisciplinary functional genomics pipeline is given short shrift and the fundamental limitations of expression microarray technologies misunderstood, the enterprise of functional genomics appears to approximate the "fishing expedition" that has been the oft–stated concern of traditional biologists regarding this nascent discipline. Consequently, even if a particular investigator participates in only a fraction of the pipeline, understanding the safe design of an entire functional genomics pipeline can maximize the yield of these experiments, or at the very least produce convincing and reproducible negative results. It is the intent of this book to point the way to investigations that provide such an understanding.

Because of the dramatically different backgrounds (at least at present) of the different contributors to the functional genomics pipeline, its social dynamics may be challenging, as described below. We pay attention to these dynamics because one aspect of an *integrative genomics* is the integration of disciplines and experts (the other side will be described before the end of this chapter).

1.3.1 Sociology of a functional genomics pipeline

Given the requirement for a multiinvestigator, multi–institutional effort, several pragmatic decisions and realities must be confronted. We describe these in the context of academic collaborations but analogies to the commercial world are obvious. The first question in academic collaborations around

the functional genomics pipeline often is: Who will get the credit for the work and for the discoveries that ensue from a particular functional genomics investigation? More concretely, who will get first and senior authorships in the publications that report on the discoveries obtained from this functional genomics discovery pipeline? Will it be the surgeons who obtained the tissue, the bioinformaticians who performed the cluster analysis, the biologist who runs the microarray facility, or the clinician who obtained the phenotypic characterization of the patient from whom the tissue was obtained?

Resolving this issue is nontrivial because of some fundamental human and cultural considerations of the various types of investigators. Within each discipline, an investigator will tend to see those outside his or her discipline as performing more of a utility function rather than making a significant intellectual and creative contribution to the research process. For example, a molecular biologist might view the bioinformatician as providing cookbook analytic algorithms to winnow out the relevant findings from a biological system that they have spent time, energy, and creativity in developing. Conversely, the bioinformatician might view the biologist as a laborer plodding along in the murky swamps of biological experimentation on which the bioinformatician can then shed light by virtue of his or her insights into the general principles of automated inference, clustering, and classification.

For those of us who have labored on both sides of this cultural divide, it is quite clear that there are plodders and creative geniuses in both fields. Successful collaborations in this arena require thoughtful recognition of relevant contributions prior to initiating a long-term collaboration around this strategy. In our own collaborations, we have adopted the following heuristics: If the bioinformatics techniques used are not novel and the interpretation is rote, then the biologists and clinicians usually assume first and senior authorships. If the experimental design is innovative and informed by the nature of the bioinformatics analyses, and the computational techniques have to be developed or customized, then the bioinformaticians usually take first and senior authorships. Often, the major contributions are mixed, as when the bioinformatics analyses are novel and the biological validation steps are creative and arduous, in which case the authorships are split accordingly. In the end, however trite as it sounds, it is true that nothing substitutes for the collegiality and good will that arise from mutual respect for different skills and contributions.

1.4 Functional Genomics, Not Genetics

We have noticed, among some of our biological collaborators, a tendency to view the massively parallel methods of functional genomics as a highly efficient large-scale application of methods that they have already applied. For instance, gene expression profiling and polymerase chain reaction (PCR) are all methods that have been used by molecular biologists for decades. What we hope the reader will obtain from this book is an appreciation of how the near-comprehensive (and soon to be truly comprehensive) nature of the functional genomics approach as permitted by expression microarrays changes qualitatively and fundamentally the nature of biological investigation. Before our potential readers with biological backgrounds become offended by or disgruntled with this assertion, let us assure them that we present an equivalent critique for the purely computationally oriented bioinformaticists and genomicists in the following section. Functional genomics is not, as some have portrayed it, a hypothesis-free fishing expedition, nor is it, even more charitably, only a hypothesis-generating enterprise requiring subsequent biological validation. It is fundamentally different in that it permits the posing of large questions that are grounded in an essential biological understanding of a particular domain. Unlike the questions posed in "traditional" genetics or molecular biology, these questions have less stringent requirements for prior supposition or claims of the role of a particular gene or metabolite in a biological process. An example of some of the broad questions that can be asked are:

- Which of all the known genes have regulatory mechanisms that appear to be similar to those regulated by the sonic hedgehog transcription factor in the cerebellum?
- Given the effect of 5000 drugs on various cancer cell lines, which gene singly is the most predictive of the responsiveness of the cell line to any chemotherapeutic agent?
- Given a known clinical distinction, such as that between acute lymphocytic leukemia and acute myelogenous leukemia, what is the minimal set of genes that can most reliably distinguish these two diseases?
- Is there a group of genes that can serve to distinguish the outcomes of patients with large B-cell lymphoma that are otherwise clinically indistinguishable on presentation?
- What distinguishes the signaling pathways of two of the substrates of the insulin receptor?

These questions are important biologically and clinically, and yet they can only be posed reasonably if they involve a comprehensive view of genomic regulation and involve the use of computational methods that can efficiently sift through the vast quantities of genomic data generated by the experiments required to answer these questions. Another way to consider functional genomics is to view it as serving as a filtered funnel through which these broad questions can be strained. The residue that remains is high-yield, detailed, and contains particular biological questions that are answerable by more traditional genetic or molecular biology techniques. This is illustrated in figure 1.6 below. The utility of this metaphor is as follows. The universe of possible participants in any given biological regulatory mechanism is finite but very large. Even with the most comprehensive in-depth expertise, a biologist may find be surprised about insights obtained through data mining of expression patterns, human genome sequences, and often from data obtained from other species. Without a genomic approach to guide her experiments, this biologist may expend several months in false leads or alternatively miss an important component to the system under study. Similarly, if the biologist is looking at a set of genes that are thought to be predictive of a given clinical condition, such as transplant rejection or cardiac disease, without the comprehensive view brought by functional genomics, elements of the diagnostic or prognostic procedure, such as the concentration of a gene transcript or a protein, may be omitted with a concomitant decrease in the sensitivity and specificity of the prognosis or diagnosis.

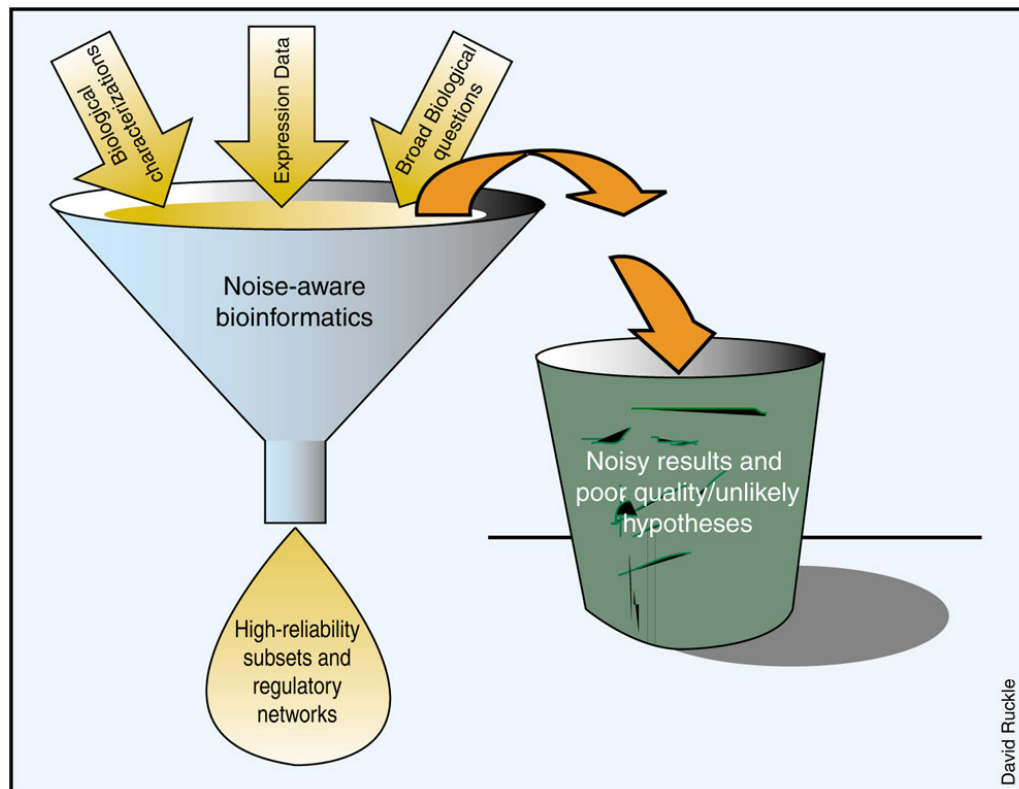


Figure 1.6: The functional genomics investigation as a funnel for traditional biological investigations. Broad questions and comprehensive data are the mix in which bioinformatics techniques are filtered to separate high-yield hypotheses or candidate genes from spurious findings and poor-quality hypotheses.

1.4.1 *In silico* analysis will never substitute for *in vitro* and *in vivo*

There is little doubt that one of the tremendous accomplishments of the Human Genome Project is that it has enabled a rigorous computational approach to identifying many questions of interest to the biological and clinical community at large.^[7] However, the danger of a computational triumphalism is that it makes several dubious assumptions. The first is genetic reductionism: At an abstract level most bioinformaticians understand that a particular physiology or pathophysiology is the product of the genetic program created by the genome of an organism and its interaction with the environment throughout its development and senescence. In practice, however, a computationally oriented investigator often assumes that all regulation can be inferred from DNA sequence, based solely on the syntax of its sequence elements. That is, it is assumed that it is predictable whether a nucleotide change in a sequence will result in a different physiology. We refer to this as "sequence-level reductionism."

The second dubious assumption is the computability of complex biochemical phenomena. One of the most venerable branches of bioinformatics involves modeling molecular interactions such as the thermodynamics of protein folding, and protein-protein and protein-nucleic acid interactions. As yet, all the combined efforts and expertise of bioinformaticians have been unable to provide a thermodynamically sound folding pattern of a protein in the heterogeneous solvent environment of a cell for even as long as one microsecond. Furthermore, studies by computer scientists over the last 10 years [23,114] suggest that the protein-folding problem is "NP hard." That is, the computational challenge belongs to a class of problems that are believed to be computationally intractable. Therefore it seems overly ambitious to imagine that within the next decade we will be able to generate robust predictive models that are able to accurately predict the interactions of thousands or millions of heterogeneous molecules and the ways in which they modulate the transcription of RNA and the translation of messenger RNA (mRNA) into protein and the subsequent functions of these proteins. We refer to this ambition as "interactional reductionism."

This is not to say that models have no useful role in molecular biology or bioinformatics. On the contrary, they are extremely useful to embody what we currently believe we know about biological systems. Where the predictive capabilities of these systems break down points to where we should guide further research. Also, the educational value of such models cannot be underestimated.

The final questionable assumption is the closed-world hypothesis. Both sequence level reductionism and interactional reductionism are predicated upon the availability of a reliable and predictive and complete mechanistic model. That is, if a fertilized ovum can follow the genetic program to create a full human being after 9 months, then surely a computer program should be able to follow the same genetic code to deterministically infer all the physiological events that are determined by the genetic code. Indeed, there have been several efforts, such as the E-cell effort of Tomohita *et al.* [179], which aimed to provide robust models of cellular function based on the known regulatory behavior of cellular systems. Although such models have important utility, our knowledge of all the pertinent parameters for these models appears grossly incomplete today. These parameters are required to describe intracellular processes, intercellular processes, and the unimaginably large repertory of possible environmental interactions with both sets of processes. This incompleteness, and the lack of knowledge of where the boundaries are between the complete and the incomplete, imply that these models will have behaviors that may diverge substantially and unpredictably from those that actually occur.

These caricatured positions of the traditional molecular biologists and the computational biologists are, of course, overdrawn. When prompted, most of these investigators will articulate the fullness of the complexities of the analytic tasks of functional genomics. In the conduct of their research or even in the discussions within their publications, these same investigators will nonetheless often retreat to the simplifications and assumptions described above. This may be because they take for granted that their colleagues and readers understand these simplifications, but such unstated assumptions can often misdirect novices in this discipline.

What we argue for, and hope that this book communicates, is the necessity for a rapid generate–and–test paradigm that cross cuts repeatedly across the disciplines of genetics, computational biology, and molecular biology. Operationally, this means that bioinformatics tools can be used to guide the investigations of an experimental biologist investigating a particular biological system or disease process. But for even the smallest assumption, rather than relying on the statistical association or predicted behavior of a system, empirical evidence has to be developed to support these. It is only in this incremental accretion of evidence that the discipline of functional genomics can become a science.

Why microarrays? Why focus on microarrays? After all, there are many other ways to impute function to genes. Using only genomic DNA easily obtainable from a peripheral blood sample or a buccal smear, genetic epidemiologists can conduct association studies using microsatellite markers or polymorphisms [43] to associate prognoses, diagnoses, and even biological function with a particular gene [67,126]. And then there are the more conventional genetic techniques of transgenic and misexpression whole–organism models of the function of various genes. Even more recently, the feasibility studies for proteomic assays suggest that in the future we will directly be able to assess changes in protein concentration at the cellular level.

In contrast to linkage and association studies, microarray studies are designed in principle to measure *directly* the activity of the genes involved in a particular mechanism or system rather than their association with a particular biological or clinical feature. An association–linkage–genetic epidemiology study relies upon a long indirect probabilistic causal chain: that a change in DNA sequence results in a change in gene regulation or protein structure, resulting in a change in cellular physiology measurable as a change in a whole–organism profile (*e.g.*, the human phenotype). For some changes in genomic sequence, particularly in the instances of multigenic regulation, the effects may be so small that any conceivable population study may not be able to detect them.^[8] Also, the cost of screening the genome of sufficiently large populations to achieve adequate statistical power has prohibited all but the most focused association studies (although this is likely to change). Unlike the current state of art and engineering of large–scale proteomic assay systems, gene microarrays are currently affordable and within many applications have acceptable reproducibility and accuracy.

Another aspect of an integrative genomics Notwithstanding these apparent advantages of expression microarray studies, as we discuss in sections 1.5.3 and 1.5.2, there are several kinds of information that we are missing by not including such measurements. Our decision to restrict the scope of this book to the exploration of functional genomics and genomic medicine from the perspective of microarray technology is then largely a pragmatic one. Expression microarrays are sufficiently well engineered and cost–effective to allow thousands of researchers to productively employ them to drive their investigations. If, in the future, as we expect, massively parallel measurements of individual proteins becomes cost–effective, large–scale, and highly reproducible, then we will certainly expand the analysis to address these methodologies. The same will be true when high–resolution (*i.e.*, every kilobase) genome–wide scans of hundreds of individuals will become economically feasible for most clinical research studies. Current estimates have these technologies available on the genomic and population scale within 5 years. A well–prepared

genomic investigator will have prepared the pipeline to take advantage of all these measurement technologies.

From the computational perspective, the measurement of any analyte, whether it be an inorganic constituent of serum, an RNA transcript, or a protein, are all simply point measurements of variables corresponding to the total state of the cell. Likewise, all clinical measurements (*e.g.*, height, blood pressure) and history (*e.g.*, age of menarche, time of cancer diagnosis) are point measurements corresponding to the total state of the organism (*e.g.*, human). It is only in the important details of the quality and meaning of these measurements that they differ. This is said both with tongue planted firmly in cheek and in all seriousness.

And indeed, it is this other aspect of an *integrative genomics* to include as many modes of data measurement as are available. Each mode reflects another aspect of cellular and organismal physiology each with its own set of specificities and sensitivities with respect to a phenomenon or process of interest. The role of bioinformatics in an integrative genomics is to both provide the glue bringing together all kinds of genomic and phenotypic data and the means to extract knowledge (or at least high-yield hypotheses for subsequent testing) from them in an efficient, large-scale, and timely fashion. It is also the inspiration for the title of this book.

^[7]Although even the most basic of the original conclusions, stated in the publications heralding the completion of the draft of the human genome, the order of magnitude of the number of genes in the human genome, now seems to be again in contention.

^[8]That is, there would be insufficient numbers of individuals with the necessary constellation of phenotypes across the entire human population. At the same time we recognize that there are several diseases, such as sickle cell anemia, where the change of one base in the hemoglobin gene results in a severe and unsubtle disease phenotype.

1.5 Basic Biology

This section is meant solely for the quantitatively trained scientist who knows essentially none of the biology developed over the last three decades. For those of you who need a significant refresher in molecular biology we list several good introductory texts and on-line resources (table 1.1). So we start from the beginning. In almost all cells making up a living organism, there is believed to be an identical set of codes describing the genes and their regulation. This code is encoded as one or more strands of the deoxyribonucleic acid molecule: DNA. That is the same in almost every cell in the human body.^[9] For instance, a liver cell and a brain cell have the same DNA content and code in their nucleus. What distinguishes these cells from one another is that portion of their DNA that is transcribed and translated into protein, as described below.

Table 1.1: Molecular biology primers

- *Genomes*.
T. A. Brown & Austen Brown. New York, Wiley-Liss, 1999.
- *Human Molecular Genetics*, 2nd edition.
Tom Strachan & Andrew Read. New York, Wiley-Liss,

1999.

- *Primer on molecular genetics*
<http://www.bis.med.jhmi.edu/Dan/DOE/intro.html>.
- *Primer on genomics with a commercial flavor*
<http://www.biospace.com/articles/genomics.primer.cfm>
- *Introductory biology course at MIT (7.01) hypertext book*
<http://www.esg-www.mit.edu:8001/esgbio/701intro.html>

The entire complement of DNA molecules of each organism is also known as its *genome*. The overall function of the genome is to drive the generation of molecules, mostly proteins, that will regulate the metabolism of a cell and its response to the environment, as well as provide structural integrity.

Structure of DNA Each molecule of DNA maybe viewed as a pair of chains of the nucleotides adenine (A), thymine (T), cytosine (C), and guanine (G). Moreover, each chain has a polarity, from 5' (head) to 3' (tail). The two strands join in opposing polarity (5' binds to 3') through the coordinated force of multiple hydrogen bonds at each base-pairing, where A binds to T and C binds to G. ^[10]

DNA is able to undergo duplication, which occurs through the coordinated action of many molecules, including *DNA polymerases* (synthesizing new DNA), *DNA gyrases* (unwinding the molecule), and *DNA ligases* (concatenating segments together).

Transcription of DNA into RNA In order for the genome to direct or effect changes in the cytoplasm of the cell, a transcriptional program may be activated for the purpose of generating new proteins to populate the cytosol—the heterogenous intracellular soup of the cytoplasm. DNA remains in the nucleus of the cell, while most proteins are needed in the cytoplasm of the cell, where many of the cell's functions are performed. Thus, DNA is copied into a more transient molecule called RNA. ^[11] A *gene* is a single segment of the coding region that is transcribed into RNA. RNA is generated from the DNA template in the nucleus of the cell through a process called *transcription*. ^[12]

The RNA sequence of base pairs generated in transcription corresponds to that in the DNA molecules using the complementary A–T, C–G, with the principal distinction being that the nucleotide uracil is substituted for the thymine nucleotide. Thus, the RNA alphabet is *ACUG* instead of the DNA alphabet *ACTG*. Each cell contains around 20 to 30 pg of RNA, which represents 1% of the cell mass. The RNA that codes for proteins is called *messenger RNA*, and the part of the DNA that provides that code is called an *open reading frame* (ORF). When read in the standard 5' to 3' direction, the portion of DNA before the ORF is considered *upstream*, and the portion following the ORF is considered *downstream*.

The specific determination of which genes to transcribe is determined by *promoter regions*, which are DNA sequences upstream of an ORF. Many proteins have been found containing parts that bind to these specific promoter regions, and thus activate or deactivate transcription of the downstream ORF. These proteins are called *transcription factors*. ^[13]

A diagram of the genetic information flow, from DNA to RNA to protein, is illustrated in Figure 1.7

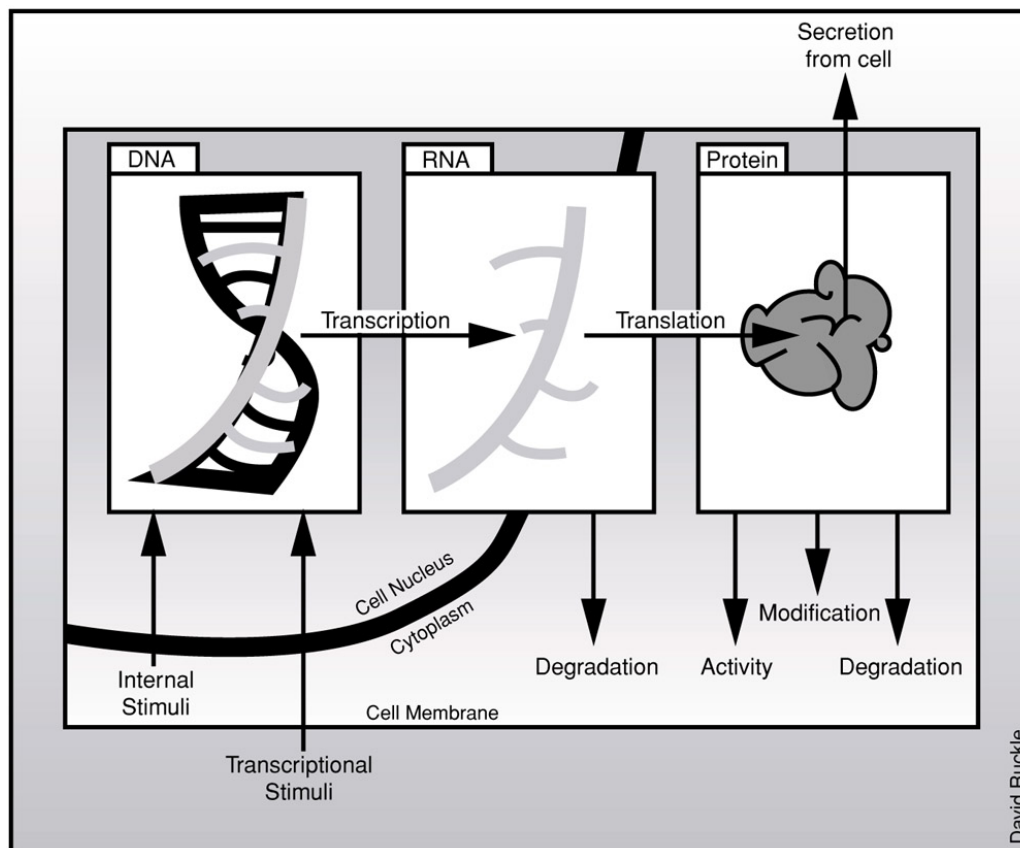


Figure 1.7: Flow of genetic information, from DNA to RNA to protein. This simplified diagram shows how the production of specific proteins is governed by the DNA sequence through the production of RNA. Many stimuli can activate the specific transcription of genes, and proteins can play a wide variety of roles within or outside cells. Note that even in this simplified model, it is obvious that since we are currently able to (nearly) comprehensively measure only gene expression levels, we are missing comprehensive measurements of protein modification, activity, transcriptional stimuli, and many other components of the state of a cell.

Prokaryotic and eukaryotic cell types Although there are many taxonomies one could use, we can essentially divide the world of organisms into two types: *eukaryotes*, whose cells contain compartments or organelles within the cell, such as mitochondria and a nucleus; and *prokaryotes*, whose simpler cells do not have these organelles. Animals and plants are examples of eukaryotes, while bacteria are prokaryotes. Most prokaryotes have a smaller genome, typically contained in a single circular DNA molecule. Additional genetic information may be contained in smaller satellite pieces of DNA, called plasmids.

The structure and processing of RNA transcripts Eukaryotic genes are not necessarily continuous; instead, most genes contain *exons* (portions that will be placed into the mRNA) and *introns* (interruptions that will be spliced out). Functions have been recently discovered for introns, such as promoter-like control of the transcription process. Introns are not always spliced consistently; if an intron is left in the mRNA, an *alternative splicing product* is created. Various tissue types can flexibly alter their gene products through alternative splicing.^[14]

Before coding RNA is ready as mRNA, the pre-mRNA must be processed.^[15] In eukaryotes, after the splicing process, the generated mRNA molecule is actively exported through nuclear pore complexes into the cytoplasm. The cytoplasm is where the cellular machinery, in particular the ribosomal complex,^[16] acts to generate the protein on the basis of the mRNA code. A protein is built as a polymer or chain of amino acids, and the sequence of amino acids in a protein is determined by the mRNA template. The mRNA provides a degenerate coding in that it uses three nucleotides to

code for each of the twenty common naturally occurring amino acids that are joined together to form the polypeptide or protein molecule. With three nucleotides there can be 4^3 possible combinations for a total of 64 combinations. Consequently, several trinucleotide sequences (also known as codons) correspond to a single amino acid. There is no nucleotide between codons, and a few codons represent start (or initiation) and stop (or termination).^[17] The process of generating a protein or polypeptide from an mRNA molecule is known as *translation*.

As an example, if an RNA transcript had the nucleotide sequence

GCT TGC AAG GCG

(the spacing and grouping by three, which obviously is not visible in the RNA transcript, was added to make this easier to read), the first three nucleotides would code for this chain of amino acids

Alanine Cysteine Arginine Alanine

Note that both the initial GCT and the final GCG code for alanine; because of the degenerate coding mentioned above, there are four codes for alanine in the standard genetic code.

Most mRNA has a terminating poly-A tail. This terminal sequence makes it easy to pick out the labeling reactions that are used before hybridization to DNA microarrays, described in section 3.1.

Processing of amino acid chains Once the protein is formed, it has to find the right place to perform its function, whether as a structural protein in the cytoskeleton, or as a cell membrane receptor, or as a hormone that is to be secreted by the cell. There is a complex cellular apparatus that determines this translocation process. One of the determinants of the location and handling of a polypeptide is a portion of the polypeptide called the *signal peptide*. This header of amino acids is recognized by the translocation machinery and directs the ribosomal-mRNA complex to continue translation in a specific subcellular location, *e.g.*, constructing and inserting a protein into the endoplasmic reticulum for further processing and secretion by the cell. Alternatively, particular proteins may be delivered after translation and chaperones can prevent proper folding until the protein reaches its correct destination.

Transcriptional programs Initiation of the transcription process can be caused by external events or by a programmed event within the cell. For instance, the piezoelectric forces generated in bones through walking can gradually stimulate osteoblastic and osteoclastic transcriptional activity to cause bone remodeling. Similarly, heat shock or stress to the cell can cause rapid change or initiation of the transcriptional program. Additionally, changes in the microenvironment around the cell, such as the appearance of new micro- or macronutrients or the disappearance of these, will cause changes in the transcriptional program. Hormones secreted from distant organs bind to receptors which then directly or indirectly trigger a change in the transcriptional process.

There are also fully autonomous, internally programmed sequences of transcriptional expression. A classic example of this is the internal pacemaker that has been found with the *clock* and *per* genes (see figure 1.8 below) where, in the absence of any external stimuli, there is a recurring periodic pattern of transcriptional activity. Although this rhythmic pattern of transcription can be altered by external stimuli, nonetheless it will continue initiating this pattern of transcription without any additional stimuli.

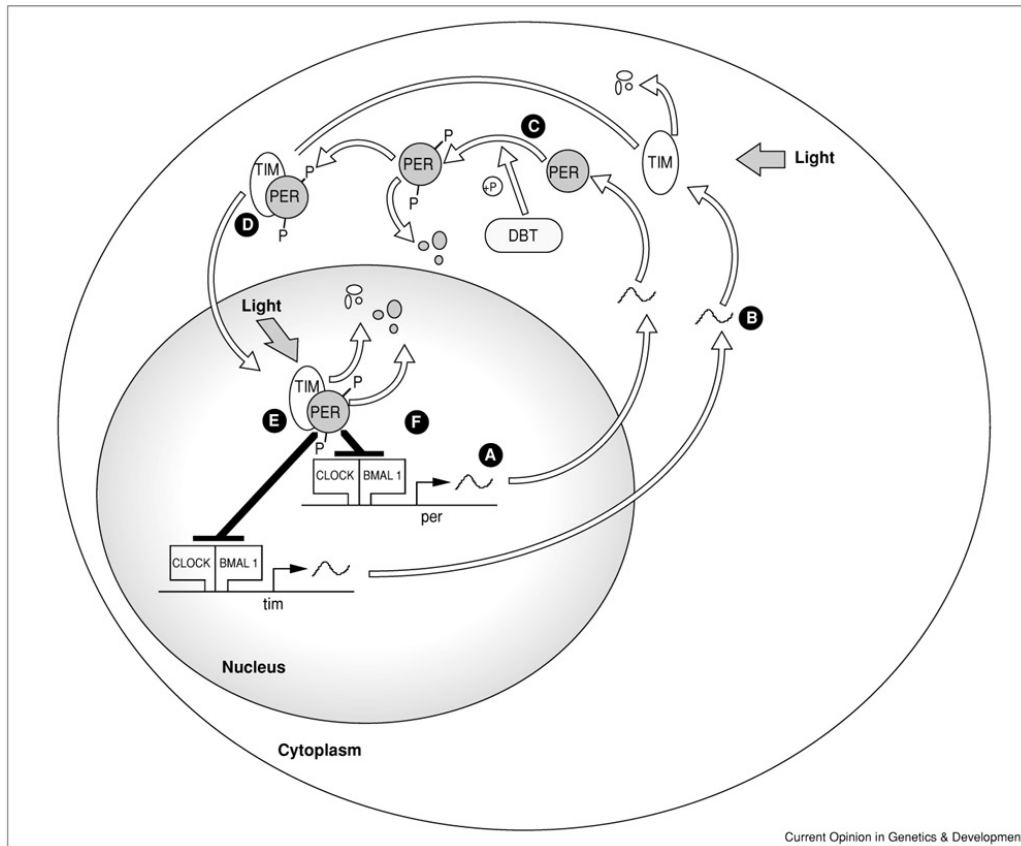


Figure 1.8: Genetic machinery of the circadian rhythm. Current molecular model of rhythm generation in *Drosophila*, from [191]. The succession of events (A–F) occur over the course of approximately 24 hours. A, CLOCKBMAL heterodimers bind the *per* and *tim* promoters and activate mRNA expression from each locus; CLOCKBMAL may also activate transcription of other circadian-regulated genes (not shown). B, *per* and *tim* mRNA are transported to the cytoplasm and translated into PER and TIM protein, respectively. C, Regulation of protein levels occurs by two mechanisms: DBT protein phosphorylates and destabilizes PER, and light destroys TIM. Light during the early subjective night can phase-delay the clock. Small "blobs" indicate degraded proteins. D, PER and TIM levels slowly accumulate during the early subjective night; TIM stabilizes PER and promotes nuclear transport. E, PER and TIM dimers enter the nucleus and inhibit CLOCKBMAL-activated transcription. F, Protein turnover (combined with the lack of new PER and TIM synthesis) leads to derepression of *per* and *tim* mRNA expression; the cycle begins again (A). Light during the late subjective night can phase-advance the clock.

Finally, there are pathological internal derangements of the cell which can lead to transcriptional activity. Self-repair or damage-detection programs may be internal to the cell, and can trigger self-destruction (called apoptosis) under certain conditions, such as irreparable DNA damage. As another example, there may be a deletion mutation of a repressor gene causing the gene normally repressed to instead be highly active. There are many clinical instances of such disorders, such as familial male precocious puberty [161] where puberty starts at infancy due to a mutation in the luteinizing hormone receptor. This receptor normally activates only when luteinizing hormone is bound, but with the mutation present, activation does not require binding.

1.5.1 Biological caveats in mRNA measurements

There are several desiderata that should be understood about transcription in order to better understand the limitations of the gene profiling techniques.

- *The set of protein-coding RNA, otherwise known as the transcriptome, should be viewed as a pool of mRNA.* Even at equilibrium, each type of mRNA is degraded at its specific rate. New transcription provides additional mRNA to replace those being degraded, maintaining the levels. Thus, a transcriptional program should more accurately be viewed as a change in the transcriptome. In addition, just because a particular mRNA is seen in the transcriptome, it does not necessarily imply that the mRNA is about to be translated. Similarly, presence of a particular mRNA in the transcriptome does not mean that mRNA was recently transcribed. To determine what genes may be in the process of being translated, one can specifically look at polyribosomal mRNA, or that mRNA found in the presence of ribosomes. To determine which genes are currently being transcribed and processed, one can construct studies looking for pre-mRNA levels instead of mRNA levels (*i.e.*, by finding intronic splice sites, *etc.*).
- *mRNA can be transcribed at up to several hundred nucleotides per minute but may be transcribed at much slower rates.* In eukaryotic organisms, genes can take many hours to transcribe. For instance, a gene found in muscle, dystrophin, can take up to 20 hours to be transcribed. The lifespan of an mRNA molecule from initiation of transcription to ultimate degradation is a complex function of the rate of transcription, the stability of the particular mRNA molecule, and changes in its processing due to other cellular events, whether internally or externally initiated. Gene expression measurements, as performed by microarrays, measure the concentration of each mRNA as a snapshot at a single point of time or relative to another sample of mRNA. If the transcription of two genes is simultaneously stimulated and proceeds at the same rate, but the mRNA samples have different lifespans, then the expression measurements (and downstream protein production) from these two genes can be quite divergent.
- *Proteins perform most cellular functions.* The lifespan of proteins is at least as variable as that of mRNA. Consequently, measurements of gene expression (*i.e.*, mRNA measurement) may not accurately correspond to the concentration or activity of the protein for which it codes. This should be a caution to any investigator imputing function to a gene based solely on gene expression patterns.
- *The mRNA generated from the transcription of a gene may differ depending on which exons and introns are or are not spliced into the mRNA molecule.* These alternate splicing products are mRNA molecules with divergent sequences. Therefore, measurements of mRNA that are designed to measure only one of these alternate splicing products will provide incomplete if not misleading information.
- *The genetic basis for organismal diversity is due in large part to differences in sequences, also known as polymorphisms, of each gene.* Most of these polymorphisms differ from one another by one nucleotide and are known as single nucleotide polymorphisms (SNPs). Due to the small portion of the genome coding for proteins and to redundancy in the mRNA code, described on page 27, only some SNPs will result in differently constructed proteins. If a gene expression measurement technology is highly specific for a particular SNP, then other variants will not be measured. When we consider that the Human Genome Project to date only includes the sequence of handfuls of individuals, the implications of such measurement specificity become apparent. Nonetheless, as common SNPs become documented for each gene, it is likely that successful expression measurement techniques will measure each of them.

Despite these *caveats*, it remains that gene profiling studies have proven to be remarkably robust in describing the functional grouping and coordinated behavior of genes. The reasons for this are discussed in section 2.2, page 60.

1.5.2 Sequence–level genomics

Sequencing or genotyping the genome of individuals allows the characterization of what distinguishes the heritable material of each individual from that of others. By matching differences in phenotype (*e.g.*, blood pressure, adult height) between individuals and this genomic characterization (*i.e.*, in association studies) genetic epidemiologists are able to impute these phenotypic differences to a small span of the genome. The smallness of the span is a function of the spatial resolution with which the genomic characterization occurs. Although there is controversy around what constitutes sufficient resolution, there is some consensus that genomic markers such as SNPs spaced every thousand bases will be sufficient to unambiguously resolve the span of the genome associated with a phenotypic difference to a single gene [112]. Currently, the cost of a single genotype is around \$0.50 and so the cost of a high–resolution genome scan of an individual is on the order of magnitude of \$1 million. At this cost, only a very few institutions can afford a comprehensive study of a population. If the recent past is to be a guide, the cost of genotyping is likely to drop by several orders of magnitude well within a decade, at which point genome–wide scans of populations will become economically feasible.

The kind of information that these studies will provide includes the contribution of particular polymorphisms to changes in phenotype, presumably via changes in gene function, pattern of expression, or both. Currently, expression microarrays capture information about gene polymorphism poorly, if at all. Presently shipping microarrays typically code for a canonical "normal" gene sequence. Departures from this sequence will result in changes in the intensity reading reported by any of the currently employed microarray platforms, as will become obvious after reading the chapter on measurement techniques (chapter 3). Undoubtedly, in the next 5 years, the continued geometric increase in the complexity and density of expression microarrays will allow specific assaying for all common and all clinically significant polymorphisms. Until then, however, polymorphisms will be a source of "noise" or unexplained variation. Functional genomics investigations designed to elucidate the role of particular polymorphisms in phenotypic mechanisms or phenotypic variation are best conducted using methods other than RNA expression detection microarrays, such as high–throughput sequencing and genotyping. These latter techniques are beyond the scope of this text (see [43, 112] for an excellent survey of sequencing and genotyping) and therefore this particular avenue of functional genomics will not be addressed below except tangentially.

1.5.3 Proteomics

There is a good deal of enthusiasm at present about the emerging discipline of proteomics. The promise of proteomics is that we will be able to measure, in a similarly comprehensive and parallel fashion to RNA microarray measurements, the concentrations of proteins present in a particular cellular system. In a very abstract sense, for the purely computational bioinformatician, the field of proteomics does not harbor any particular novelty in that all it provides is another 100,000 variables, or more, to describe the state of cellular processes. In that perspective, a proteomics data set reduces simply to another array that has distinguishing noise characteristics (*i.e.*, sources of biological and measurement variability as described in section 3.2.5) and is just as amenable to the clustering and classification techniques of chapter 4 as any set of microarray expression measurements. Less abstractly, however, proteomics offers a set of insights that are quite different and divergent from those of expression microarrays. The assumption underlying expression microarray measurements is that by capturing the patterns of expression management, we will capture the basic irregularity rhythms of the cell [34]. Although these assumptions may hold at times and have done remarkably well in helping biologists elucidate some fundamental biology and to classify clinical phenomena, there are several persuasive reasons why these assumptions should not always hold. First, we know that most of the effector molecules in cellular metabolism are

proteins. To the extent that the timing of protein synthesis and the half-life of proteins is not closely coupled to that of RNA expression, the assumption of the representativeness of RNA levels does not hold. As outlined in section 1.5, these assumptions do not hold in many instances. Nonetheless, in proteomics, we will be bedeviled by a new set of assumptions that will be equally problematic and challenging. Assuming that we have high reproducibility and compact systems for assessing the concentration of tens of thousands of proteins, we will be faced with the following challenges:

- Similar concentrations do not imply co-regulation. Given that proteins have hugely different half-lives, even within a single cell (e.g., a structural protein in a bone osteoblast and a parathyroid hormone receptor in the same cell), then the concentrations of protein molecules in a cell may only remotely reflect joint regulation. This problem also haunts the analysis of RNA expression microarray data because of the wide range of the stability-degradation rate of mRNA.
- Conversely, repeatedly different concentrations of two proteins imply co-regulation. At any given sampling time, the two proteins could have quite variable concentrations and different mutual relationships. Yet, there is nothing about this to preclude important functional interactions between these proteins.^[18]
- Localization heterogeneity. Unlike transcription of genes, which occurs within the nucleus, protein activity has very distinct and heterogeneous functional significance in different parts of the cellular compartments, and therefore an essential part of understanding protein function and regulation from proteomic data will require detailed localization to subcompartments of organelles in order to be meaningful. This problem also exists with RNA expression microarray data, because of the differing biological implications of RNA concentrations measured before splicing, after splicing in the cytoplasm, and during translation.

These challenges of proteomics will eventually be addressed by novel ways of looking at protein activity over time and in different spatial locations. Nonetheless, at present, the basic mechanism for cheaply and reliably obtaining large numbers of parallel measurements of protein activity have yet to be worked out and industrialized, so that these developments are not likely to occur on a large scale for at least 1 or 2 years. When these challenges have been resolved, then indeed the arrays of proteinomic data will be amenable to the same techniques of analysis as described in this book for RNA expression. For thoughtful and comprehensive insights into the challenges of proteomics and data analytic techniques, we refer the reader to the following papers [25,69,141] and websites <http://www.expasy.ch/>, <http://www.hip.harvard.edu/>.

^[9]There are several exceptions to this rule, such as mature red blood cells, which lack nuclei and therefore the organism's genome, and gametes (spermatozoa or ova), which have half the usual complement of DNA. However, for the purposes of this overview, the above generalization will suffice.

^[10]These pairings are present in all DNA and are the most thermodynamically stable of all possible pairings of nucleotides, which accounts for the high specificity with which complementary strands of nucleotide polymers bind to each other.

^[11]That portion of the entire DNA molecules that is transcribed into RNA is called the *coding region*.

^[12]Transcription involves unwinding a DNA molecule so that the particular gene that is to be transcribed is sufficiently exposed to the transcriptional machinery, notably RNA polymerase.

^[13]Not all RNA codes for proteins, however. In fact, only 4% of total RNA is made of coding RNA. Of the noncoding RNA, ribosomal RNA (rRNA) and transfer RNA (tRNA) are used in various components of the protein translational apparatus mentioned below, and are not themselves translated into proteins. Eukaryotes also contain small nuclear RNA (snRNA), which is part of the splicing apparatus (see below); small nucleolar RNA (snoRNA), which is involved in methylation of rRNA; and small cytoplasmic RNA (scRNA), which can play a role in the expression of specific

genes.

^[14]In fact, some cells can use the ratio of one alternative splicing to another to govern cellular behavior.

^[15]Modifications to the RNA include the addition of a cap at the 5' end and a tail made of repeated adenine nucleotides (the poly-A tail)

^[16]A complex containing hundreds of proteins and special-function RNA molecules.

^[17]There are notable exceptions: the code for the naturally occurring amino acid selenocysteine is identical to that for a stop codon, except for a particular nucleotide sequence further downstream.

^[18]One way to circumvent this problem is with multiple serial measurements of all proteins and then to subject them to a dynamics analysis.

Chapter 2: Experimental Design

2.1 The Safe Conception of a Functional Genomic Experiment

Experimental biologists and biomedical scientists who are considering the use of microarrays in their research programs have often asked us questions of the following form: "How do I best design an experiment with the following mutant mouse?", or, "We have a budget for 20 microarrays to analyze this system. How can we complete the experiment within budget?" Unlike the typical biomedical study design, many of these investigations that the experimentalist are contemplating are driven not by the testing of a particular hypothesis, but by the hope of generating hypotheses that will then lead to a directed and relevant experiment that will subsequently yield an interesting biological or biomedical insight.

We find it even more challenging when we are asked secondary questions related to the subsequent microarray data analyses such as, "What is the right machine-learning technique or clustering technique to apply to a set of data to best obtain an answer about [some biological mechanism]?" The questioner often goes on to ask whether one particular software package works better than another, is more reliable, or whether it is worth its cost. We view these questions as well-intentioned but fundamentally flawed and ill-posed. Although a particular machine-learning algorithm may be slightly superior to another in one set of circumstances (see section 2.2.2 on the relative merits of different clustering algorithms), the key to a successful functional genomic investigation is to start with an experimental design that maximizes the possibility of observing gene expression patterns that are relevant to and informative of the biological aspect or question under investigation. If the experimental design is such, then *most* of the common machine-learning algorithms will reveal those patterns.

So how does one proceed to an effective experimental design for the expression of thousands of genes in order to, hopefully, answer a biological question? We have found that it useful to frame our answers within the concepts of *experiment design space* and *expression space*. Let us start with the experiment design space.

2.1.1 Experiment design space

Experiment design space is the space of potential combinations of stimuli or conditions to which one can subject the regulatory apparatus of the genome. These stimuli include exposure to pharmaceutical agents such as mitogens, environmental effects such as temperature differences, withdrawal of nutritional sources, overexpression of a particular gene such as a transcriptional factor in a transgenic animal, or conversely, an engineered misexpression or "knockout" model. Other common experimental conditions include the passage of time, such as from a newborn to an old mouse; the observation of the environment (organisms from one particular condition vs. another); the observation of organismal processes (organisms with a particular morphology, or afflicted with a particular pathophysiologic condition vs. another); physical manipulation of the cell membrane; exposure to infectious agents; and changes in osmotic conditions, all of which can potentially affect the genomic regulatory mechanisms which maintain cellular existence and the regulation of the cellular response to the environment.

Experimental conditions can be represented as combinations of values on several scales:

- *Binary* scales, *e.g.*, the presence or absence of a diagnosis.
- *Continuous* scales, *e.g.*, such as the gradients of exposure to a chemical. These can have value ranges, say, 10.0, 1.0, 0.1, 0.01, representing the chemical concentration or 0, 5, 10,

15, 20, representing time in minutes from the initial state.

- Unordered *discrete* scales, e.g., the type of mutant mouse model created or which of the multiple mammalian muscle tissue types were sampled.

When different experimental conditions from these different scales are combined, then the experiment can be described as being located in a particular locus of the multidimensional experiment design space. Shown in figure 2.1 below is the simple example of an experimental space over three sets of conditions. In this case, the first dimension represents whether the mouse model being studied is an *irs1* [138] knockout, an *irs2* knockout [192], or a wild-type mouse. The insulin receptor signaling proteins *irs1* and *irs2* are thought to be part of the signaling cascade through which multiple cellular events^[1] are triggered upon binding of insulin to the cellular receptor. The "knockout" is an engineered mouse in which a particular gene is no longer sufficiently expressed. The second dimension is the amount of insulin to which the mouse circulatory system is exposed. The third dimension represents the three types of mouse tissues being studied in this particular set of experiments, namely, the liver, brain, and fatty tissue. Each of these tissues has distinct metabolic responses to insulin exposure. Within this three-dimensional experiment design space, each set of possible experiments can be uniquely located. This notion obviously generalizes to the much larger number of possible dimensions in any experimental system. Note that the first and third dimensions have unordered discrete scales, whereas the second dimension is continuously scaled.

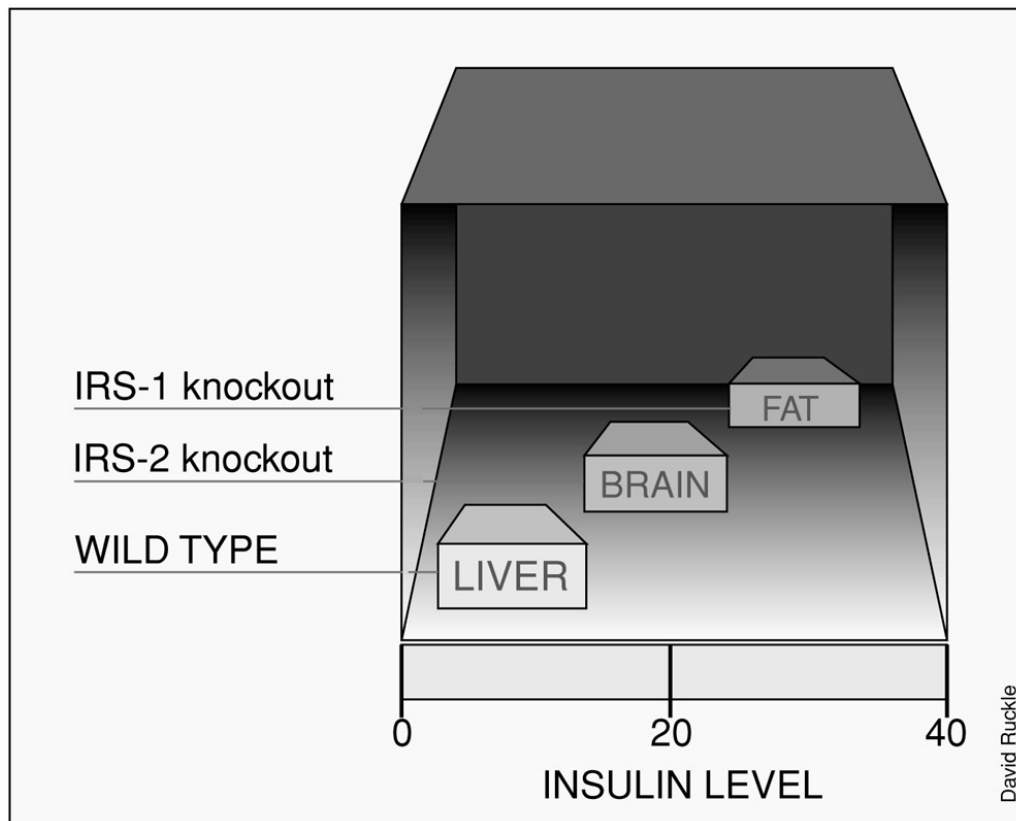


Figure 2.1: Three experiments within the multi-dimensional representation of experiment design space. Experiment design space defines all the possible stimuli or conditions to which a particular biological system could be subjected. Shown here is an experiment design space that is concerned with insulin signaling in different tissues, in different mouse "knockout" models, with different levels of insulin.

A functional genomic study might involve a trajectory across the experiment design space where the mouse models are studied across multiple insulin levels and multiple tissues. As the number of experimental dimensions increases, the number of experiments required to cover the experimental

design space grows exponentially. However, the concept of an experiment design space by itself does not suffice to determine which kind or number of experiments is adequate to assess the genomic physiology of a tissue or system.

2.1.2 Expression space

Now that we have defined the experiment design space, we define the expression space to be the space of potential expression values of all genes in a given genome. In the context of microarray measurements, this is a large multidimensional space—with each dimension corresponding to a single gene—and it can be immediately partitioned into several overlapping subspaces. For example, coordinated gene expression measurements made from normal muscle tissue under normal conditions can be represented by a subspace of expression space which will most likely have a nonempty intersection with the subspace of coordinated gene expression measurements made from diseased muscle tissue. We often repeat mantra-like to ourselves the phrase "A cell is a cell," by which we mean that in order to survive as living and organized entities, most cells share a large repertory of biological processes, whether they be components of muscle tissue, liver tissue, or white blood cells. That is, the expression subspaces of most cell types have significant overlap.

Expression space contains clues about the interactions between two or more clusters of genes. As an example, we start with the simple case of three arbitrary genes: insulin-like growth factor 1 (*IGF-1*), insulin-like growth factor binding factor-1 (*IGF-BP1*), and fibroblast growth factor 2 (*FGF2*). The regulatory relationship between these three genes^[2] is, in some tissues, that one gene partially determines the expression level of the other two. This will cause the three genes to move in a coordinated fashion across the range of their respective possible values. That is, if this is a particularly tight regulation system, then the expressions of this triplet will only be found in a small subspace of all possible loci of the expression space. If this were a loosely regulated or unregulated system, it is possible that the gene expression values might be found in all parts of the expression space, as shown in figure 2.2 below. The totality of the potential loci that these genes can occupy constitutes the potential set of all coordinated gene expression levels, or expression space. The subset of the expression space wherein a cellular system has its expression under all stimuli is called the *transcriptome* [41, 184]. More precisely, the transcriptome encompasses the expression space of all putative 30,000 genes in the human genome (or 100,000 gene products if current alternative splicing estimates are correct). Each point in the human transcriptome corresponds to the possible values that each gene's expression might attain under any physiological or pharmacological stimulus.

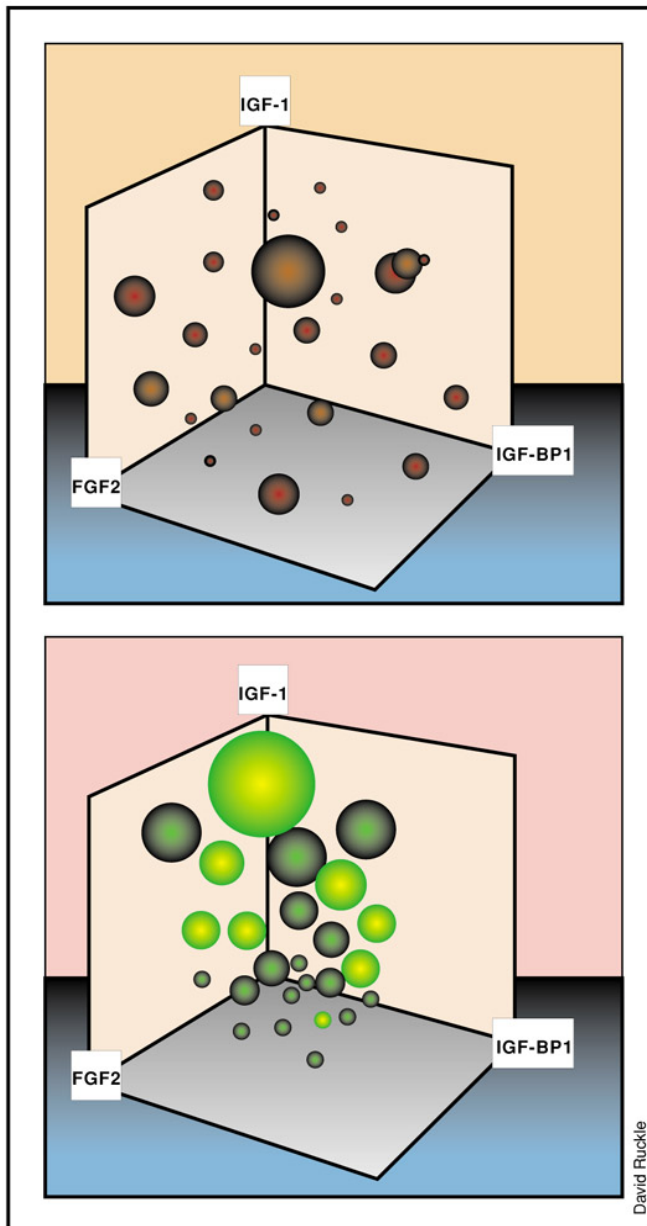


Figure 2.2: Expression space in which gene expression is loosely versus tightly coupled. If genes are tightly coupled in the expression space, they will tend to occupy a small subspace of expression space over any set of experiments. If, however, these genes are loosely coupled or even causally unrelated, then their expression levels will have little relation to one another and therefore tend to be scattered over a greater volume of the expression space.

The meta-goal of the experimentalist/investigator then is to define the experiment design space that provides the best and most complete picture of the entire transcriptome *or* a subset of the expression space of interest. This is the point in experimental design where a great deal of domain-dependent knowledge, *i.e.*, specific or *a priori* biological knowledge, is of significant utility. The investigator must ask herself or himself: "What is the minimal set of experimental conditions, *i.e.*, the smallest subspace of the *experiment design space*, that will cause the largest set of relevant subspaces of the *expression space* to be sampled?" In order to answer this question, the investigator must first understand how each experimental condition could potentially affect the expression of potentially interesting genes and be able to quantify its effect. Second, the investigator must consider all possible subspaces or interactions between genes that are of interest. For example, let us say that we are only interested in the interactions between *IGF-1* and *IGF-BP1* in the example earlier in the section and we wish to determine how the relationship between these

two genes changes under both physiological and pharmacological conditions. Then we would obtain those sets of experiments in the experiment design space which would most likely cause one or both of these genes to be expressed over the broadest range of their possible values so that all the evidence of the interactions would be captured, *i.e.*, the subspace of experiments that covers the largest plane in the *IGF-1* and *IGF-BP1* expression subspace. A researcher might be interested in the signaling mechanisms of *IGF-1* once it has bound to IGF-BP1 and how it affects the liver's expression of other genes. These genes may code for many proteins involved in postreceptor signaling, including possibly *IGF-BP1* itself. With a finite budget and a finite source of tissue or animals, the investigator will have to choose between alternative strategies. A "mutant screening" strategy might be to assay expression levels in the liver over a set of knockout mice in which each element of the set constitutes a knockout of a different putative element of the *IGF-1* signaling pathway, whereas a "physiological stimulus" strategy, would be to take a single element of the set (one of the knockout strains or a wild-type animal) and subject it to a sequence of different concentrations of extracellular *IGF-1*. Determining which of these alternatives provides the best exploration of the expression space will require an understanding of whether the range of mutants or physiological stimuli engages the portion of the genetic machinery that is relevant, and how close to the normal homeostatic behavior or pathological behavior of interest is the experiment intended to explore.

Figures 2.3, 2.4, and 2.5 illustrate the pitfalls of insufficiently exercising the regulatory mechanisms of gene expression. In figure 2.3, we observe the joint behavior of two genes in which there exists a strong component of randomness in the relationship between the two genes—as evident from the low Pearson's correlation coefficient that is calculated between these two genes within the current range of expression levels. This is also summarized by the Pearson's correlation coefficient for the linear regression calculated between these two genes, which is calculated to be very low. However, if an additional stimulus is applied to the regulatory system such that gene 1 has a wider range of expression, then despite the randomness in the relationship between the two genes observed in the previous range of expressions, we begin to see what appears to be a linear relationship with a much larger Pearson's correlation coefficient (figure 2.4). If the expression level of gene 1 is perturbed to an even wider range, it becomes apparent that the relationship is not linear but, in fact, curvilinear, possibly even a quadratic relationship, as shown in figure 2.5. If the experiment space was incorrectly selected, then these and many other pairs of genes like it would be assumed to be poorly or randomly related when, in fact, with the right stimulus, a tight curvilinear relationship would be determined, and *vice versa*.

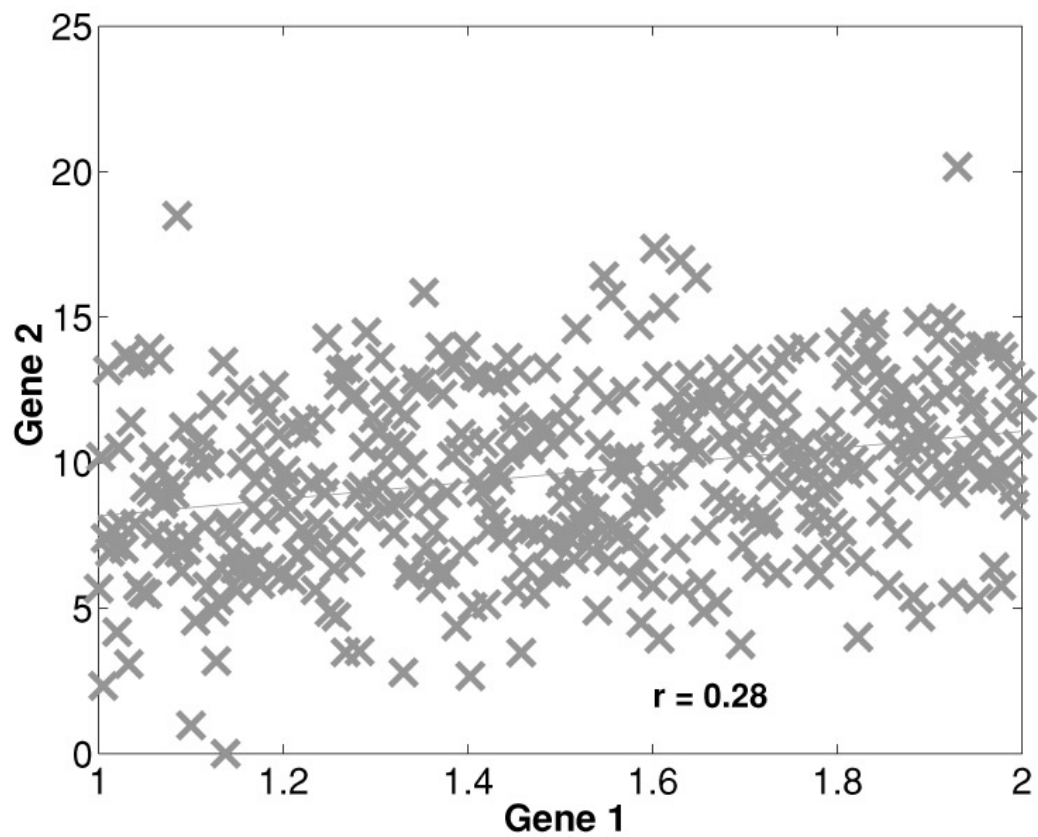


Figure 2.3: Apparently random relationship in expression space between two genes.

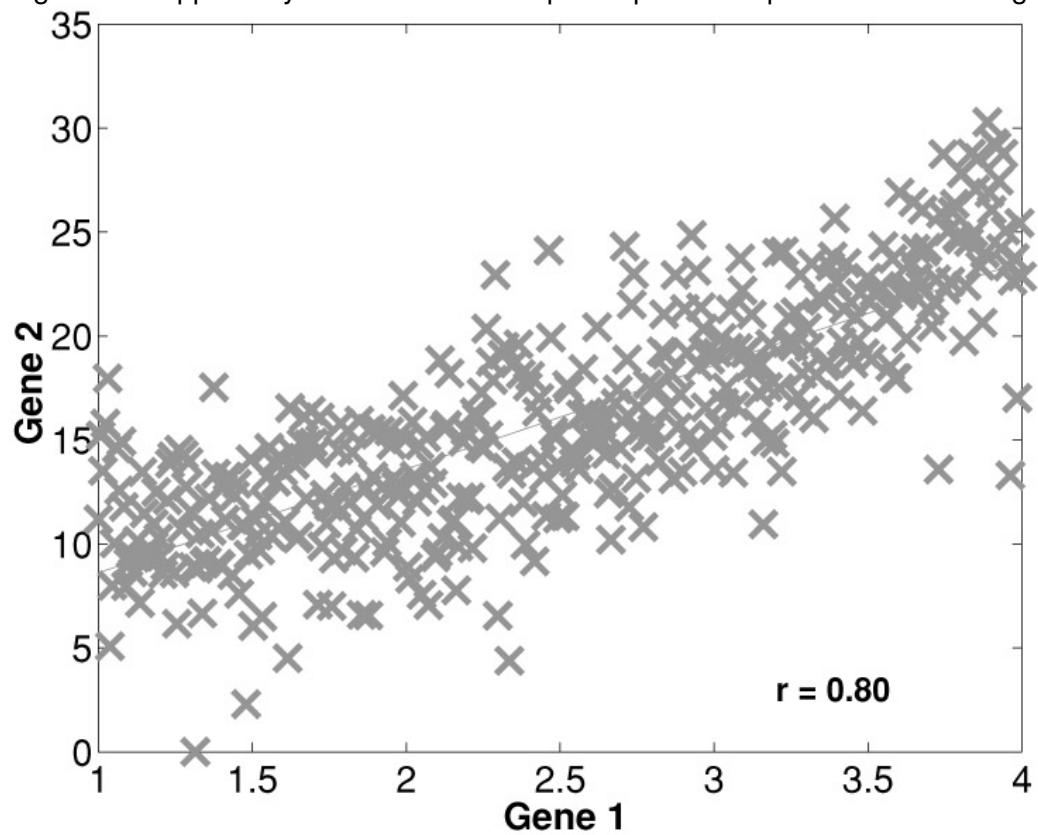


Figure 2.4: Apparently linear relationship in expression space between two genes.

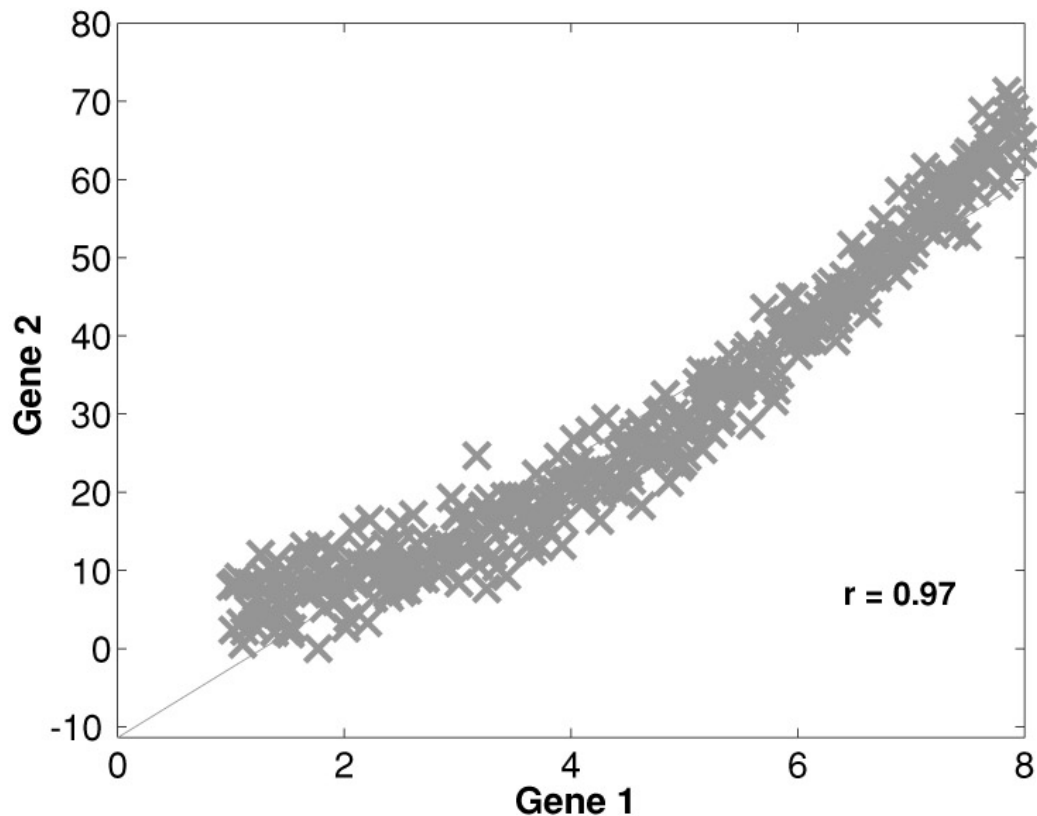


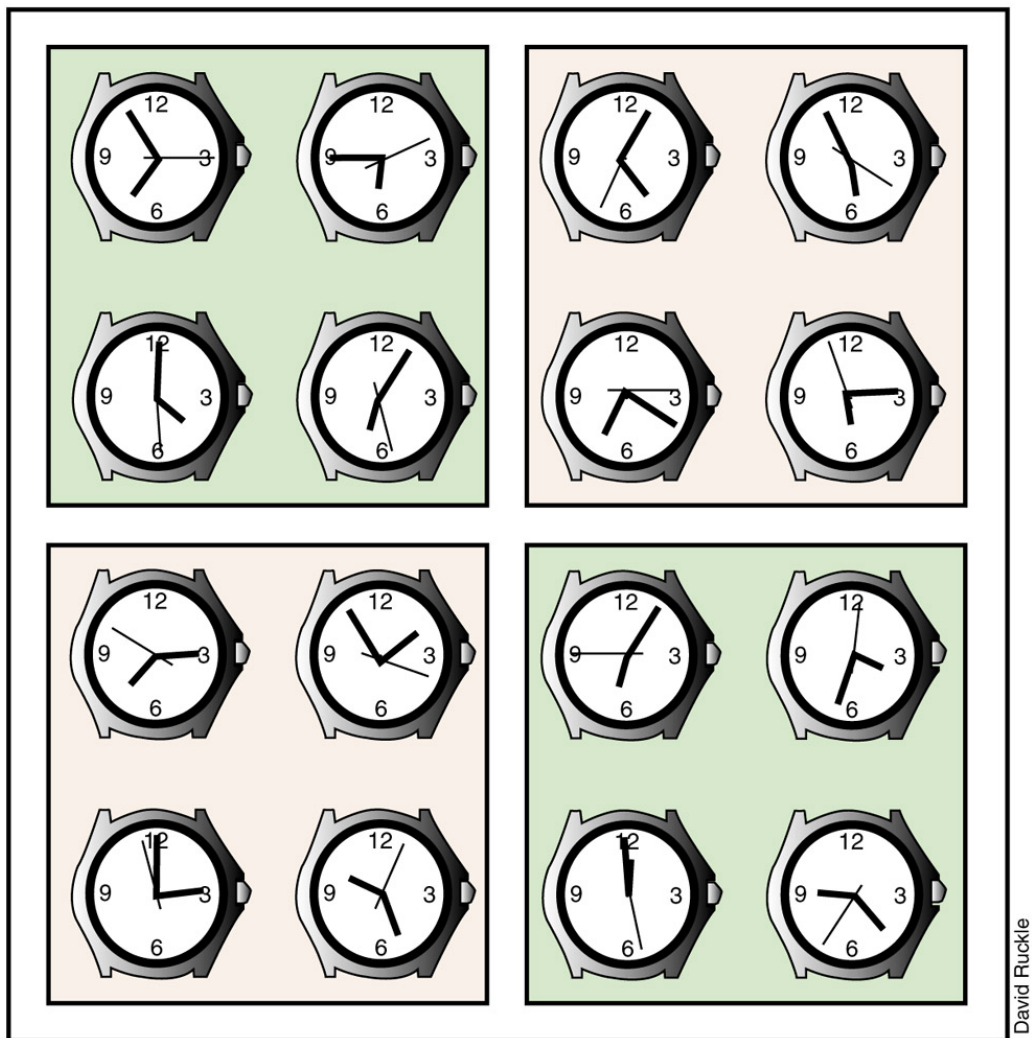
Figure 2.5: Apparently curvilinear relationship in expression space between two genes. This curvilinear relationship would be obscured if the expression space had been insufficiently exercised.

2.1.3 Exercising the expression space

In section 2.1.1, we described how each experiment can be viewed as an exploration of the space of all possible expression patterns. In exploring this expression space, the investigator is constrained to the space of possible experiments. Informally, the goal of experiments should be to *maximally exercise the genome* bringing out the couplings or correlations between genes that are of greatest relevance to the biological process being studied. We have found the following "watch metaphor" to be of some use in explaining the interaction between expression space and experiment design space. We leave it to the reader to judge whether it is helpful, or simply a confusing distraction.

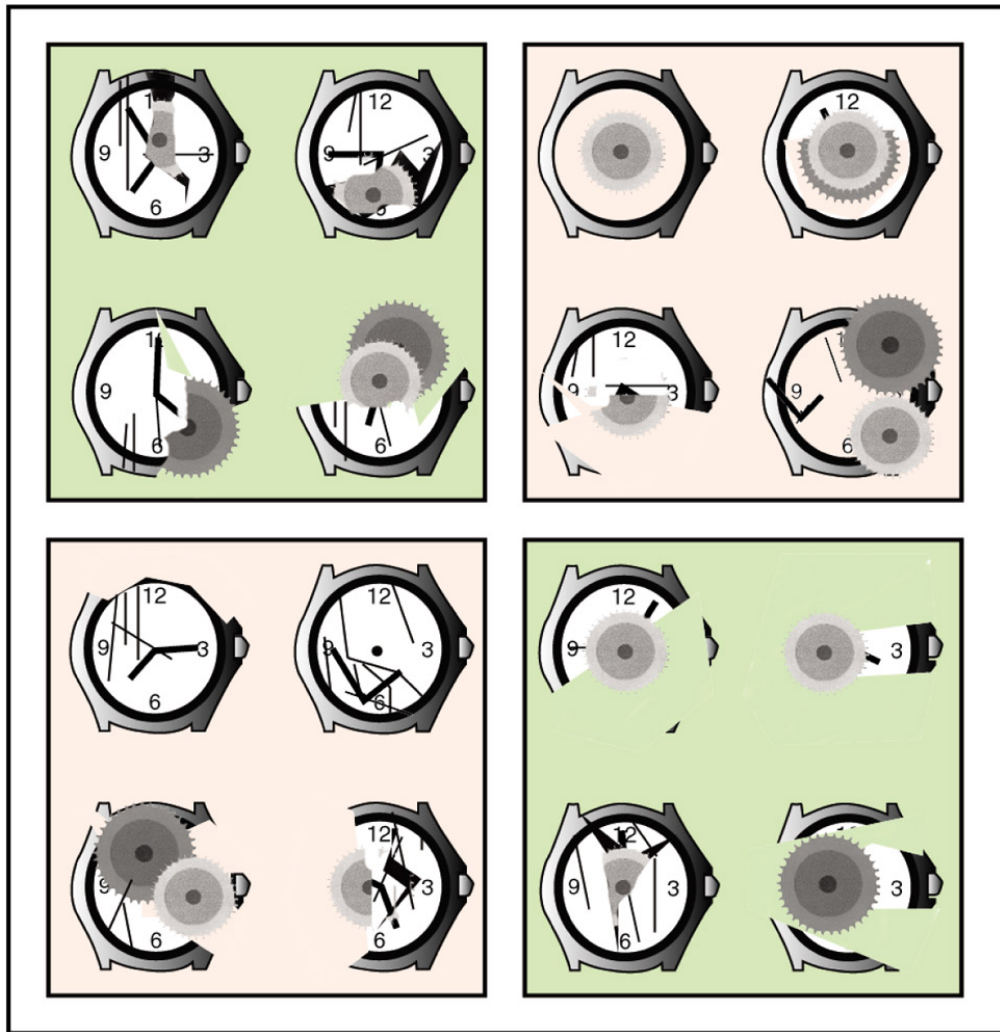
The watch metaphor Suppose, that we were interested in discovering how an old analog watch of the sort illustrated in figure 2.6 worked. Simply watching the hands of the watch going around might allow us to generate some hypotheses about the mechanisms that generate this motion or behavior. Empirically, we could derive the relationship between the movement of the minute hand and the hour hand, but the total sum of mechanisms that could explain the behavior would be large. A more invasive investigation would be to take a hammer to the watch and examine the pieces that fall out of the watch's case (figure 2.7). The combination of the prior observed empirical behavior and the examination of these pieces might give us a better, if still grossly incomplete hypothesis (not to mention the loss of a costly watch). That is, the number of possible mechanisms explaining the observed behavior of the watch hands would have been substantially reduced. Furthermore, if one had no other similar watch, these hypotheses might not be verifiable on account of its destruction. A more sophisticated approach might be to pry open the back of the watch and observe the movement of the gears and escape mechanisms. With these observations, and one or both of the prior kinds of investigations, an even better understanding can be obtained of how the various components of the watch are mechanistically related to one another. All of these experiments are observational studies, and at best we could generate some fairly accurate hypotheses without ever

demonstrating causality. Causality would be most convincingly demonstrated by intervening in the operating mechanism, *e.g.*, winding the watch (partly or until the mechanism breaks), shaking the watch, immobilizing one of the gears, or replacing a gear with one of different-sized teeth. These interventions are progressively more technically challenging but also provide better insights and often confirmation of the hypotheses generated during the earlier observational studies.



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Figure 2.6: Noninvasive monitoring of mechanistic behavior. By observing the concerted action of the watch hands, several competing *weak* hypotheses about the underlying mechanism can be generated. The more observations made at different times, the smaller the number of possible hypotheses generated.



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Figure 2.7: Decomposition of the watch. An invasive exploration of mechanism will reveal most of the components of that mechanism. In the process of the invasive investigation some of the components will be damaged. Also, the relationship of these components inside a working watch must be inferred and is not directly observed because of the invasive nature of this investigation. The analogy with expression studies, with the indulgence of the reader, can be made as summarized in table 2.1. These describe different kinds of experiments— distinct points in the potential experiment space. Some of these experiments allow the components of the watch to work in their usual fashion, *i.e.*, within their usual "expression space," whereas others bring them into an operating mode or expression space in which they do not usually function. The question the watch investigator would have to ask himself or herself is what part of the expression space of the watch will best inform him or her of the normal function of the watch. It may be that smashing the watch is the only part of the experiment space that is cost effective or feasible for the investigator. The smashed or overwound state of the watch may not accurately reflect the normal intact and possibly more interesting state of the watch's expression space.

Table 2.1: Exploring the experiment space with a watch

Experiment Space in Watch	Experiment Space Biological Analogy
Observing the movement of the hour, minute hands	Hypothesizing cellular function based on gross cellular behavior
Smashing the watch, examining the pieces	Sequencing some of the genes in the cell or measuring a few proteins or RNA transcripts

Seeing all the gears working from the back of the opened watch	Parallel measurement of thousands of genes at a time Time series measurements
Partially winding up the watch	Physiological stimulus
Winding the watch until it is near breaking or until it breaks, thereby bringing the internal gears of the watch into behavior they would not typically exhibit	Pharmacological stimulus that brings the gene expression patterns into an operating region outside the usual physiological range
Removing a gear	Creating an organism with a gene "knocked out"
Replacing a gear with one of different weight	Creating a transgenic organism

To take these analogies one step further, there are multiple arrangements of gears that could account for the observed interactions between these gears and their effect on the hour hand and minute hand. Similarly, although perturbations of the genome can provide insight into the interactions among the major genes involved in a given pathological pathway, they do not reveal the entire pathway. Rather they expose a significantly large family of possible mechanisms that could account for the observed behavior. We address this issue in greater depth in the section on pathway re-engineering (section 4.13.2). Nevertheless, as suggested earlier, there are some misconceptions about what can be done on the basis of microarray expression studies implicit in some of the current genomic and bioinformatics literature.

Misconceptions in "reverse engineering" To date, there have been several efforts to try to reconstruct pathways from gene expression measurements [57]. Given the success of recapitulating the first few steps of the glycolytic pathway from substrate measurements [12], it would appear plausible to do this. However, without careful consideration of the pathways that one is trying to reconstruct, there is significant risk of a methodological and metaphorical flaw in this analysis.

It is important to realize that pathways could exist on several molecular and physiological levels. The genes that are regulated in one pathway may play a role in another pathway. As an example, let us consider the genes coding for lactic acid dehydrogenase: The gene *LDHA* is expressed mostly in muscle and is located on chromosome 11p15.4, while *LDHB* is located on chromosome 12p12.2 and is expressed mostly in heart. *LDHA* is known to have binding sites for HIF-1, CREB1/ATF-1, and p300/CREB binding protein [62], as well as binding sites for NRE, NREBP/c-FOS, AP1, AP2, and other transcription factors in its promoter region [104]. *LDHB* is thought to have a binding site for SP1 [159]. There is also likely to be a post-transcriptional level regulation of these two genes. Because of the differences in promoter regions, it is safe to assume that both *LDHA* and *LDHB* participate in their own gene expression regulatory networks.

Both *LDHA* and *LDHB* code for protein subunits. The final protein product is made up of four subunits. Various combinations of the two types of subunits are assembled into five different lactic acid dehydrogenases, from LDH-1 (containing four *LDHB* subunits), to LDH-5 (containing four *LDHA* subunits). These five enzymes are found in a binomial distribution in mammals. Additionally, the final protein itself is an enzyme that assists in the conversion of lactic acid to pyruvate, called lactic acid dehydrogenase, and defined by the Nomenclature Committee of the International Union of Biochemistry and Molecular Biology as Enzyme Commission (EC) 1.1.1.27. Lactic acid and pyruvate happen to be substrates in the glycolytic pathway of anaerobic energy production.

Thus, in this one simple example, one finds at least three pathways, as shown in figure 2.8. The operator at the *enzyme level pathway* participates as one enzyme in a major biochemical substrate pathway, that of glycolysis. That operator, however, is assembled stochastically in the *assembly pathway* from subunit components. Finally, those subunits are produced within their own *genetic regulatory pathways* involving a multitude of transcription factors. Considering the three layers of

pathways, it should now become apparent that it may not be possible to "reconstruct" the glycolytic pathway simply from measuring multiple gene expressions. However, on a more optimistic note, given enough gene expression data, one should be able to reconstruct the first layer of these pathways, *i.e.*, that of gene expression regulation and involving transcription factors and promoter regions.

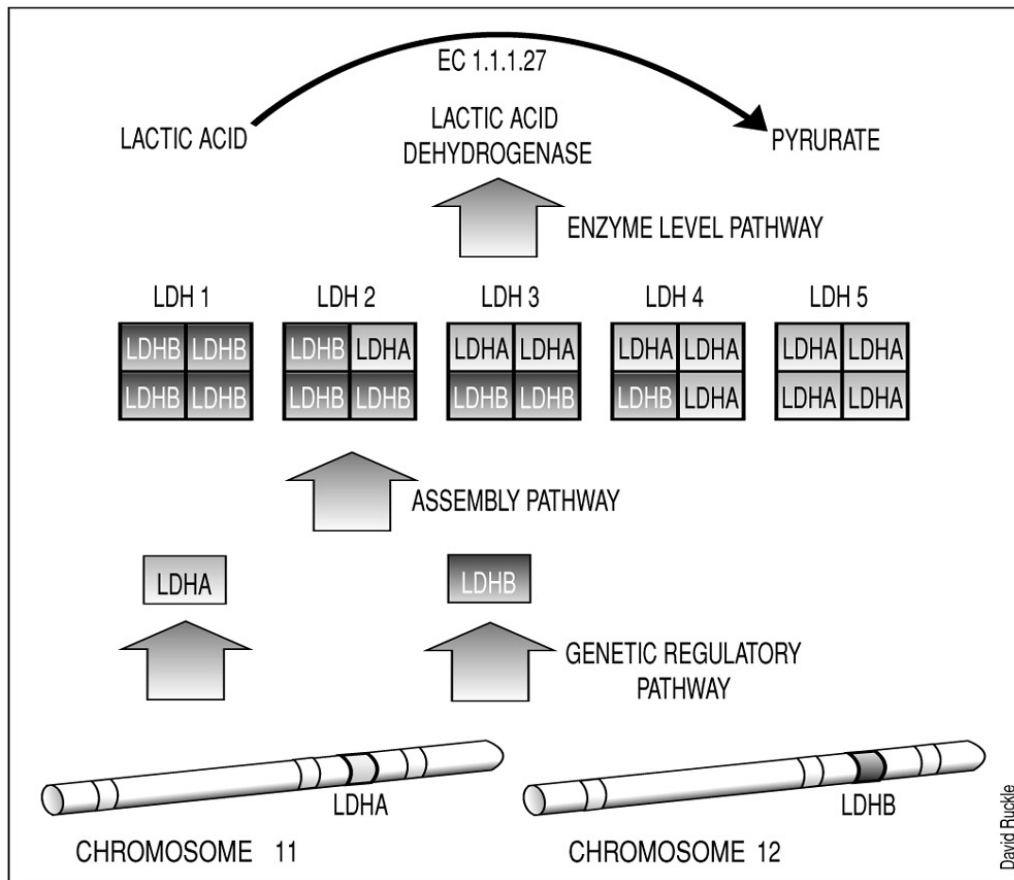


Figure 2.8: An example of why discovery of pathways solely using gene expression measurements is difficult. At least three "pathways" are involved in the conversion of lactic acid to pyruvate. The highest enzyme level pathway involves the action of an enzyme called lactic acid dehydrogenase, designated EC 1.1.1.27. However, when one traverses to a lower-level pathway, one learns that the role of this enzyme is performed by five separate protein products, which appear in a binomial distribution. When one traverses lower, through the assembly pathway, one learns this distribution is present because of the various possible combinations of *LDHA* and *LDHB*, two individual protein subunits, of which four must be put together to make an active enzyme. Then, when one traverses even lower, through the genetic regulatory pathway, one learns that each of these subunits is regulated differently, and appears on different chromosomes. Not shown are the pathways involved in transcription, translation, movement of proteins to the appropriate locations, and movement of substrates into and out of the cells and organelles, and many other pathways.

How vigorously should a model system be stimulated or perturbed? Let's revisit the phrase "maximally exercising the genome." What constitutes a maximum perturbation will in part be defined by the goals of the investigator. If she or he wishes to understand the interactions between the genes under normal physiological conditions, then it may be that only relatively minor perturbations followed across time will be informative. Even though extreme perturbations are easier to measure, they might not accurately represent interactions that occur over the normal dynamic range of activities of the RNA species, structural proteins, and enzymes programmatically generated by the genome. There are, however, several circumstances when it may be advantageous to perturb the relationships between the various constituencies of the transcriptome much more violently and

dramatically than is found under normal physiological or homeostatic conditions. An example of such a motivation might be to measure the effect of pharmacological doses, rather than physiological doses, of a pharmaceutical. The intended effect may lie outside the normal operating range of the genetic machinery, as is often the case in pharmacology. Other times, the goal may be to examine biological systems which naturally fall far outside the usual operating parameters of the genomic machinery. This includes a variety of disease states such as a malignancy or diabetes, where there are single or multiple massive perturbations of the usual homeostatic system, and therefore a perturbation of the model system that corresponds to the pathological condition would be appropriate. It is worthwhile considering some extreme examples to illustrate the importance of this decision.

For example, if one is trying to understand the relationship between a set of genes thought to be involved in the effect of insulin signaling, it may not be particularly helpful to start with a sample population of human adipose tissue (fat) which is known to be insulin sensitive, but obtained from individuals who are all in the fasting state when insulin levels are low or absent. That is, there may be *insignificant* insulin exposure or stimulating effect of the insulin-responsive cellular machinery. Consequently, the experiment design subspace sampled will be too small to adequately explore the expression space of interest. Furthermore, if the sample population is ethnically homogeneous or, in the case of a mouse experiment, shares the same genetic background, then these samples will only inform us of the reproducibility and noise of the biology and the measurement system. In contrast, if the same population were exposed to varying degrees of insulin, then the behavior of the genetic program in response to insulin might be better understood. Alternatively, taking a population known to have varying degrees of insulin resistance would also provide a more diverse range or heterogeneity of interactions in insulin signaling, which would enable the investigator to explore the space of possible relationships between the genes. This would then allow the application of a machine-learning algorithm to elucidate these dependencies. If the degree of insulin resistance is clinically measured (e.g., as obtained in a carefully performed hyperinsulinemic euglycemic clamp^[3]) in close temporal proximity to the expression measurements, then the investigator will probably have the most accurate control of the experiment design space. The goal then is to stress or stimulate the cellular regulatory system, which in this instance is glucose homeostasis and insulin signaling, so that the genes "show us" how they interoperate under normal, and pathological or pharmacological circumstances. There are several other obvious variations of these kinds of stimulations such as:

1. Examine the expression profile in adipose tissue at different points in time after the glucose exposure to understand the timing of the genetic program. We should not expect all genes to act at the same time or rate, and therefore sampling the changes in the transcriptome across multiple time points at the appropriate frequency will be much more revealing than just one or two points in time.
2. Assess gene expression at different fixed concentrations of insulin.
3. Assess gene expression in ethnically diverse populations. The difference in genetic backgrounds of these populations will give a sense of how much variation in the signaling maybe due to genes outside the set that is being studied in a particular expression experiment. This implies that we would need to hypothesize that changes in the genetic background affect the functioning and thus the functional genomics of the system. This presupposes that
 - ◆ insulin production and insulin effect is the combined result of the activity of several genes;
 - ◆ polymorphisms in these genes are distributed heterogeneously across human populations;
 - ◆ such polymorphisms result in significant changes in gene expression or function, or

both.

4. Explore the pharmacological effects of insulin. After a certain dosage level, increasing levels of insulin may no longer result in increased effects of glucose transport. However, increasing insulin may affect other physiological systems within the same cell, as, *e.g.*, transporting ions across the cell membrane. Taking tissue samples at pharmacological concentrations of insulin and comparing their expression profile to those obtained with insulin concentrations in the physiological range would allow determination of genetic mechanisms that respond to the pharmacological dosing.

In summary of our discussion, we note that an appropriate set of (microarray) experiments is that in which the cellular response to the stimuli involved in the experiments will exercise the interaction between genes over the range of interest and relevance to the biologist–investigator. This critical aspect of the experimental design will trump any subsequent choice of bioinformatic analytic technique. Conversely, *no* bioinformatics analysis will be sufficient to extract functional genomic knowledge if the expression space is not adequately explored.

2.1.4 Discarding data and low–hanging fruit

After an investigator obtains his or her first microarray data sets, often the first question he or she will ask is: "Which of the observed up or downregulation of genes represents biologically significant changes in expression?" This and similar questions are addressed in detail in section 3.3. Here we provide a context and motivation for an answer that biologists often find discomfiting: that *very few genes are in fact significantly changed in expression in a way that is distinguishable from biological and measurement variation and noise*.^[4] We should emphasize at this point that there is an important distinction between mathematical or statistical significance, and biological significance. The former has to do with analytically quantifying the difference between two or more different sample sets of numbers, *i.e.*, microarray expression data, for which there exists standard statistical tests like the Student's *t* or χ^2 . These tests typically ignore the rich biological meaning and structure in these numerical representations of expression. The latter is a more complex matter which we shall discuss here.

It is likely to be the case that there are thousands of genes with important biological outcomes effected by small relative changes in the expression levels. In the present state of the microarray technology and with the typically small number of replicate experiments for any given condition, these small changes cannot be reliably and reproducibly distinguished from noise. This may mean that even though we measure tens of thousands of genes in a microarray experiment, we would only obtain hundreds of genes that we are reasonably convinced are involved in a particular biological system. On the other hand, not all statistically significant gene expression changes lead to a significant change in the biological or physiological state of a system. This answer is most disconcerting to those biologist–researchers who have had substantial experience in investigating the expression of only a few genes at a time. Their investigations typically involve one or two highly likely candidates based on many prior investigations, and where multiple measurements (all of which are carefully checked) have been made of the expression levels. Consequently, the researchers have well–grounded ideas of what constitutes a biologically relevant or significant change in expression for these few genes. They do not have the benefit of this sort of knowledge for each of the thousands of genes that are represented on a microarray but nonetheless are uncomfortable with discarding from further analysis thousands of genes that appear to have large numerical changes in expression.

In contrast, due to the large number of genes that can be measured in a single experiment, computer scientists and computational biologists are comfortable— perhaps too comfortable—with generating exhaustive lists of genes that are possibly involved or interacting in a particular biological

process. This kind of exhaustive list generation has been quite common in publications of microarray experiments from 1997 until today. Unfortunately, such lists are not that helpful to the biologists who wish to determine which elements in these lists are worth pursuing, *i.e.*, are biologically significant for their investigations. This is often the reason for questions of what constitutes a significant fold change, where significance can mean anything from "present in the tissue," "associated with the system being studied," "causative of the changes in the system being studied," or even simply meaning "worthy of further study." Thus, an analyst for a typical functional genomics study will simply draw two boundaries: one covering increases in expression and the other covering decreases in expression, as shown in figure 2.10. Such an analyst will then declare that any gene found beyond these lines (the red points) is significant or relevant to the biological process being studied. Unfortunately, there is no single threshold number or even function that we can provide that would generate a particular number of candidate genes that is worth pursuing for any arbitrary experiment and experimental design. Fortunately, there are well-founded decision analytic procedures that can be followed in order to come up with the appropriate number of candidate genes for any given investigation. These are described in detail in section 3.2. The underlying motivation for picking a threshold is straightforward.

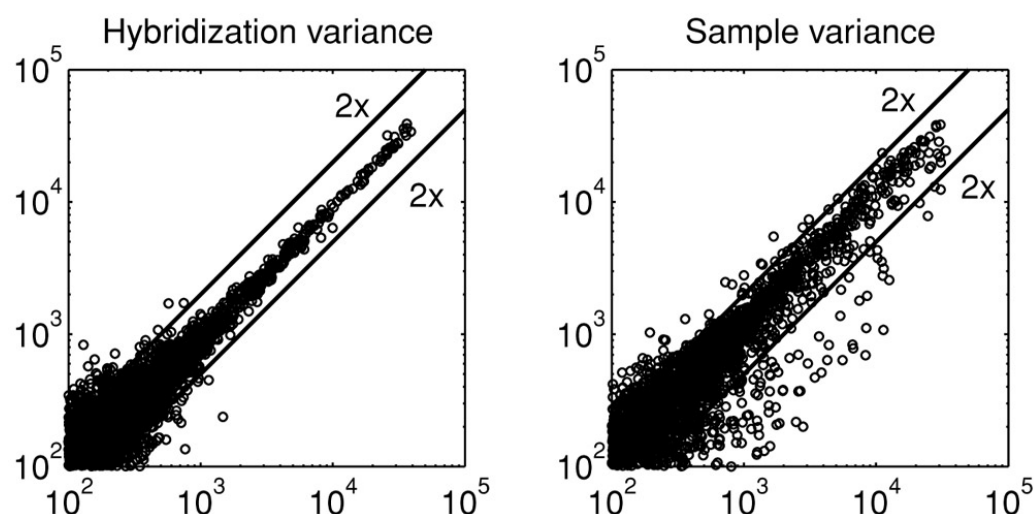


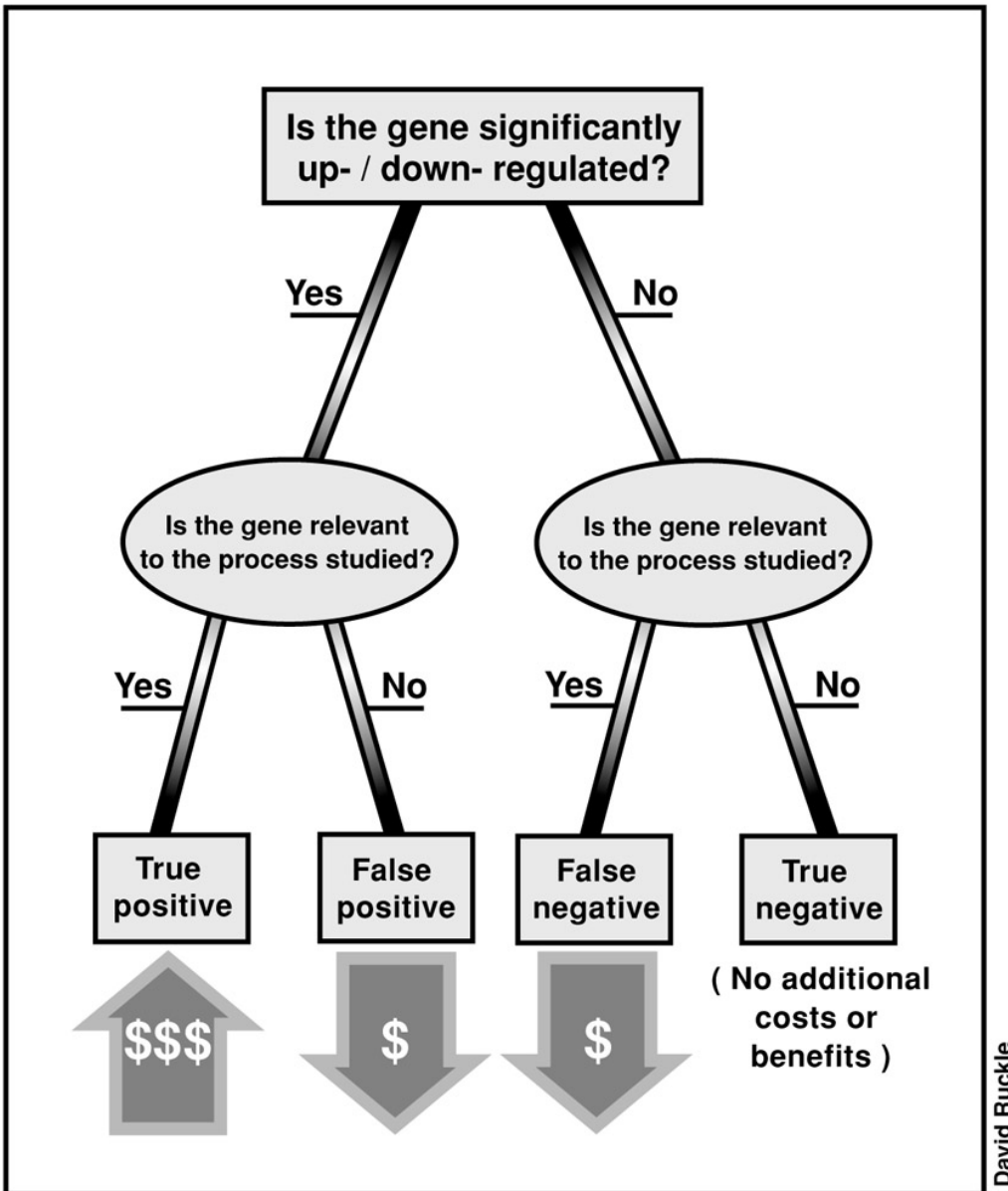
Figure 2.10: Typical uses of fixed fold thresholds. Reproducibility scatter plots. In each of the experiments, samples were hybridized to identical oligonucleotide arrays containing probes for 6800 human genes. The message abundance in arbitrary units is plotted. Left panel: A single biotinylated RNA target was divided in two and each half hybridized to two arrays. Fifty-nine genes (0.8%) judged to be expressed "present" by the Affymetrix GeneChip software differed by more than twofold, and 0.3% differed by more than threefold. Middle panel: A single biotinylated target was hybridized to one array, the sample removed and then rehybridized to a second array. Of "present" genes 2.6% and 0.4% differed by more than twofold and threefold, respectively. Right panel: A single total RNA sample was converted to biotinylated cRNA in two independent labeling reactions, and the cRNA then hybridized to two arrays. Of the "present" genes 2.2% and 0.4% differed by twofold and threefold, respectively.

The first question one has to consider is the cost of a *false positive*. A *false-positive* gene is a gene which, in reality, is not biologically significant to the process under investigation, but which the analytic technique deems to be significant, statistically or biologically. That is, what is the cost involved in the follow-up procedure to confirm the function of a gene in a particular biology process? Whether this confirmation process, or biological validation, involves quantitative PCR, *in situ* hybridization, the generation of a transgenic mouse, a transient-transfection assay, or transfecting a cell line, its cost in time and money will be substantial. This cost thus limits the number of genes that can be investigated. The threshold that will then be picked will be determined in part by the disutility or cost of the biological validation step for false positives. The tens of thousands of genes present on any microarray now ensure that the number of false positives could

potentially overwhelm the typical time and financial budgets of most (academic) research laboratories.

Of course, there is the second and converse question of the cost of missing a *false negative* from the same system. A *false-negative* gene is a gene which, in reality, is biologically significant to the process under investigation, but which the analytic technique deems not to be significant, statistically or biologically. There are likely to be several genes involved in any signaling pathway of interest. Not all genes in a pathway are equally amenable to biological validation. Furthermore, some of the chosen genes may be more suitable targets than others for diagnostic assays or therapeutic interventions. If the threshold picked for considering a gene to be significantly^[5] changed in expression excludes one of these targets, then the cost of that false negative will be quite high, e.g., a missed scientific opportunity or lost opportunity for commercial development.

With this in mind, it becomes clearer how a threshold should be picked. First, the investigators should ask themselves how many false-positive and false-negative leads they can tolerate. Then they can conduct a series of replicate experiments as discussed in section 3.2. and determine where they will have to draw the thresholds to attain the required sensitivity and specificity.^[6] So that if one is about to embark on a functional genomics investigation, an integral part of the experimental design must be the decision analysis diagrammed in figure 2.9. This requires that the cost to one or one's enterprise of missing the one or more genes that are likely to be involved in the genetic regulatory pathway of interest be made explicit. Similarly, the cost of having to follow up on false leads must be estimated. These two costs define the principal disutilities that one is trying to minimize. In order to complete this simple decision analysis, the probabilities corresponding to the sensitivities and specificities that are diagrammed will also have to be obtained. After this is complete, one will then know whether it is possible to engage in a productive high-throughput functional genomics strategy, or whether one has to increase the sensitivity or specificity of the measurement techniques, or whether one needs to increase one's biological validation budget, or all three. Given the current level of reproducibility of expression measurements in microarrays, most investigators will choose a high level of specificity to reduce the false-positive rate. This will inevitably lead to a high false-negative rate. At this point the biologist will find that among the false negatives are genes that are known to have changed expression (but did not meet the significance threshold computed by the bioinformatician) which will lead naturally to the following worry: Many other genes of relevance to the biological system under investigation are being discarded from the analysis and follow-up. This worry is likely to be well-founded but given the state of the measurement technology, and the typical sample sizes employed, it cannot be easily remedied. It remains, that the *low-hanging fruit*, while only numbering in the hundreds, are likely to shed new light on the processes studied. Already this represents several orders of magnitude more hypotheses to be tested as compared to the investigations of the pregenomic era. For those fruit higher up the tree, the above decision analysis suggests the need to wait for more accurate and cheaper microarray technologies.



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Figure 2.9: A decision analytic procedure for picking a threshold for selecting genes from a functional genomics experiment. Because of the large number of probes on current microarrays, it is all too easy to underestimate the cost and practical intractability caused by even a moderate false positive rate. Investigators are advised to perform this simple decision analysis in order to determine what false positive and false negative rate they can afford before they proceed with any experiments.

It goes without saying that the preceding discussion is an oversimplified decision analytic procedure. Attempting to assess the utilities of false-negative findings will be particularly difficult for most investigators, but the constraints of realistic budgets will be a driving factor in considering this analysis. However, the procedure we have described does have the merit of providing a rational basis for deciding upon the number of genes that one should seek to obtain from one's pipeline. Even a coarse decision model can be useful in avoiding unpleasant surprises after the initial excitement of obtaining a list of putatively interesting genes wears off. The reader who is interested in more sophisticated decision analyses is referred to the textbook by Weinstein *et al.* [187]. It should be noted that the raw specificities and sensitivities of the expression microarrays used in a functional genomics pipeline need not be the ones that are used in this decision analysis. As we demonstrate in sections 3.2.2 and 3.2.8, the application of appropriate noise models over repeated experiments can lead to improved sensitivity and specificity. The cost of such models is typically in

the increased number of microarray experiments required to develop those models.

Dynamics Perhaps the best glimpse of the interactions between the various genetic components, and in particular their causal dependencies, is obtained through analysis of the dynamics of the system. This is elaborated considerably in the section on dynamics (section 3.6.3, page 146). Suffice it to say here that understanding the trajectory over time of various RNA component species rather than their absolute values at any given point in time provides far more information about the operation of the underlying system. To return to our watch metaphor, observing how the gears turn inside the watch while we smoothly move the minute hand or the hour hand will give us much more information about how the gears move together than a few (unordered) snapshots in time of the different positions of these hands on the face of the watch. In most analyses to date, even when expression data are obtained as a time series, the information buried in the specific ordering and timing of these measurements is rarely exploited. For example, many of the clustering and classification methods employed (and introduced below) will generate the same answers even if the order of the measurements were shuffled. In fairness, the number of time points and the sampling intervals of many of the existing time-series expression data sets is minimal. They provide insufficient information for most of the standard armamentarium used to analyze time-oriented data. As the price of microarrays falls, the quantity and quality of time-series experiments will undoubtedly increase, at which point many of the techniques for analyzing trends and periodicity that have been developed in statistics and the signal processing literature will become applicable. At that time, the temporal relationships between every single expressed gene with every other gene measured at all time points will provide a qualitatively improved set of insights into causal relationships in the genetic regulation of cellular physiology.

^[1]For example, cellular proliferation and differentiation during development.

^[2]All of these genes have been implicated in somatic growth and bone repair.

^[3]This involves giving the subject a continuous intravenous infusion of insulin and glucose, while maintaining a relatively constant blood glucose concentration by varying the rate of the glucose infusion.

^[4]This is particularly true given the noisy nature of most microarray-based expression measurement systems.

^[5]For the discussion here, we define "significant" as "worthy of further study."

^[6]Sensitivity can be described as $(1 - \text{false negatives})/\text{all positives}$; specificity as $(1 - \text{false positives})/\text{all positives}$.

2.2 Gene-Clustering Dogma

There is a central underlying assumption in all gene-clustering techniques for expression analysis. Simply put, the assumption is that genes that appear to be expressed in similar patterns are in fact mechanistically related. Furthermore, the corollary to this assumption is that although genes may distantly affect the function of other gene products, they fall into groups of more tightly regulated mechanisms.^[7] For instance, the genes that govern chromosome function or meiosis may be more tightly linked to each other than they are to the genes involved with another function, such as apoptosis. This has been the basis of our collective experience in biological investigations over the last century: that there are groups of proteins which interact together more closely than others. Often they have been organized into pathways such as glycolysis, the Krebs cycle, and other metabolic pathways in which the gene products called enzymes have to work in concert. Other more obvious functional clusters are those of structural proteins which have to come together in a conserved and reproducible fashion in order to serve their purpose, whether they are the components of the ribosomal unit or the histoproteins which are essential to maintenance of the structure of chromatin. On this basis, if we can find genes whose expression patterns approximate one another, we can possibly impute that they are functionally clustered together. That is, they have functions that are related.

Several important caveats are worth noting here. First of all, it remains unclear just how discrete the functional groupings of gene function are in the cellular apparatus. It may be that individual gene products have so many different roles under different circumstances that several of them partake of essential roles in significantly different functions.^[8] The second caveat is that the term "functionally related" is itself ill-specified. If the pattern of expression of one gene is similar to that of another, it could signify all kinds of relationships, ranging from "two genes having gene products that physically interact," to "one gene encoding a transcriptional factor for the other gene," to "two genes having different functions but similar promoter sequences," to "two genes both with promoter sequences bound by repressors which are knocked off when a nuclear receptor is activated, even though the two genes have widely disparate functions." Of course there is a level of abstraction at which all genes are functionally related in their role of keeping the cell alive and producing whatever components are needed for the rest of the organism. But below this level of abstraction, there are many alternative and, by their nature, sloppy definitions of clustering. Therefore, we should be somewhat wary of the claim that similarity in expression corresponds to similarity in function. Nonetheless, it is a useful starting point for many analyses of a genome whose function remains by and large unknown at this time. Additionally, as we discuss in the chapter on dissimilarity measures (chapter 3), the question of what constitutes a similar expression pattern is itself poorly defined, or at least has multiple alternative definitions. For example, similarity could mean having similar patterns of change over time. It could mean similar absolute levels of expression at any given point in time, or it could mean perfectly opposite but well-choreographed patterns of expression. Just which dissimilarity measure is chosen for looking at patterns of expression will influence the kind of functional clusters that we expect.

2.2.1 Supervised versus unsupervised learning

In the introductory chapter, we discussed why the computational techniques applied in the analysis of gene expression are qualitatively different from those of traditional biostatistics: the data sets are of high dimensionality and yet the number of cases are relatively small. Consequently, the number of solutions that could explain the observed behavior is quite large. For this reason, the machine-learning community has recognized the potential role for their techniques specifically designed to explore high-dimensional spaces (such as those of voice or face recognition) and have also recognized the enormous need to apply these techniques to genomic data sets. To this effect, the first reaction of a computer scientist with a background in machine learning when he or she becomes aware of the new challenges created by genomic data sets is to pull out the tools of the standard armamentarium of machine learning. He or she then begins to explore informally, and subsequently evaluates formally the results obtained when these tools are applied to genomics data sets. We provide a framework for the armamentarium in the genomic data mining chapter (chapter 4, page 149) but here we only provide enough of an overview to discuss these tools with respect to experimental design.

Two useful broad categorizations of the techniques used by the machine-learning community are *supervised learning techniques* and *unsupervised learning techniques*. These are also commonly known as classification techniques and clustering techniques, respectively. The two techniques are easily distinguished by the presence of external labels of cases. For example, labeling a tissue as obtained from a case of acute myelogenous leukemia (AML) or acute lymphocytic leukemia (ALL) is needed first before applying a supervised learning technique to create a method to learn which combinations of variables predict or determine those labels. In an unsupervised learning task, such as finding those genes that are co-regulated across all the samples, the organization or clustering of the variables operates independently of any external labels. The kinds of variables (also known as *features* in the jargon of the machine-learning community) that characterize each case in a data set can be quite varied. Each case can include measures of clinical outcome, gene expression, gene sequence, drug exposure, proteomic measurements, or any other discrete or continuous

variable believed to be of relevance to the case.

What kinds of questions are answered by the two types of machine learning? In supervised learning, the goal is typically to obtain a set of variables (e.g., expressed genes as measured on a microarray) on the basis of which one can reliably make the diagnosis of the patient, predict future outcome, predict future response to pharmacological intervention, or categorize that patient or tissue or animal as part of a class of interest. In unsupervised learning, the typical application is to find either a completely novel cluster of genes with a putative common (but previously unknown) function, or more commonly to obtain a cluster or group of genes that appear to have similar patterns of expression to a gene (*i.e.*, they fall into the same cluster) already known to have an important well-defined function. The goal there is to find more details about the mechanism by which the known gene works and to find other genes involved in that same mechanism either to obtain a more complete view of a particular cellular physiology or, in the case of pharmacologically oriented research, other possible therapeutic targets. Although the distinct goals of supervised versus unsupervised machine-learning techniques may appear rather obvious, it is important to be aware of the implications for study design. For example, an analyst may be asked to find classifiers between two types of malignancy, as was done in the Golub *et al.* investigation of AML and ALL [78]. However, the lists of genes that reliably divide the two malignancies may have little to do with the actual pathophysiological causes of the two diseases and may not represent any particular close relationship of those genes and function. Why might this be? It is quite possible that small amounts of change of some gene products such as transcriptional activators and genes such as *p53* may cause large downstream changes in gene expression. That is, with only a subtle change, an important upstream gene may cause dramatic changes in the expression in several pathways that are functionally only distantly related but are highly influenced by the same upstream gene. When applying a classification algorithm directly on the gene expression levels, the algorithm will naturally identify those genes which change the most between the two or more states that are being classified. That is, a study design geared toward the application of a supervised learning technique may generate a useful artifact for classification, diagnosis, or even prognosis, but it will not necessarily lead to valuable insights into the biology underlying the classes obtained.

Let us consider the more general cases where gene expression values are not the only data type. For example, as illustrated in figure 2.11 below, a given case can include several thousand gene expression measurements but also several hundred phenotypic measurements such as blood pressure, a laboratory value, or the response to a chemotherapeutic agent. Here again a clustering algorithm can be used to find those features that are most tightly coupled in the observed data. When designing an experiment that includes the various data types, it is worthwhile thinking ahead of time whether some kinds of features are more likely to cluster together, separately from the genomic data. That is, after application of a clustering algorithm the data set may reveal relationships between the nongenomic variables that are much more significant and stronger than any of those that involve gene expression or sequence. While that is not necessarily a bad outcome, it will not help the investigator who is trying to understand the particular contribution of genetic regulation to the observed phenomenon. As an example, if one looks at the effect of thousands of drugs on several cancer cell lines, then it should not be surprising if these drug effects were most tightly clustered around groups of pharmaceutical agents that were derived from one another through combinatorial chemistry. Similarly, phenotypic features which are highly interdependent, such as height and weight, will cluster together. The strength of these obvious clusters will often dominate those of heterogeneous clusters that contain phenotypic measurements as well as gene expression measurements. This suggests that careful use of feature reduction to only include those features that are nonredundant and only truly independent phenotypic measures for each case should be used. We refer the reader interested in systematic approaches to feature reduction to the excellent text by Sholom Weiss and Nitin Indurkha [188].

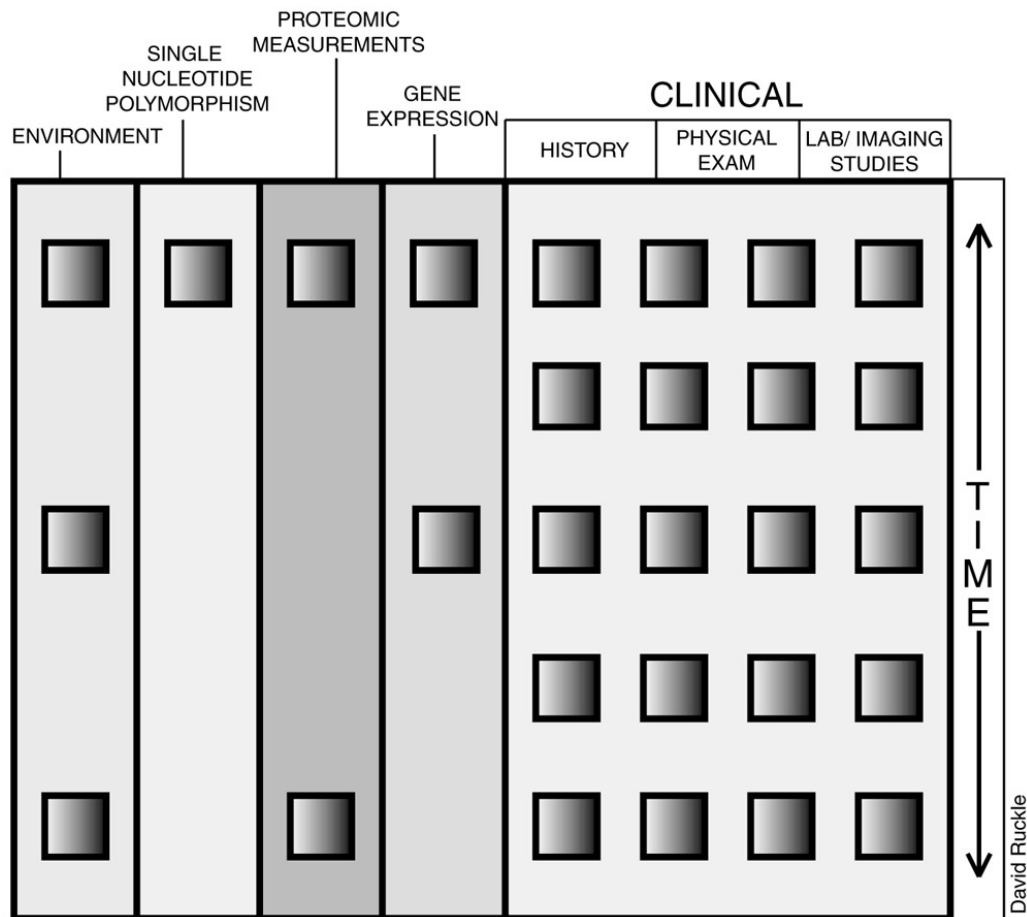


Figure 2.11: Clustering more than only expression values. A full-edged functional genomic study with clinical relevance will involve multiple and quite heterogeneous data types. The realities of clinical research and clinical care will ensure that there is a requirement for the handling of missing data.

2.2.2 Figure of merit: The elusive gold standard in functional genomics

Whereas the discussion above was motivated by the questions posed by our collaborators who are primarily biologists or clinicians, it parallels similar questions coming from those colleagues with a computer science background. One question is, how can we determine whether a particular methodology that we are applying is successful or not? Is one machine-learning algorithm better than another, or does one clustering method provide more robust clusters than another? In other words, how do we know how successful a particular functional genomics investigation is? What is the *figure of merit* that we are trying to obtain?

The figure of merit is probably the most ascertainable for the case of a known classification. Take, for instance, the case of the task of the classifying whether a particular tissue belongs to one of two tissue types using a supervised learning technique. Suppose also that a test sample is available. Then, various classification algorithms can be run against one another and the sensitivity and specificity of these algorithms can be compared. More generally, the receiver operating characteristic (ROC) curve of each of these classification algorithms can be defined. A good example of such a comparison is provided by Michael Brown *et al.* [33]. Several classification algorithms were employed and compared and ranked based on a weighted measure of true and false positives and true and false negatives. The best performance turned out to be in the application of support vector machines. The test that was used for this "bake-off" was the *Saccharomyces cerevisiae* data set from Stanford involving expression data of 2467 genes across 79 different hybridization experiments.^[9] Because the classification is known in advance, the

performance of these algorithms can be measured. However, there are several limitations which should be recognized. The classification algorithms' performance only pertains to the particular population or set of experiments on which it was originally tested. That is, a classification algorithm working on a yeast data set may not necessarily work as well relative to other algorithms on distinguishing two different types of leukemia or two different types of diabetes mellitus in humans. Even less dramatically, the same algorithm trained on one set of patients with leukemia may have a different relative performance compared to other classification algorithms on another set of leukemic patients drawn from a different population or obtained with a different ascertainment bias. For example, one set of patients may have been selected because of particularly refractory disease that caused them to be referred to a tertiary care hospital and another set of patients may have been treated in community hospitals. It is possible that the underlying diseases of these populations may be different and therefore the possibility of underfitting or overfitting the underlying classes of disease in these populations will differ. The figure of merit, therefore, in a classification experiment is the correct classification performance of an algorithm or measure for that particular population. Its applicability or merit for use in even related populations is uncertain or problematic.

Figure of merit in a clustering experiment What is the figure of merit for a clustering analysis? That is, how can we evaluate the degree to which the cluster of genes obtained are in fact correct or relevant? For anybody who has performed a clustering procedure on microarray expression data, it will seem obvious that it is altogether too tempting when one obtains a cluster of genes through any machine-learning algorithm to come up with a *post hoc* justification for why those genes may have fallen or risen together. This temptation is all too evident in the publications of the last 3 years in which clusters of genes reported by investigators are described typically as falling into one of three different categories. The first is well-known published associations, either mechanistically verified or empirically known, that have been obtained through other more conventional techniques. Second is associations that appear plausible to the authors, and presumably the reviewers, but have not been proven in the literature. The third is associations for which the authors are hard-pressed to find support. For the biological investigator who wishes to further investigate the fundamental biology of genes and their function, the last two categories constitute a fairly unsatisfactory basis on which to invest several months, if not years, understanding the basis for the imputed associations or potential functional dependencies.

The challenge then for the functional genomist using microarray data is to come up with means to validate the clusters obtained. There are two levels of validation. The first is a statistical or methodological validation with techniques such as permutation or cross-validation as described elsewhere in this book (section 4.12.1 and section 4.12.2). These techniques can be used to ensure, within a specified degree of certainty, that for the particular data set being studied, the clusters are neither the result of serendipitous coordination of gene behavior nor that the samples are apparently inadequate^[10] to estimate the reliable coordinated behavior of gene expression. An alternative method is to estimate the probability of particular clusters. This methodology is still only in its infancy and the relevant literature is only now beginning to be generated for functional genomics.[15, 178]. As with other aspects of functional genomics, much larger data sets will allow more effective use of these methods. Ultimately however, we should remember that the clusters or classifications that we obtain through machine-learning techniques are only reflections on the measurements made in a particular system. If we are to make broad claims about the empirical or scientific truth of the relationships or classes that we infer from expression data, we will have to submit analyses to the same kind of tests that have been developed for other experimental sciences with longer histories. Specifically, we need to at least come up with the microarray equivalent of Koch's postulates. In 1890, Robert Koch set out to develop criteria for judging whether a given bacterium was the cause of the disease. The criteria are summarized in table 2.2.

Table 2.2: Koch's postulates

1. It (the suspected pathogen) should be present in every instance of the disease.
 2. It should be isolated from the diseased host and grown in pure culture.
 3. The specific disease must be reproduced when a pure culture of the agent is inoculated into a healthy susceptible host.
 4. The same agent must be recoverable from the experimentally infected host.
-

It has been recognized there are several problems with Koch's postulates and there is a great deal of discussion and disagreement about this in the literature of the history of science [143]. However, the postulates are a good first approximation of the requirement for validation of functional genomics experiments. Therefore the analogs to Koch's postulate in the domain of microarrays may be useful. These are illustrated in table 2.3. The criteria describe the test for an inferred regulatory dependency obtained through microarray analysis.

Table 2.3: Functional genomics analogs to Koch's postulates

1. If gene A is found to be correlated with gene B then this relationship should be reproducible through northern blots and other quantitative expression measurement techniques [151].
 2. Furthermore, in all pathological conditions in which the gene is thought to play a critical role, the predicted level of that gene should be found at the right time and in the right part of the tissue during the disease process.
 3. If a gene thought to be involved in a pathway is underexpressed or overexpressed in a model system, then the process controlled by that pathway will be affected.
-

It is only with these biological tests that we can "rate" the performance of clustering algorithms—it is the only durable figure of merit. It is not surprising that very little of this kind of hypothesis testing has been reported in the literature. After all, for each gene in a cluster, there is implied a laborious sleuthing effort in which the bioinformatician and biologist must closely collaborate to verify the results using the analogs to Koch's postulates in table 2.3. If we are to avoid a functional genomics "meltdown," hinted at in the introductory chapter, then the interdisciplinary nature of the functional genomics pipeline (diagrammed in figure 1.3.1) will have to be substantively adopted. All gene associations or regulatory relationships suggested by bioinformatics techniques should undergo at least a minimal validation by the biologist. Conversely, experimental design without the involvement of the bioinformatician is likely to lead to wasteful, expensive, and unrewarding analyses.

^[7]In other words, there are several distinct and statistically distinguishable processes.

^[8]Well-known examples of these are the transcriptional factors such as sonic hedgehog (*shh*) which in some tissues and at some times are involved in cell proliferation and in others in cell differentiation processes.

^[9]Although this paper is one of the better examples of comparison and classification techniques, it does not compare the entire ROC curve but only one single point on the curve. Therefore, it remains unclear what would be the overall performance of these algorithms under circumstances in which specificity and sensitivity were valued differently. See page 201 for a brief discussion of using

the ROC curve to evaluate a classification test.

^[10]Insufficient numbers of biological samples, insufficiently strong effects, or sufficiently distinct biological processes, or any combination of these.

Chapter 3: Microarray Measurements to Analyses

3.1 Generic Features of Microarray Technologies

The gene expression assaying technologies that will be discussed in this chapter are RNA detection microarrays, variously known as DNA chips, biochips, or simply chips. It is easiest to think of the utility of microarrays in functional genomics as a five-step process.^[1]

- *Probe*: This is the biochemical agent that finds or complements a specific sequence of DNA, RNA, or protein from a test sample [8]. It can include pieces of cDNA amplified from a vector stored within a bacterial clone, oligonucleotides synthesized in a fluid medium, oligonucleotides built on a solid phase base pair by base pair, or nucleotide fragments of a chromosome. This can be extended to incorporate the database of expressed sequence tags (ESTs) for serial analysis of gene expression (SAGE), (see Glossary, page 277).
- *Array*: The method for placing the probes on a medium or platform. Current techniques include robotic spotting, electric guidance, photolithography, piezoelectricity, fiber optics, and microbeads. This step also specifies the type of medium involved, such as glass slides, nylon meshes, silicon, nitrocellulose, membranes, gels, and beads.
- *Sample probe*: The mechanism for preparing RNA from test samples. Total RNA may be used, or mRNA may be selected using a polydeoxythymidine (poly-dT) to bind the polyadenine (poly-A) tail. Alternatively, mRNA may be copied into cDNA, using labeled nucleotides or biotinylated nucleotides.
- *Assay*: How is the signal of gene expression being transduced into something more easily measurable? For the microarrays in common use, gene expression is transduced into hybridization. For SAGE, gene expression is transduced into oligonucleotides via restriction enzymes and ligation. For PCR, gene expression is transduced into amplified pieces of cDNA.
- *Readout*: How is the transduced signal going to be measured? How is information about the signal represented? For the microarrays in common use, hybridization is typically measured either using one or two colored dyes, or radioactive labels. For SAGE, the constructed oligonucleotides are measured through sequencing. For PCR, the amplified pieces of cDNA can be measured using gel electrophoresis.

For the microarrays in common use, one typically starts by taking a specific biological tissue or system of interest, extracting its mRNA, and making a fluorescence-tagged cDNA copy of this mRNA. This tagged cDNA copy, typically called the *sample probe*, is then hybridized to a slide containing a grid or array of single-stranded cDNAs called *probes* which have been built or placed in specific locations on this grid (see table 3.1 for alternative terminology). Similar to the general hybridization principles behind Southern or Northern blots (see glossary, page 277), a *sample probe* will only hybridize with its complementary *probe*. Fluorescence is typically added to the sample probe in one of two ways: Either (1) fluorescent nucleotide bases are used when making the cDNA copy of the RNA, or (2) biotinylated nucleotides are first incorporated, followed by an application of fluorescence-labeled streptavidin, which will bind to the biotin. Depending upon manufacturer-specific protocols, the probe-sample probe hybridization process on a microarray typically occurs over several hours. All unhybridized sample probes are then washed off and the microarray is lit under laser light and scanned using laser confocal microscopy analogous to the phosphor imager in the traditional blot procedures. A digital image scanner records the brightness level at each grid location on the microarray corresponding to particular RNA species.

Table 3.1: Common probe nomenclature

	Immobilized nucleic acid on microarray surface	Free nucleic acid that is being interrogated
General microarray	Probe	Sample Probe
Robotically spotted microarray	Probe	Probe
Affymetrix microarray	Probe	Target

Studies have demonstrated that the brightness level is correlated with the absolute amount of RNA in the original sample, and by extension, the expression level of the gene associated with this RNA [157]. At a coarse level, one microarray experiment may be thought of as N many Northern blots that simultaneously assay a total RNA sample on a small common medium or substrate for as many different mRNA species— N being the total number of unique RNA species probes located on the physical chip. Note that the amount of total RNA required for a typical Northern blot is more than sufficient for one microarray experiment in the current technologies. However, this analogy breaks down in that only a single hybridization condition (*e.g.*, temperature, time) is used in hybridizing all N assays, and unless the probes are carefully chosen, this may not be the optimal condition for the assay of all RNA species.

A prominent characteristic of microarray technologies is that they enable the comprehensive measurement of the expression level of many genes simultaneously on a common substrate. In this regard, typical applications of microarrays include the comprehensive quantification of RNA expression profiles of a system under different experimental conditions, or expression profile comparisons of two systems under one or several conditions. The former embraces the comparison of expression profiles of a system under a control and a test condition, whereas the latter includes contrasts between different strains of organisms, as, for instance, between a normal (*e.g.*, wild-type), and a constructed (*e.g.*, knockout) organism. Another intriguing use of microarrays is to compare expression levels between neighboring cells within the same microscopic field, as demonstrated in [123]. Aside from their widespread utility in functional genomics, oligonucleotide microarrays have also been used for single nucleotide polymorphism (SNP) analysis since many probe SNPs can be placed on the microarray for comprehensive parallel SNP detection, and in much the same way one can also perform DNA sequence analysis.

Regardless of the gene expression technology to be adopted, almost all of them have performance factors that depend critically on the general validity of certain fundamental biological assumptions outlined below:

1. *There is a close correspondence between mRNA transcription and its associated protein translation.* As noted by Brown and Botstein [34], one would ideally like to measure the final products of every gene, such as proteins, or even better, the biochemical activity of these products, which are more directly related to biological functionality. Such quantitation would provide a link between chemical DNA bases at microscopic levels with biological aspects that are manifest at macroscopic scales such as phenotype and physiology. However, there is no practical generic tool to do this yet. The assumption that there is a principle of parsimony—akin to Hamilton's principle of least action in the physical sciences—which drives the close relationship between gene expression and biological function was most clearly articulated by P. O. Brown and D. Botstein in [34]:

"The second reason [for using DNA microarrays to study gene expression on a genomic scale] is the tight connection between the function of a gene product and its expression pattern. As a rule, each gene is expressed in the specific cells and under the specific conditions in which its product makes a contribution to fitness. Just as natural selection has precisely tuned the biochemical properties of the gene product, so it has tuned the regulatory properties that govern when and where the product is made and in what quantity. The logic of natural selection, as well as experimental evidence, provides part of the basis for our belief that there is a sensible link between the expression pattern and the function of its gene product. Thirty years of molecular biology have provided numerous examples of genes that function under specific conditions and whose expression is tightly restricted to those conditions."

Biologists can quickly find exceptions to this assumption. For example, proteins that make up the cellular matrix can considerably outlast the lifetime of their associated mRNA, or conversely, they may be metabolized or degraded much more rapidly than the mRNA from which they were transcribed. Nevertheless, the initial successes in the applications of gene expression microarrays in investigations of expression and function suggest that this assumption holds true more frequently than not.

2. *All mRNA transcripts have identical lifespans.* Again, there are several well-known exceptions. For instance, we know that length of the 3' poly-A tail of an mRNA appears to be related to its stability. Furthermore, there are examples of mRNA that have longer- or shorter-term stability within specific cells or after its transcriptional event such as dystrophin mRNA from patients with certain types of muscular dystrophies. As a tangential comment on temporal effects: The probe-sample probe hybridization rate is known to be a function of the guanine-cytosine (GC) content of a transcript. In general, this rate is proportional to GC richness.
3. *All cellular activities and responses are entirely programmed by transcriptional events.* At a meta-systems level, this assumption may indeed be true, but in terms of direct mechanistic coupling, there exist many examples in which external stimuli cause changes in the biochemical program within the cell *without* engaging the transcriptional machinery. Figure 3.1 is an illustration of a response of free intracellular calcium to aldosterone exposure. Aldosterone is a steroid hormone that typically acts through binding with receptors that are translocated to the nucleus and which then initiate or modify a transcriptional program. Here, the time scale of the acute response suggests a nongenomic mechanism; in other words, the response does not require transcriptional activation. This example demonstrates how a molecule that usually works via modulation of transcription (*i.e.*, steroid hormones) may also affect bioprocesses at the nongenomic level. There is also a much larger class of biological processes that do not primarily operate at the transcriptional level. These include muscular contraction, nerve excitation, and hormonal release. Eventually, all these events will cause some change in transcriptional activity, *e.g.*, replenishing stores of neurotransmitter, but the patterns of gene expression would probably not reveal the *control processes* that govern them at the subgenomic time scale.

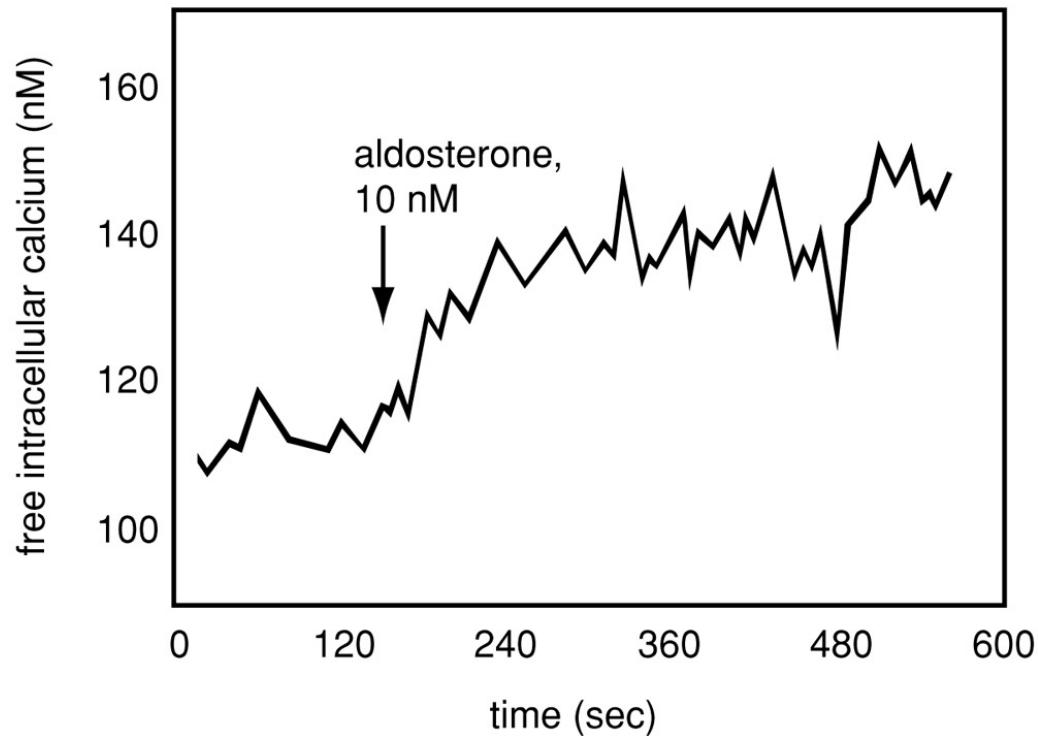


Figure 3.1: The nongenomic time scale response in aldosterone exposure. The response (in seconds) shown here is *much* faster than any known receptor-to-transcription response that steroid hormones are usually thought to act through. (Derived from Gamarra *et al.* [74].)

There are presently two types of chip technologies in common usage: *robotically spotted* and *oligonucleotide* microarrays. It is important to note that these two microarray technologies are the very earliest and are currently the most widely used. Several competing technologies are emerging that may prove to be more cost-effective, reliable, and versatile (see chapter 7). Next, we briefly describe these two most popular microarray technologies.

3.1.1 Robotically spotted microarrays

Robotically spotted microarrays (figure 3.2), namely, the robotically spotted cDNA glass slide, were introduced into common use at Stanford University and first described by Mark Schena *et al.* in 1995 [157]. They are also known as cDNA microarrays.

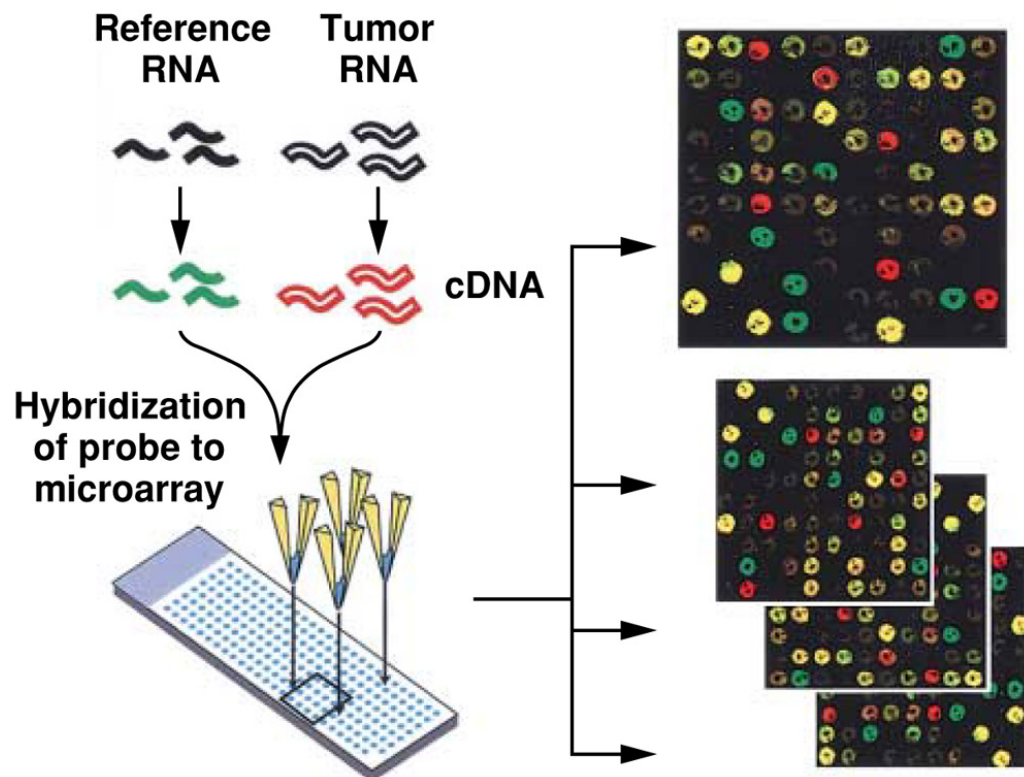


Figure 3.2: Robotically spotted microarray hybridized to two samples, each stained with two colored dyes. An overview of procedures for preparing and analyzing cDNA microarrays and tumor tissue. Reference RNA and tumor RNA are labeled by reverse transcription with different fluorescent dyes (green for the reference cells and red for the tumor cells) and hybridized to a cDNA microarray containing robotically printed cDNA clones. The slides are scanned with a confocal laser scanning microscope, and color images are generated for each hybridization with RNA from the tumor and reference cells. Genes upregulated in the tumors appear red, whereas those with decreased expression appear green. Genes with similar levels of expression in the two samples appear yellow. Genes of interest are selected on the basis of the differences in the level of expression by known tumor classes (e.g., BRCA1 mutation-positive and BRCA2 mutation-positive). Bioinformatics analysis determines whether these differences in gene expression profiles are greater than would be expected by chance. (Derived from Hedenfalk *et al.* [90].)

In making the array, a robotic spotter mechanically picks up specific cDNA sequences, typically amplified from vectors in bacterial clones using PCR, from separate physical containers and deposits them in specific locations in the grid on the glass slide to create specific probes. Each cDNA drop should ideally be equal in quantity. This fabrication approach epitomizes the do-it-yourself tendency in microarray measurement, even though there are several commercial ready-to-use versions available. The frequently home-grown quality of these arrays has led to the production of highly localized and customized microarrays which pose specific challenges in the dual tasks of background noise reduction and foreground RNA signal amplification during subsequent data analysis stages. These challenges are discussed in section 3.2. Designer-definable parameters exist to control spot-basing size, the amount of time for drying, as well as experimental parameters concerning the glass slide material to be used.

There are several advantages and disadvantages in using a robotically spotted microarray:

- The first advantage is customizability. A large subsequence ($\sim 2 \times 10^3$ base pairs long) complementary to the actual sequence that is to be probed is laid down on the chip by the designer who has full control over the species of probes that are to be used. This means, e.g., that if one wishes to make customized chips for a specific purpose, such as for probing

the expression of specific RNA in certain cell types, or specific polymorphisms of those genes, one can design a layout and can direct a robotic spotter to make these microarrays. The setup cost of this approach is on the order of \$20,000 and detailed guidelines may be found at the Brown laboratory microarraying website.^[2] Additionally, companies such as Affymetrix, Amersham Pharmacia, BioRobotics, and Cartesian Technologies sell robotic spotters—arrayer units. Glass slides are available from a variety of vendors specifically for microarray construction. Commercial aspects of the manufacture of such chips appear to be expanding. The notable disadvantage with greater customizability is that it may lead to more possibilities for errors. For instance, poor quality control or nonuniformity in the construction of different species probes such as spot-basing size will complicate the subsequent analysis and interpretation of the resulting chip data.

- The second advantage of robotically spotted microarrays is that larger pieces, or entire cDNAs, are placed on the chip, thus reducing the likelihood of nonspecific hybridization of labeled sample probes to the probe that was laid on the chip. Typically, a designer who wishes to have a particular gene probe on a chip will create a clone with the 5' and 3' ends of that particular gene of interest. The substrings of these cDNAs of the original gene are on the order of 100 to 200 base pairs long. A distinct weakness here is that even though a long probing subsequence ensures a sufficiently confident representative substring of the original gene, it does not mean that hybridization conditions will be fully and equally optimized for all species of cDNA subsequences. As we have noted earlier, the probe-sample probe hybridization rate is known to vary depending upon the GC content of a transcript.
- The third advantage is that RNA from two different samples (typically a test and a control condition) can be hybridized onto a common cDNA microarray substrate at the same time. The two separate RNA samples are typically labeled with different fluorescent dyes, such as Cy3 and Cy5. The two-dye system allows for the excitation of the microarray by laser light at two different frequencies and thus the image of hybridized RNA abundance can be scanned for both colors (corresponding to distinct samples) separately. Since the hybridization conditions, and thus the brightness, of any one spot is not the same as another spot, the individual signals are not typically used separately. Instead, the calculated signal is the *ratio* or *fold* difference in the brightness of the hybridized RNA of one sample versus another, specifically the intensity of Cy3 versus Cy5, *i.e.*, Cy3/Cy5. If a background intensity for each color is measured and controlled for, the ratio becomes $\frac{\text{Cy3} - \text{Cy3 background}}{\text{Cy5} - \text{Cy5 background}}$. This in turn provides a measure of relative abundance of the RNA from one sample with respect to the RNA from another sample, and hence the relative expression level of the corresponding gene in the samples. On the other hand, it has been shown that not all cDNA sample probe sequences label symmetrically with Cy3 and Cy5 [106] and paired dye-swapping experiments are performed (*i.e.*, if the first hybridization is control (Cy3) versus test (Cy5), then switch dyes for the second hybridization). If the labeling were symmetrical, then the plot of $\left(\frac{\text{Cy3}}{\text{Cy5}}\right)_{\text{Hybridization 1}}$ versus $\left(\frac{\text{Cy3}}{\text{Cy5}}\right)_{\text{Hybridization 2}}^{-1}$ for every probe would be line of slope one through the origin.

In summary, each robotically spotted microarray experiment has its own built-in control, and results are given in terms of fold change or difference from a control situation. However, the measurement of *absolute* quantities and variations are more challenging in this technology.

3.1.2 Oligonucleotide microarrays

The second popular class of microarrays in use has been most notably developed and marketed by Affymetrix. Currently, over 1.5×10^5 oligonucleotides of length 25 base pairs each, called *25-mers*, can be placed on an array. These oligonucleotide chips, or oligochips, are constructed using a photolithographic masking technique similar to the process that is used in microelectronics and

integrated circuits fabrication, first described by Stephen Fodor *et al.* in 1991 [70]. Currently, these commercially available microarrays are not produced individually, but instead are made in parallel. An entire wafer (containing between 40 and 400 microarrays) is constructed, tested, then broken apart to create the individual microarrays. At this time, commercially produced oligochips exist that are disease—as well as species—specific, such as rat neurobiology and yeast genome arrays, and custom microarrays can be ordered with a 4-week turnaround or less.

The manufacturing technique for an Affymetrix oligochip is markedly different from the more mechanical process of making robotically spotted arrays. Each wafer starts out as an empty glass slide. On this substrate, 25-mer probes are built base by-base by placing single DNA bases on the glass and then on top of a preceding base. All 25-mer probes are constructed in parallel, with high precision, by selectively masking specific coordinates or locations on the glass surface and exposing the entire ensemble to ultraviolet light in between laying on the additional bases adenine (A), thymine (T), guanine (G), and cytosine (C) separately. Each applied photolithographic mask generates different areas of photodeprotection on the solid glass substrate. The combination of these masks with an intervening chemical coupling step allows the incorporation of additional nucleotides to existing strands only where desired. This entire process is known as *light-directed oligonucleotide synthesis*.

One advantage of the oligonucleotide microarray is that its higher density of probe pairs allows for more genes to be screened or assayed on a single chip, as compared to robotically spotted arrays. Consequently, there is less need for *a priori* restrictions on the number of genes that are to be scanned. A disadvantage is that the current technology only allows for at most one experiment to be run on a single chip at any one time. Thus, for example, one does not obtain meaningful data from placing the control sample and test probes on an oligonucleotide microarray simultaneously; instead these two samples are measured on two separate oligochips. This means, in turn, that one typically has to apply a suitable normalization transformation across separate microarray data sets (*i.e.*, *inter-array*) at the subsequent data analysis stage in order to make meaningful comparisons of reported expression changes from a control to a test condition. Additionally, if a scientist is interested in studying a specific species for which no appropriate oligochip exists, then this technology is not presently available. Oligochips are not as easily customizable at the user's end as robotically spotted microarrays.

Since each probe is limited to 25 base pairs in length, a question immediately arises as to how each gene can be screened uniquely using only 25 base pairs. For each gene that needs to be represented, or whose expression needs to be measured on Affymetrix's oligochip, a set of sixteen to twenty 25-mers are chosen which uniquely represent that particular gene, and would hybridize under the same general conditions.^[3] The Affymetrix literature calls the sample probe that is to be interrogated by cDNA probes on its microarray, the *target*. Every set of *perfect match* (PM) probes for an mRNA has a corresponding set of *mismatch* (MM) probes. An MM probe is constructed from the same nucleotide sequence as its PM probe partner, except that the middle (usually the 13th) base pair has been switched to result in an alphabet mismatch. For example, the following two 25-mers may be associated PM-MM probes to assay for the following sample probe:

Probe on chip	{	PM :	ATCGACTGATGCATGCATCCATCAT
		MM :	ATCGACTGATGCCTGCATCCATCAT

Sample Probe	TAGCTGACTACGTACGTAGGTAGTA
--------------	---------------------------

The combination of a PM and its associated MM oligonucleotide probe is called a *probe pair*. There are two principal reasons for the use of MM probes. First, at low concentrations of the target or sample probe when the PM probes have already reached their lower limit of sensitivity, the MM probes display greater sensitivity to changes in concentration. Second, MM probes are thought to

bind to nonspecific sequences at the same rate as the PM probes. Thus, MM probes serve as an internal control for *background* nonspecific hybridization (see section 3.1.2 for details). However, depending upon the total RNA sample, it could turn out that the PM probe is already highly specific and the MM probes are simply binding to differently-specific labeled subsequences in the sample. One has to be careful to distinguish between *nonspecific* hybridization and *differently-specific* hybridization with regard to the use of MM probe data.

For target preparation, sufficient amounts of sample probe are first synthesized by reverse-transcribing the total RNA using an oligo-dT primer containing a T7 polymerase site for 52 to 32 transcription. Amplification and labeling of the cDNA sample probe is achieved by carrying out an *in vitro* transcription reaction in the presence of biotinylated deoxynucleotide triphosphates (dNTP), resulting in the linear amplification of the cDNA population (approximately 30- to 100-fold). This linearity assumption becomes increasingly weak with decreasing quantities of total RNA and increasing number of amplification cycles. The biotin-labeled cRNA probe generated from the sample is then hybridized to the oligonucleotide arrays, followed by binding to a streptavidin-conjugated fluorescent marker. Laser excitation of the hybridized sample, confocal microscopy, and image acquisition by an optical scanner is performed. This results in an image file in which each oligonucleotide species is represented by a small rectangular area ($\sim 50 \frac{1}{4}m^2$), called the probe cell, which is itself composed of several image pixels, each occupying an area from 3 to 24 $\frac{1}{4}m^2$ (figure 3.4). The image file is processed so that the intensity of each probe cell is reported as the 75th percentile of the intensities of all the pixels in a probe cell, excluding the pixels at the border of the cell. These probe cell intensities are stored in a `.cel` file, which therefore reports a measure of hybridization per contiguous oligonucleotide surface on the microarray. We describe the contents of a typical `.cel` file below.

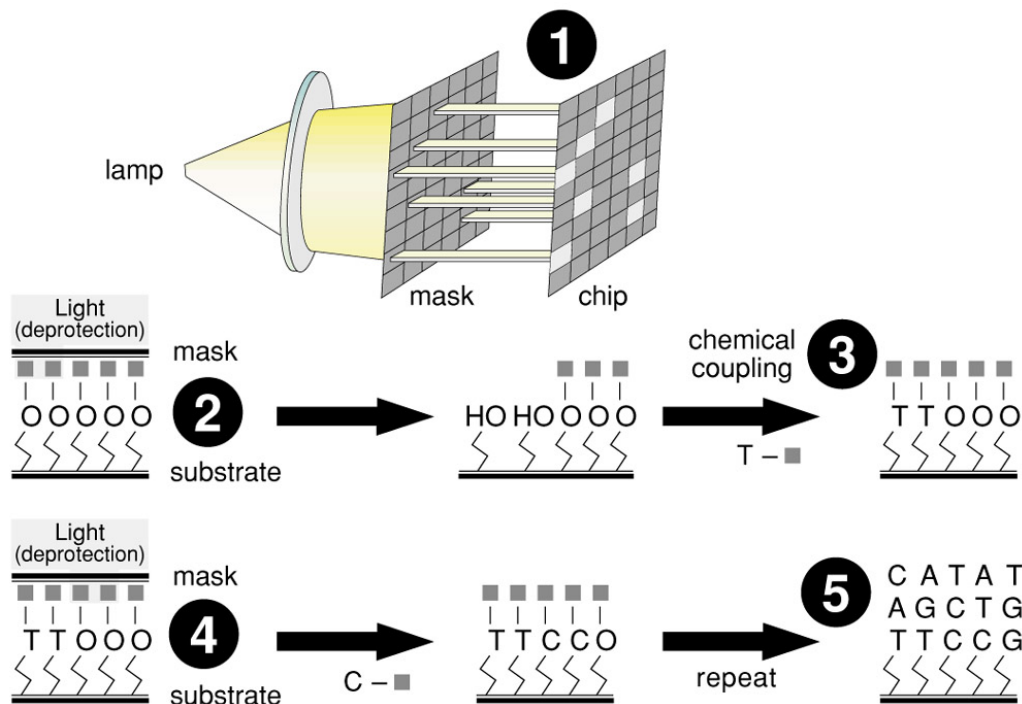


Figure 3.3: The photolithographic construction of microarrays. Synthesized high-density oligonucleotide microarray manufacturing with photolithography. Using selective masks, photolabile protecting groups are light-activated for DNA synthesis (1, 2); photoprotected DNA bases are added and coupled to the intended coordinates (3). This cycle is repeated (4) with the appropriate masks to allow for controlled parallel synthesis of oligonucleotide chains in all coordinates on the array (5). (Derived from Lipshutz *et al.* [121].)

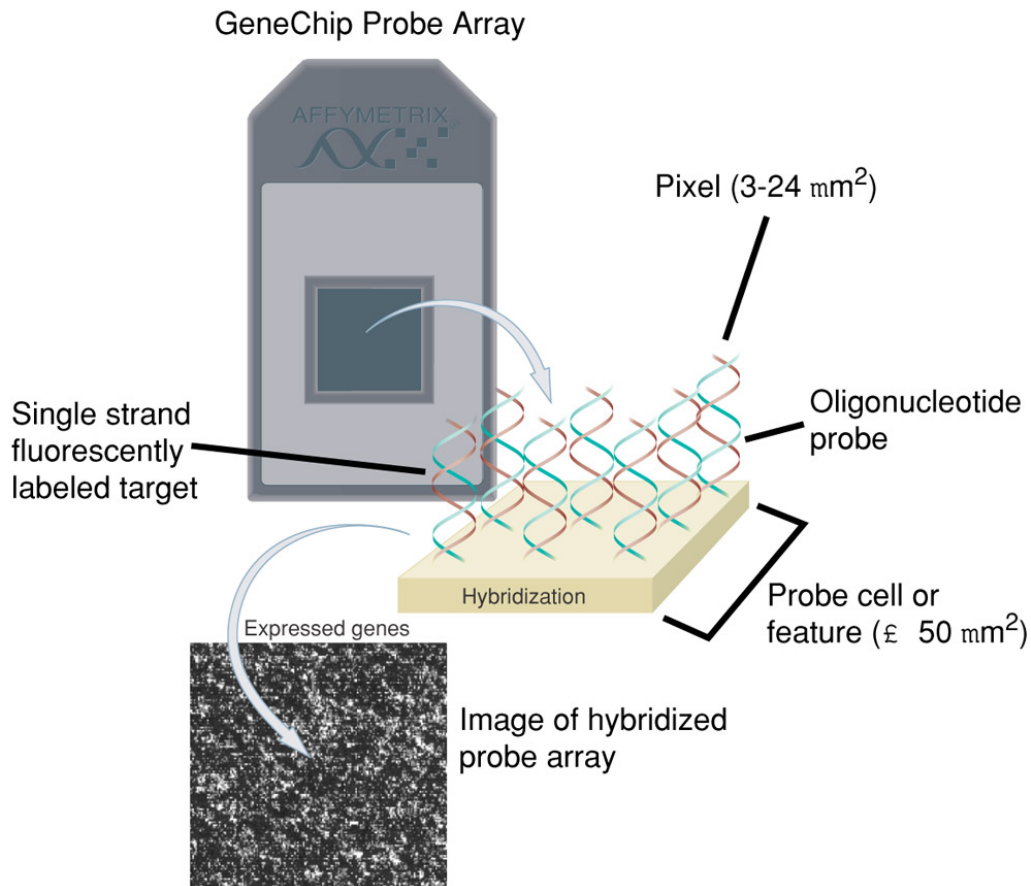


Figure 3.4: Pixel, probe cell, and Affymetrix scanned image. Schematic showing how images are composed of probe cells, which contain probes that appear as pixels. (Derived from Jain [102].)

Affymetrix microarrays and background intensity calculations The Affymetrix GeneChip analysis protocol uses the term *absolute analysis* to describe the algorithm for determining whether transcripts represented on the probe array are detected and the intensity of expression. Briefly, for *each* microarray that has been hybridized with a prepared target, washed, and scanned, one obtains the following sequence of files as the Affymetrix software attempts to translate that particular chip experiment into numerical data representing detectable RNA intensity levels:

- A grey-scale `.tiff` image file of the physical microarray where a lighter or darker pixel indicates a respectively stronger or weaker hybridization of a cDNA fragment to the probes at the particular coordinate marked by the pixel
- A `.dat` text file containing coordinates and intensity levels for individual pixels
- A `.cel` text file with probe cell coordinates and intensity calculated as a trimmed average of pixel intensities—on average a probe cell is made up of $8 \text{ pixels} \times 8 \text{ pixels}$
- A `.chp` text file with 11 columns that contain the statistic, calculated by the Affymetrix software, for sets of 16 to 20 probe cell pairs, PM and MM, which interrogate particular transcripts

An important consideration in all microarray experiments is the contribution of the intensity of the *background effects* on each microarray. Since this can vary from one array to another, Affymetrix has developed its own methodology for subtracting the *background* from the hybridization signals. *Background effects* refer to brightness that ends up in the reported measurement reading for a probe cell, even though these effects did not originate from the probe cell. It typically includes *intra*-chip phenomena such as localized physical changes (*e.g.*, temperature during hybridization) in a probe cell that diffuse into its neighboring probe cells, or nonuniformity in ambient brightness levels in localized regions on a microarray surface. Background effects are components of the more

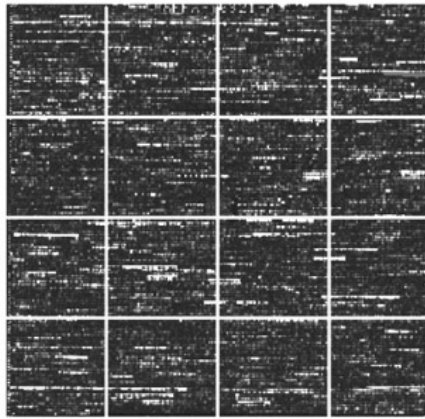
general phenomenon of *noise*, which we discuss in section 3.2.

The entire microarray surface is divided into 16 sectors. Within each sector an average statistic is calculated from the lowest 2% of probe cell intensity values. This is the background intensity for that sector and this value will be subtracted from the average intensities of all image features in that sector. Consequently, the number of background probe cells (*i.e.*, probe cells that are used to calculate background intensity) will depend upon the number of probe cells in the array. In theory, calculating background on a per-sector basis minimizes the effect of changes in the microenvironment across different parts of the array.

In addition to calculating average background intensity, background probe cells are also used to compute the effects of background *noise* variations on the reported measurements. As shown in figure 3.5, the mean intensity of these background probe cells is obtained and the distribution around the mean is calculated. In other words, it is assumed that all the background probe cells should be the same near-zero intensity, and the variation around this is considered noise. Intuitively, a wider distribution around the mean for the background probe cells implies a more pronounced noise component for all expression measurements in the probe array.

The calculation corresponding to this intuition is given in figure 3.5(b). This is the formula that Affymetrix uses to calculate the significance of a difference in intensities between the PM probe cells and associated MM probe cells. Both the ratio and the difference between the PM and the MM probes, PM/MM and PM-MM, for each probe pair are computed. These values are then compared against two thresholds respectively: the *statistical ratio threshold* (SRT) and the *statistical difference threshold* (SDT)— see figure 3.5C —which are themselves functions of the background for that probe array. If both thresholds are exceeded in the negative direction, the probe pair is considered "negative," and if they are exceeded in the positive direction, the probe pair is "positive." These assignments of probe pairs as negative or positive are then incorporated into more aggregate measures described below.

A.



Take the average of the lowest 2% of cell intensity values in this sector. This is the background of the sector. Subtract the Background from the average intensities of all cells in this sector.

The number of **Background Cells** depends on the number of probe cells in the array:

B.

$$Q = 1 / N \sum_{i \in \text{allbgcells}} \frac{\text{stdev}_i}{\sqrt{\text{pixel}_i}} * SF * NF$$

↓ ↓ ↓ ↓ ↓ ↓
 Noise for a given probe array Total # of background cells - lowest 2% for each sector Standard deviation of the intensities of the pixels making up background cell *i* Total # of pixels in cell *i* Scaling Factor Normalization Factor

C.

	Intensity		Intensity	
PM	2,346	This probe pair is most likely Positive since the intensity of PM is much higher than MM	PM	293
MM	684		MM	1,872
				This probe pair is most likely Negative since the intensity of MM is much higher than PM

The significance is determined by calculating both the ratio (PM / MM) and the difference (PM - MM) associated with each probe pair. These values are then compared against two threshold values: the **Statistical Difference Threshold (SDT)** and the **Statistical Ratio Threshold (SRT)**. This is expressed mathematically as follows:

A probe pair is **Positive** if:

1. PM - MM ≥ SDT

And

2. PM / MM ≥ SRT

A probe pair is **Negative** if:

1. MM - PM ≥ SDT

And

2. MM / PM ≥ SRT

Note: not all probe pairs will be scored as Positive or Negative.

Figure 3.5: Background noise on Affymetrix arrays. **A**, A diagram showing how background noise variance is obtained from the background cells. **B**, The calculation of the background noise. **C**, The definition of SDT and SRT and how they define "positive" and "negative" probe pairs. (Derived from [10].)

For each gene whose expression the microarray has been designed to measure, there are between 16 and 20 probe cells representing PM probes and a same number of cells representing their associated MM probes. Collectively, these 32 to 40 probe cells are known as a *probe set*. A `.cel` file contains all the probe cell intensities for all the probe sets represented on a microarray. The `.cel` file is used, in turn, to generate derived or aggregate statistics for each probe set (e.g., a measure of expression or a particular gene). These aggregate statistics are stored in a `.chp` file.

Theoretically, each Affymetrix probe set of 16 to 20 25-mers is designed to be uniquely

representative of a particular gene or EST, and to no other known gene or EST. However, it is often the case that a probe set cannot be found for a particular EST which fulfills these rules, and thus must be designed by relaxing the rules. In practice, nonspecific hybridization of individual 25-mers is more significant than in robotically spotted microarrays, and this is one form of *noise* (see section 3.2 for more coverage of noise). In addition, nonspecific hybridization with other RNA species may result in an entire probe set for a gene fluorescing with little or no noticeable contrast between the PM and MM, as shown in figure 3.6:



Figure 3.6: Nonspecific hybridization on an oligonucleotide microarray. The two rows of probe cells represent the probe set for a gene transcript. The PM probes are on the top row and the MM probes are on the bottom row. Even if there is a lot of specific hybridization to the PM oligonucleotides due to the presence of the targeted gene transcript, if there is significant nonspecific hybridization, then the amount of transcript cannot be estimated accurately. In this illustration, the number and intensity of dark probe cells on the bottom row is higher than that in the top row and therefore most software packages would report that the reported intensity for the gene is unreliable—an "Absent" *Absolute Call* in Affymetrix parlance.

Recall that in theory, an MM probe is designed to bind to nonspecific sequences at the same rate as its associated PM probe. Due to the low specificity of individual oligonucleotide probes as compared to the lengthy cDNA sequences used in most robotically spotted microarrays, Affymetrix's goal has been to use all 32 to 40 probe cells to increase the aggregate specificity. In order to meet this goal, Affymetrix has developed several measures across the PM and MM to increase specificity and generally improve the signal-to-noise ratio of the hybridization measurements. These derived values provide measures of quantified gene expression and the reliability of gene expression. These measures are summarized in table 3.2.

Table 3.2: Affymetrix aggregate or derived measures per probe set

<i>Positive Fraction</i>	$\frac{\# \text{ positive probe pairs}}{\text{total } \# \text{ probe pairs}}$. Pairs used are equal to the number of probe pairs in the probe set minus the masked-out probe pairs. For a variety of reasons, the user may decide to mask out one or more probe pairs. Also, probe cells reporting intensities greater or less than 3 SD from the mean probe cell intensity in the probe set are masked out for being outliers.
<i>Positive-to-Negative Ratio</i>	$\frac{\text{Pos}}{\text{Neg}}$ ratio or $\frac{\# \text{ positive probe pairs}}{\# \text{ negative probe pairs}}$.
<i>Log Average Ratio</i> (Log Avg Ratio)	A number describing the hybridization performance of a probe set by determining the ratio of the PM to MM intensities for each probe pair, taking the logarithm of each of the resulting values, then averaging those across the probe set. This is a slight simplification as the reported Log Avg Ratio is also corrected to exclude the extremal outlier probe pairs in a probe set, <i>i.e.</i> , pairs with the largest and smallest contrast in intensity. It indicates random cross- or nonspecific hybridization with the higher values suggesting a higher likelihood that a transcript is detected.
<i>Average Difference</i> (Avg Diff)	The Avg Diff is a number calculated by taking the difference between the PM and MM of every probe pair and averaging the differences over the entire probe set. It corresponds to the absolute expression level of a transcript.
<i>Absolute Call</i> (Abs Call)	Has values A (Absent), M (Marginal) or P (Present) regarding the presence of a transcript and is determined from a decision matrix combining the Log Avg Ratio, $\frac{\text{Pos}}{\text{Neg}}$ Ratio and Positive Fraction.

The two most commonly quoted measures are the *Average Difference* (Avg Diff) and the *Absolute Call* (Abs Call). The Avg Diff is an aggregate measure of the difference between the PM and MM probe cell intensities per probe set. In the simplest case, a highly specific hybridization will "light up" all the PM probe cells and none of the MM probe cells. In this case, the Avg Diff would be a positive number which increases with the quantity of that particular RNA species present in the sample. In practice, the pattern of figure 3.6 is more common. The greater the intensity of the MM probe cells, the lower the Avg Diff. If the intensities of the MM probe cells exceed those of the PM probe cells, then a negative Avg Diff can be reported. Therefore, even if a gene is expressed highly in a particular biological sample, the Avg Diff could be negative if there is also a lot of differently-specific hybridization reported by the MM probe cells.

The various techniques in the literature that have been employed to handle negative values of Avg Diff in an analysis are all controversial. Most often, genes with negative Avg Diffs are simply omitted from the analysis because the common log transformation used for robotically spotted array data is not defined for negative numbers. This loss of information is obviously not desirable. Other approaches such as thresholding negative numbers to a positive constant create a systematic artifactual bias in the distribution of Avg Diff results in the data set that affects all subsequent analyses that depend on these quantities.

Regarding the Abs Call per probe set, Affymetrix has empirically developed a decision table for each probe set of 16 to 20 probe pairs that is used to determine whether there was a "Absent," "Marginal," or "Present" call based upon the Positive-to-Negative Ratio, Positive Fraction, and Log Average Ratio. Note that this decision table will have different values for these three parameters for each chip set. The need for such a decision table stems from the fact that the specificity of the

different probe pairs will vary across probes and across samples. In theory, the Abs Call provides a qualitative measure of reliability of the reported expression level (see figure 3.7). The naming convention of the Abs Call values is unfortunate as the report of an Absent Abs Call does not necessarily mean that the gene is not expressed in a sample; rather it signifies that the gene expression measurement on that microarray and for that experiment is not reliable, *e.g.*, its signal was below background effects.

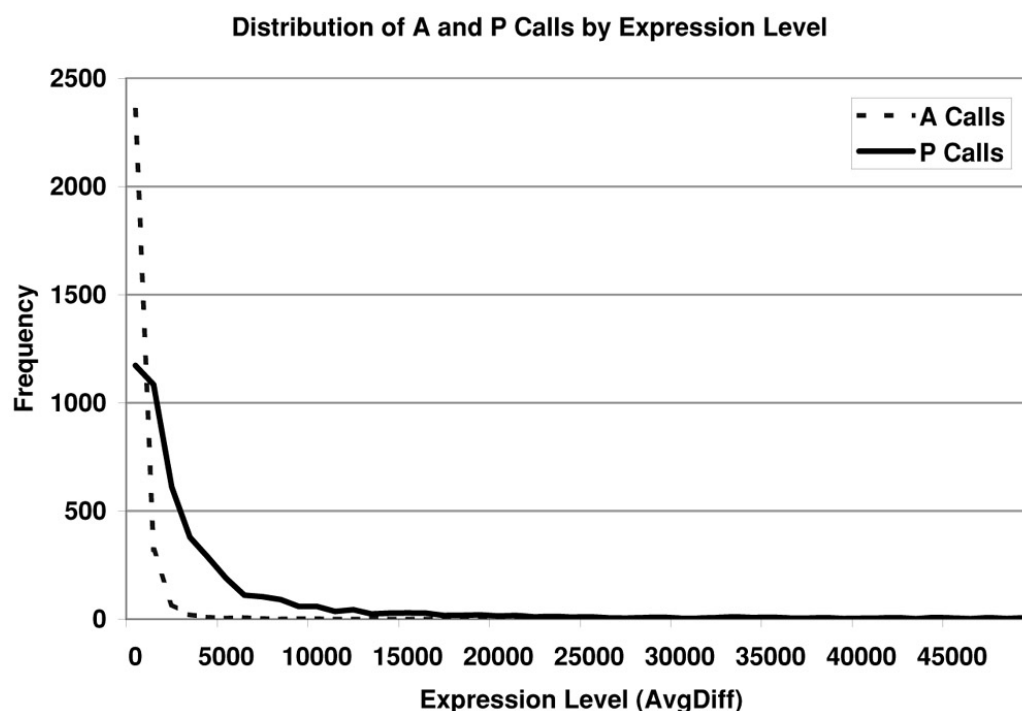


Figure 3.7: The Affymetrix *Absolute Call*. The relationship between Absolute Call and expression level for a microarray. Far more Absent (A) calls than Present (P) calls are at lower expression levels, but significant numbers of Absent calls are found even at expression levels in the low thousands.

A general understanding of these aggregate measures will make the reader aware of some major implications for analysis. First, the aggregate probe set values may not be reliable for all purposes. For example, in contrasting reported measurements from separate arrays, the aggregate measures Log Avg Ratio and Avg Diff may not be directly comparable *as is* because of the differences in specificity and sensitivity of each probe set, which is not captured by the aggregate measures. In contrast, Affymetrix software exploits probe-specific knowledge within the probe set in developing the decision table so that analyses and computations occur at the level of the `.cel` file. As described above, these files report the intensity of the individual probe cells, though at the time of writing, the oligonucleotide sequences for each probe cell are not known. Since most microarray analyses reported in the literature mainly employ Avg Diff values and not individual probe cell intensities, the comparability and reproducibility of these published results are less than optimal.

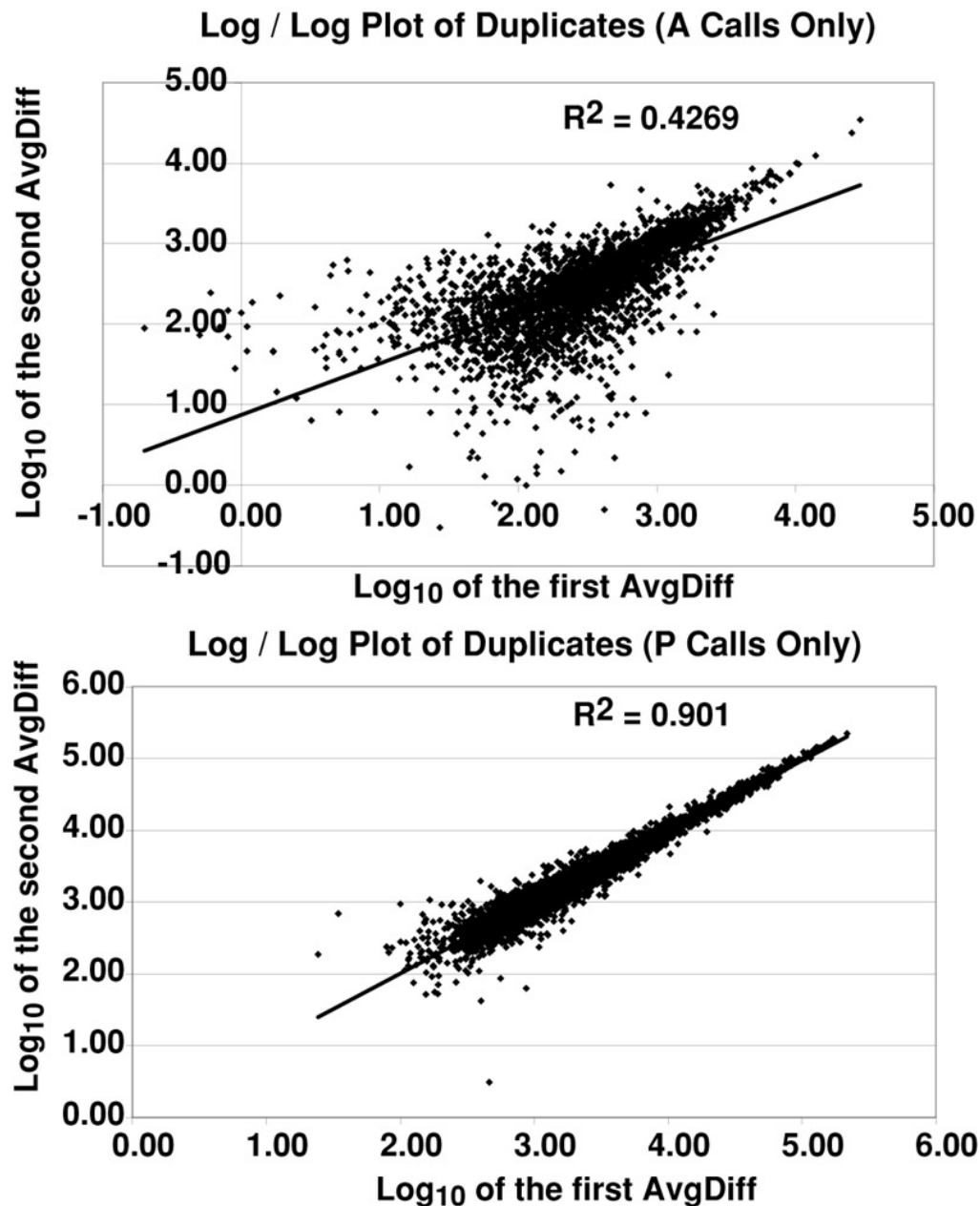


Figure 3.8: Graphical display of how Affymetrix Absolute Calls predict reproducibility. Correlation of expression levels from the same hybridization "cocktail" measured on two microarrays. Top: the correlation using only those genes for which an Absent call was reported by both microarrays. Bottom: only the Present called genes.

Another factor complicating the use of these aggregate measure values is that several settings, such as the noise threshold, are tunable by the operator of the Affymetrix scanner and software unit such that different reported aggregate measures may not be comparable across different microarray data sets. Despite the preceding concerns regarding aggregate measure values, the standard in the literature has typically been to not publish the `.cel` files with individual probe measurements but instead to release the aggregate statistic, *e.g.*, `.chp` files with all their attendant problems with comparability evolving from these.

At present and in sharp contrast to robotically spotted microarrays, the use of oligochips runs counter to the do-it-yourself ethos of many biologists for three primary reasons. First, except for large pharmaceutical companies, the construction of oligonucleotide microarrays occurs offsite and completely under the control and supervision of the manufacturers (essentially not peer-reviewed). Second, current oligochip manufacturers charge what the market will bear, and oligochips are

typically more expensive than robotically spotted arrays. Third, the specific oligonucleotide sequences that are used by the manufacturers of oligonucleotides to interrogate an RNA transcript are proprietary information. It is likely that as the market for microarrays broadens, the second and third problems will resolve themselves. Regarding the first reason, in this genomic era in which we see the rapid industrialization of biological investigations, an ever-increasing fraction of the biologist's tools will become commodities available from vendors at diverse price-performance trade-offs. From that perspective, industrial fabrication of whole transcriptome microarrays will likely become common in the near future (see section 7.1).

^[1]This schema is adapted from Leming Shi's website <http://www.gene-chips.com/>.

^[2]<http://www.cmgm.stanford.edu/pbrown/mguide/index.html>.

^[3]Early reports at the time of writing indicate the next generation of Affymetrix oligochips may use 11 probe pairs per probe set.

3.2 Replicate Experiments, Reproducibility, and Noise

A ubiquitous and underappreciated problem in microarray analysis is the incidence of microarrays reporting nonequivalent levels of an mRNA or the expression of a gene for a system under *replicate* experimental conditions. This phenomenon of microarray data *irreproducibility* is widely attributed to *noise* in the bioinformatics literature. For example, a common sample probe or target pool that is split and hybridized simultaneously to two separate chips of the same make and following the same scanning protocol will almost surely result in two, not exactly similar-looking, graphic representations of RNA abundance. These images will subsequently translate into two quantitatively different expression data files (e.g., `.cel` or `.chp` files on the Affymetrix platform) after they are fed into image-processing and statistical software. This situation is akin to saying that two Northern blot assays for a particular RNA species in a split total RNA sample, having been processed under an identical protocol and conditions, led to different X-ray film images of the RNA intensity, thereby resulting in two different reports of expression intensity by a phosphor imager, which is clearly undesirable.

The *irreproducibility* of a measurement reading—that is not entirely attributable to poor experimental design or practice—is neither new nor endemic to the microarray technology. One encounters this same problem in any measurement of a physical quantity, especially when these quantities exist on microscopic scales or when the measurement device is highly sensitive. Let us consider a thought experiment illustrating the latter point: Get 500 ml of well-mixed NaCl solution (*i.e.*, salt water). Split this solution into two samples *A* and *B*, and measure the NaCl concentration in both samples with a series of k measuring devices of increasing sensitivity which report concentrations up to an accuracy of 10^{-k} kg mL⁻¹ for $k = \{1, 2, \dots\}$ respectively. Let $C_A(k)$ denote the NaCl concentration in sample *A* as reported by device k , and likewise define $C_B(k)$. When the device sensitivity is low, say, when $k = 0, 1, 2$, the device would probably not detect any concentration differences between *A* and *B*, *i.e.*, $C_A(k) = C_B(k)$ or more concretely, $C_A(0) = C_B(0) = 5$, $C_A(2) = C_B(2) = 5.01$, say. As the device sensitivity increases, say, for $k > 9$, typically negligible physical factors such as small pockets of uneven NaCl concentration in *A* and *B* might be within the device detection limit and be recorded so that $C_A(k) \neq C_B(k)$ or more concretely, $C_A(10) = 5.0100000001$ $\neq$ $C_B(10) = 5.0100000002$, say.

The usual solution in these situations is to perform as many repeat or *replicate* measurements as are feasible and practical, with the hope that a statistic of these repeats converges asymptotically to the true measure of that quantity with the number of *replicate* measurements following the law of large numbers—provided certain conditions concerning the stochastic independence and distribution of these repeats are met. This approach may minimize, on average, certain systematic *noise*, such as measurement errors, but it cannot practically resolve more diverse manifestations of *noise* such as ones originating from biological variation in the model samples under investigation. Furthermore, there is the added issue of material cost with regard to replicate microarray

measurements. Researchers today rarely perform more than 3 replicate chip assays per experimental condition due to the relatively high cost per microarray experiment.

In studies involving microarrays, *replicate* experiments are especially important for, but not limited to, these following reasons:

- Microarray data are often employed during the earliest stages of studies that are not hypothesis driven and are primarily exploratory in nature. In such cases, the attendant analyses and conclusions will typically provide hypotheses (*e.g.*, candidate ESTs) for more focused laboratory investigation at the later stages of the study. For these endeavors, further pursuit of a false–positive conclusion can potentially be costly in terms of resources and time.
- It is generally good scientific experimental practice, whenever possible and feasible, to replicate a experiment to verify an earlier, unconfirmed quantitation.
- Data from *replicate* experiments provide a better quantitative understanding of the extent of *noise* (see section 3.2.5) inherent in both the system under investigation and the measurement device [116]. The effect of *noise* is typically proportional to the level of sensitivity of the measuring device. As we have previously noted, the amount of total RNA for a Northern blot assay of a single RNA is more than enough for a typical microarray assay of more than 10^4 different mRNA species levels in today's technology. Furthermore, the *noise* effect might not scale linearly with the detected expression level and may be sequence dependent.

The terms *replicate*, *reproducibility*, and *noise* are explained in the next section.

3.2.1 What is a replicate experiment? A reproducible experimental outcome?

No one can step into the same river twice. Heraclitus, c. 540–480 B.C.

Nature does not make leaps. G. W. Leibniz, 1646–1716

Consider the following five thought experiments that aim to compare the levels of blood glucose in 6–month–old mice, 1 hour before and 1 hour after an intravenous insulin treatment at noon.

- **E1.** A 6–month–old mouse M has its glucose levels measured and recorded by a glucometer G , 1 hour before and 1 hour after an intravenous insulin treatment at noon. Call these readings $R(M, G, \text{Monday}, 11)$, $R(M, G, \text{Monday}, 13)$, respectively.
- **E2.** A 6–month–old mouse $M2$, not the same mouse as M , is subject to the same experimental protocol as for M in **E1** on the following day and here we call the readings $R(M2, G, \text{Tuesday}, 11)$, $R(M2, G, \text{Tuesday}, 13)$, respectively.
- **E3.** The same 6–month–old mouse M in **E1** is subjected to the same experimental protocol the following day as it had undergone in **E1** with glucometer readings $R(M, G, \text{Tuesday}, 11)$ and $R(M, G, \text{Tuesday}, 13)$, respectively.
- **E4.** Another 6–month–old mouse $M2$, not the same mouse as M , is subject to the same experimental protocol as for M in **E1** simultaneously as M but using another glucometer $G2$ to obtain readings $R(M2, G2, \text{Monday}, 11)$, $R(M2, G2, \text{Monday}, 13)$, respectively.
- **E5.** During the time **E1** was carried out, M has its glucose levels simultaneously measured and recorded by another glucometer $G2$ to produce readings $R(M, G2, \text{Monday}, 11)$ and $R(M, G2, \text{Monday}, 13)$.

Questions: Which of experiments **E2** through **E5** qualify as a *replicate* of **E1**? Which is "the best" *replicate* of **E1**? Observe that **E2** replicates the experimental protocol, except for the biological

variation between mice M and $M2$ and the difference in days. **E3** replicates **E1** in terms of experimental protocol and the use of the same mouse, except for the difference in days and perhaps more important, the initial condition (physiological) of mouse M on Tuesday could be measurably different following its Monday treatment. **E1** and **E4** are replicates up to experimental protocol and simultaneity in running both experiments, modulo the biological variation between M and $M2$, and the use of separate glucometers which may be calibrated differently. Finally, **E1** and **E5** are simply repeat measurements of the glucose levels of M following the same protocol but with different glucometers. Typically in the study above, the absolute values $R(\text{mouse, glucometer, day, 11})$ and $R(\text{mouse, glucometer, day, 13})$ are themselves not important; rather, it is the relative change (difference or fold) from $R(\text{mouse, glucometer, day, 11})$ to $R(\text{mouse, glucometer, day, 13})$ that is of interest.

The point of the illustration above is to demonstrate that the question above is not well-posed until we specify what we mean by a *replicate*. Clearly, there are different levels or gradations for deciding how one experimental setup replicates another. *The way that one chooses to define a replicate experiment will be driven by, and context dependent upon the biological question that these experiments were designed to answer in the first place.* Only after one has made this definition can one sensibly discuss (*ir*)reproducibility of the attendant experimental outcomes. The range of parameters that could potentially enter into, be manifested, and detectable in a complex biological system is arguably wider and less easily characterized or controlled than in systems in the physical sciences. In our example above, the day of the experiment or the individual physical condition of the mouse may additionally affect its blood glucose level. Without entering into the epistemologies of Bacon [14] or Popper [143], the experimentalist normally attempts to resolve this difficulty by designing appropriate control conditions for the experiments and hope that – to paraphrase Leibniz's aphorism that measurable processes in nature do not change abruptly – the biological processes are not dramatically different from one normal mouse to another. Specific to our thought experiments, if the experimentalist has decided to replicate **E1** in the sense of **E4**, then she or he would reasonably expect the mice physiologies (and by extension, their blood glucose level in reaction to insulin) in **E1** and **E4** to not be radically different so that relative changes such as

$$\Delta(M', G') \doteq R(M', G', \text{Monday, 13}) - R(M', G', \text{Monday, 11}),$$

$$\Delta(M, G) \doteq R(M, G, \text{Monday, 13}) - R(M, G, \text{Monday, 11}),$$

are "close" to one another. A measure of similarity such as a *metric* is used to determine and quantify the "closeness" of $(M2, G2)$ to (M, G) , and thereby the *reproducibility* of the experimental results. Detailed discussions of measures of similarity are found in section 3.6. While it is highly unlikely in a real laboratory situation to find $(M2, G2) = (M, G)$, the experimentalist might expect that with a weaker measure of similarity—such as $(M2, G2) \approx (M, G)$ when both $(M2, G2)$ and (M, G) have the same sign—the experimental results are *reproducible*. Additionally, a statistical approach may be incorporated into the idea of reproducibility. For instance, if we carried out 1000 experiments **E1** with a different mouse each time and followed the same protocol, we may find that 990/1000 or 99% of the mice that were sampled exhibited a decrease in glucometer readings after treatment so that we can reasonably state that *most* — assuming a fair and random enough sampling — 6-month-old mice experience a drop in glucometer reading after an insulin intake at noon. Postulating (and *checking*) further properties about the data readout distribution, standard (non-)parametric statistical tools such as the Student t and Kruskal–Wallis tests may be used to determine the statistical significance of this conclusion, specifically the average drop in blood glucose reading.

3.2.2 Reproducibility across repeated microarray experiments: Absolute expression level and fold difference

Akin to situation E5 above, suppose that a common total RNA sample is split and the expression levels of the different RNA transcripts within each split sample are measured on a separate microarray of the same lot. For such situations, the current bioinformatics literature typically reports the Pearson's correlation coefficient of expression intensities of one chip reading versus the other as being close to $r = 1.0$, *i.e.*, the separate readings are highly correlated, and used this to conclude that the experimental assay is highly reproducible. We recall that in order to obtain any sort of meaningful statistic out of the Pearson's correlation coefficient, one typically has to assume that the paired expression data point for each gene is independent of those of the other genes. But how are we to account for individual genes whose expressions as reported by each chip are not exactly equal?

To date, few studies focus on the reproducibility of microarray measurements [116]. In a publicly available document [9], Incyte Pharmaceuticals demonstrated a high concordance between RNA expression measurements on cDNA chips using Cy3 and Cy5 dye signals. Based on this finding, Incyte estimates the limit of detection of fold differences at 1.8, meaning to them that 95% of fold differences between samples of 2.0-fold and higher are significant. From measuring the expression levels of 120 genes in various cancer cell lines using cDNA spotted filters, Bertucci *et al.* [24] showed that close to 98% of the measurements showed less than a 2.0-fold difference upon repeat. Richmond *et al.* [152], in their study of differentially expressed genes in *Escherichia coli*, filtered out genes below a minimum absolute expression threshold and ones with less than a 5.0-fold difference. Geiss *et al.* [75] used a Cy3/Cy5 system to measure genes differentially expressed during human immunodeficiency virus (HIV) infection in which they determined that fold differences of as little as 1.5 were statistically significant. This minimal fold threshold was determined by excluding 95% of the expression measurements that were reported, and not by using information-theoretic methodologies. Publications citing fold differences between control and test groups which are as low as 1.7-fold continue to be published, *e.g.*, [117].

The foregoing comparative studies do not address the reproducibility of fold differences if *entire* microarray experiments were repeated. Butte *et al.* [40] investigated this topic using Affymetrix Hu35K oligonucleotide microarrays containing 35,714 unique probes to measure RNA expression levels of muscle biopsies from 4 patients: P_1 , P_2 , P_3 , P_4 . Duplicated measurements were made from each patient sample: P_{21} , P_{22} , P_{23} , P_{24} .

A linear regression normalization procedure—outlined in section 3.4.1—was applied on all the chip data with respect to the data from patient P_1 . *Intra*-patient logarithmic fold differences (LFDs) were then calculated between the duplicated measurements for each of the four patients. Furthermore, *inter*-patient LFDs were calculated between all 6 possible pairs of patients (P_1 vs. P_2 , P_1 vs. P_3 , *etc.*) and their duplicate measurements (P_{21} vs. P_{22} , P_{21} vs. P_{23} , *etc.*). The LFD was used throughout this analysis so that the levels of up and down fold regulation were numerically equal in magnitude but oppositely signed.

Ideally, if the oligochips assays were perfectly reproducible across the duplicates, one would expect that Pearson's correlation coefficients at 1.0 between the four repeated expression measurements, and *intra*-patient LFDs calculated on each of the duplicated measurements to be the same, *i.e.*, if gene A is noted to be 10 times higher in P_2 versus P_1 , then it should ideally and similarly be 10 times higher in P_{22} versus P_{21} . However, the Pearson's correlation coefficients between duplicate expression measurements of the four patients across all 35,714 probes were .76, .84, .78 and .82 (see figure 3.9). Furthermore, the cross-probe Pearson's correlation coefficients for the six paired (replicate) *inter*-patient LFDs were near .0 as shown in figure 3.10 where the first graph shows the plot of $\log(P_2/P_1)$ versus $\log(P_{22}/P_{12})$. Further analysis showed that this poor correlation coefficient in the replicated LFD was primarily due to small absolute expression values: When folds are calculated using a pair of numbers where the denominator quantity is small (*i.e.*, the gene was

found by the microarray to be not highly expressed), a high fold difference is typically the result. This is particularly problematic as the effect of noise in microarray assays today appears to be more pronounced at lower absolute expression levels.

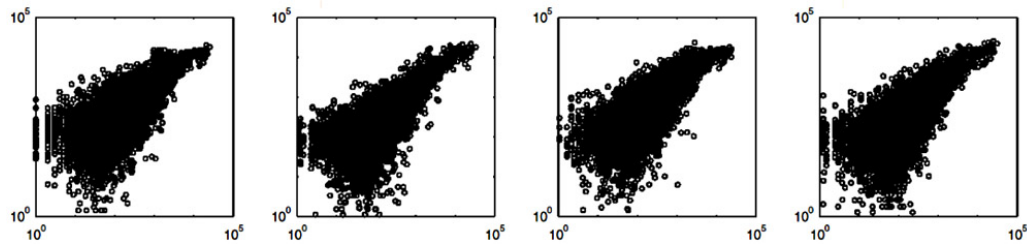


Figure 3.9: Expression measurements made in duplicate from the same RNA samples do not correlate well all the time. RNA samples from four human samples were placed on duplicate oligochips and the expression of 35,714 ESTs was measured. Each point represents an EST. The duplicate expression measurements are plotted here on a log-log scale (base 10). $r = .69, .73, .73, .69$. (From Butte *et al.* [40].)

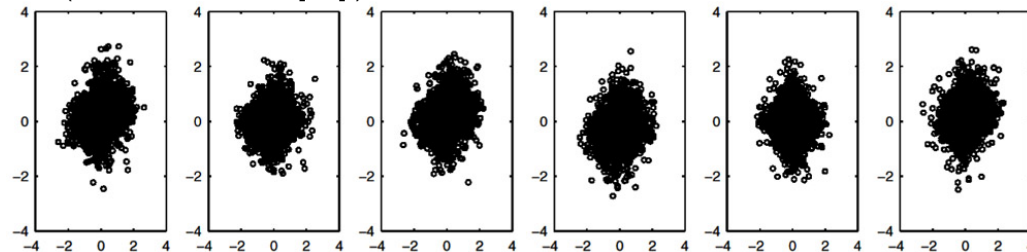


Figure 3.10: When expression measurements do not correlate well, fold differences correlate even poorer. Fold differences of 35,714 ESTs were calculated between the six possible pairings of the four patients. Fold differences are expressed in base 10 logarithm, so that ESTs that did not change between models are plotted in the center of each graph. Fold differences from the duplicated measures are shown on the x - and y -axes. Even though the correlation coefficients were high between original and repeated expression values, the correlation coefficients were very low between original and repeated calculated fold differences. (From Butte *et al.*, [40].)

It is worth noting that even when the reported absolute expression measurements are reasonably reproducible between replicate measurements, the fold differences calculated from these reported measurements may not be as reproducible across replicate experiments. As mentioned above, the fold or ratio of these expression quantities is very sensitive to small numerical perturbations, especially in its denominator quantity. This is best explained with an illustration: Say that we carry out a pair of experiments P_1, P_2 , and that the expression levels of 2 genes G_j and G_k reported in each experiment are (1.0, 2.0) for G_k and (500.0, 1000.0) for G_j , so that the fold change for both genes from P_1 to P_2 are both 2.0 ($= 2.0/1.0 = 1000.0/500.0$). Suppose we further carry out a pair of replicate experiments P_{21}, P_{22} , and that the reported expression levels are now (1.5, 2.0) for G_j and (500.5, 1000.0) for G_k . Note that the denominator quantity in the replicate P_{21} experiment is perturbed by +0.5 from the reported measurement in P_1 —an effect one typically sees in the presence of noise. Let us suppose that the overall absolute expression measurements P_1 to P_{21} and P_2 to P_{22} are reasonably reproducible, *i.e.*, the intensity–intensity Pearson's correlation coefficients between the duplicate experiments are greater than .99. Note, however, that the resulting fold changes calculated using these new values are now 1.333 ($= 2.0/1.5$) for G_j and 1.998 ($= 1000/500.5$) for G_k . Clearly, the robustness of the fold is dependent upon the absolute expression measurement, as we see here that the fold of gene G_j is more stable to noise than gene G_k . For a detailed discussion of the appropriateness of using the fold as a measure of expression change apropos microarray data, we refer the reader to section 3.5.

The key points of this subsection are as follows:

- When entire microarray experiments are not replicated, one would not be aware of, or be able to quantify, the extant irreproducibility in the experimental data and design, and false biological conclusions may potentially be generated.
- Even if the gene expression measurements from a single sample placed on two microarrays appears to have reasonable correlation coefficient, the fold differences calculated using those reported measurements may not reproduce as well between replicates at [40].

Based on this experience, we now recommend at least two to three replicates for each data point. Triplicates are certainly preferable for in worst-case scenarios, it is unclear which of the duplicate measurements is more correct. Preferably, the RNA sample for each replicate should be generated separately to maximize noise inclusiveness in the replicate model.

3.2.3 Cross-plat form (technology) reproducibility

The process of comparing gene expression information from a broad range of conditions has been widely adopted as an approach to functional discovery. As each individual laboratory may use different expression assaying plat forms to profile a common subset of genes, one obvious question arises as to how one may combine and compare expression measurements across diverse technologies. Assuming that such operations makes sense, the problem comes down to determining the reproducibility of cross-plat form measurements. Analyses by Kuo [113] suggest that for spotted cDNA probe microarrays and microarrays with Affymetrix-synthesized oligonucleotides circa 1999, the reproducibility may be poor. The mRNA expression measurements from oligonucleotide microarrays from Butte *et al.* [39] were paired with measurements from the cDNA microarrays from Ross *et al.* [154] by matching the corresponding cell lines and RNA transcript probes. When the oligonucleotide data (from the Avg Diff column) were compared to the normalized Cy5/Cy3 ratio (from the RAT2N column in the ScanAlyze output file) of the cDNA data, the overall linear correlation across all 162, 120 paired measurements was poor ($r = .0326$) as illustrated in figure 3.11. Corrections for GC content, cross-hybridization, and probe length all did not materially improve the correlation coefficient.

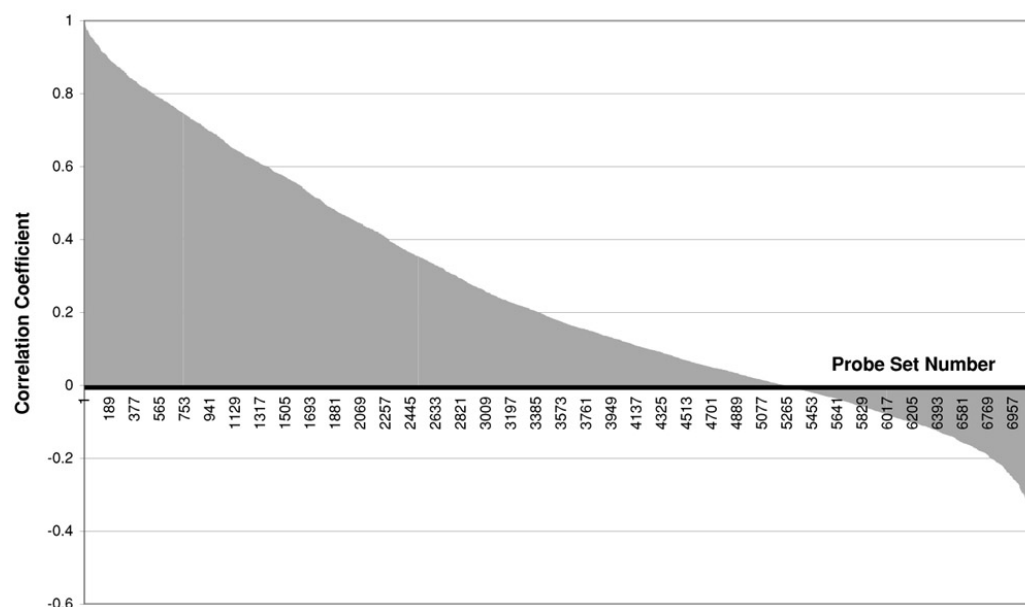


Figure 3.11: Overall correlation coefficients for each matching probe and probe set across two microarray technologies. Great variance is seen in reproducibility of measurements across probes and probe sets, with some probes showing correlation coefficients near 1.0, and some even showing a negative correlation (*i.e.*, a gene which is reported as highly expressed in one technology has an opposite expression report in the other technology).

It does not seem likely that the reported measurements from these data sets can be normalized or transformed into a single standardized unit of gene expression. Moreover, even the qualitative correlation was poor. For instance, high measurements in the oligochip did not correspond well with high measurements in the cDNA chip, and *vice versa*. At the very least, the results of this study suggest that greater (pre)caution should be taken in making cross-plat form comparisons. While complementing existing assay technologies, new techniques for expression profiling such as optically coded beads and ink-jet arrays (see section 7.1) will add to the challenge of comparing and combining expression data across different plat forms, and are likely to require similar efforts for normalization and standardization.

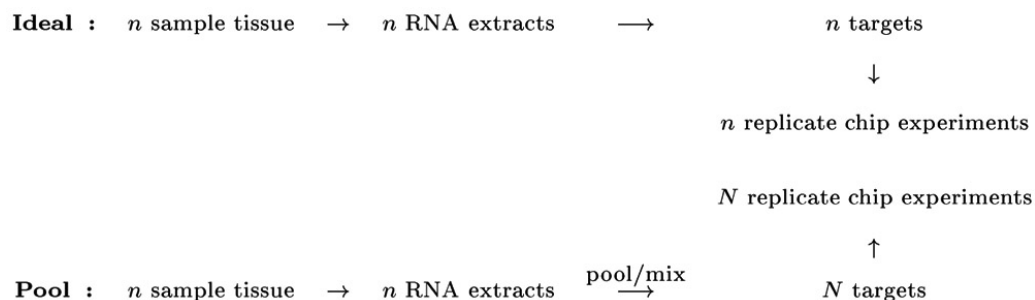


Figure 3.12: Pooling total RNA extracts for replicate experiments. Two strategies for making replicated measurements from samples, pooled and not pooled. N may not necessarily equal n .

3.2.4 Pooling sample probes and PCR for replicate experiments

A typical microarray hybridization experiment today requires on the order of 10^{-4} g of total RNA to generate the labeled sample probe compound for each array (spotted cDNA array, see [167]; oligochip, see [122]). Depending upon the experimental design, namely, how one defines a replicate experiment, it might not always be possible to obtain enough total RNA from a system, *e.g.*, specific tissue from a test animal, to prepare sample probes or targets for one or more (as repeat measurements) chip experiments. For example, suppose that a biologist wishes to obtain the gene expression profile of the mouse heart during embryonic day 12 (E12) and finds that the total RNA extracted from one embryonic mouse heart is not sufficient for generating enough labeled target for even a single microarray experiment. In this situation, an alternative solution is to pool together the total RNA extract from several, say n many, different E12 mouse hearts to create a sample probe or target compound which is then split into N , $N < n$, many targets for hybridizing on N separate chips, and hence resulting in N replicate experimental data.

When these tissue samples are drawn from n embryos that do not share a common parent, the pooling procedure will average out *intra*-species biological variations between the mice. This is independent of and in contrast to the subsequent splitting of one pooled sample into individual targets for N separate microarray assays and takes into consideration measurement error that inevitably arises in any empirical experiment.

Regarding the use of PCR to amplify these and any RNA extracts, the reader should note that beyond a small number of amplification cycles, a heterogeneous RNA mixture that has been amplified via PCR may not retain the same specific RNA percentagewise composition as in the original mixture [17]. Furthermore, one should also note that experimental data generated by re-using a target that has undergone a previous hybridization should not generally be considered a replicate assay because the biochemical composition of the target may have changed irreversibly after a hybridization procedure, due to both the protocol and time differences. Typically, a target needs to be incubated on a microarray for 12 hours or more during hybridization.

3.2.5 What is noise?

Noise is an informal term widely used in microarray data analyses and experiments to refer to:

- **N1**. Physical effects and parameters extraneous to the aspect of a system one is investigating, which have entered into the system or the measurement device that quantifies this aspect, and which, if detectable, would lead to,
- **N2**. Experimental outcomes which contradict *a priori* known facts, or more specifically and commonly, variations in experimental outcomes that should be "identical."

Strictly, **N1** and **N2** are origin (or cause) and effect respectively, *i.e.*, **N1**, if detectable by the measurement device or microarray, is manifested to the laboratory observer as **N2**. Most studies do not distinguish the former from the latter in their use of the term *noise*. Typically, the presence of noise in a system is deduced from comparing an actual measurement outcome of the system against another comparable reference quantity or a known and confirmed fact. For instance, if we use a ruler to measure the length of a pencil twice and obtain slightly different readings each time, then we know that there must be "noise" in our measurement activities because the indisputable *a priori* fact is that we are measuring one and only one pencil, with a single fixed length. Here, **N1** might include parallax error and **N2** is the different recorded lengths. Recalling our prior discussion on replicate experiments, the range of parameters that affect a biological system study is vast and not completely controllable so that, in general, **N1** may encompass any combination of conditions that are present at the time of a microarray assay, whether they are inherent in the measurement device or in the biological system.

For partly cultural reasons, there is a lack of appreciation for the challenge presented by noise in microarray expression data analysis. Historically, genomicists were fortunate to have had their first and successful foray into understanding patterns in the genome to have occurred at the level of the DNA sequence. DNA sequence components, being drawn from an alphabet of 4 characters (A, T, C, G), are unambiguous—and furthermore, when one assumes that these sequences code for function deterministically, one finds that the representation of information in a DNA sequence is essentially *digital*, or discrete. By contrast, the measure of mRNA species levels, and by extension the expression levels of associated genes, is intrinsically *analog*. Because of this, there is more room for measurement error, and a greater and generally unfavorable dependency on the measurement technology. RNA level is a continuous quantity and its measurement is no different from the measurement of any other biochemical processes or products, such as the rate of lactic acid oxidation in muscle or endorphin levels in the bloodstream. Specifically, in the context of microarrays, each chip measures microquantities of upwards of 10^5 different chemical species at a time on a hybridization medium with an area of less than 10^{-4} m². Bioinformaticians who make the transition from sequential to expression genomics are often surprised by the massive increase in the ambiguity and noise levels in the genomic data.

It is worth noting that noise is not entirely a negative phenomenon. In some situations, it could inform the genomicist to reconsider *a priori* assumptions, postulates, or questions for a system which may not be entirely correct or well posed. For instance, referring to the earlier thought experiments for replicating the measurement of blood glucose levels in mice in section 3.2.1, while differences (noise) in the repeat measurements **E5** point to measurement errors which are unquestionably undesirable, differences in **E1** and **E4** (that are not measurement errors) might imply the existence of significant physiological variations between *M* and *M2* that may suggest that not all mouse glucose levels respond similarly to intravenous insulin and this knowledge inevitably refines or changes the study question and objective.

3.2.6 Sources and examples of noise in the generic microarray experiment

Before delving into details of how one can distinguish chip-reported expression data that arise from a genuine measurement of mRNA levels in the presence of noise from reported data that come from noise alone, it is informative to review the ways in which noise could enter a typical microarray experiment. Noise can roughly be classified, by point of origin, into two categories:

- **Intra-chip noise** encompasses the previously discussed *background effect* on the chip surface, where localized physical and chemical changes in one probe feature diffuse into neighboring probe features. Improper techniques in scanning are also included, such as having one area of the chip brighter than another. Chip manufacturing defects such as quantitative non uniformity in laying probe sequences on the array surface would also be covered (see figure 3.18).
- **Inter-chip noise** includes biological variation in the samples (e.g., note that even pure cell populations are unsynchronized in their cell cycle phase). Also included are protocol variations in the hybridization procedure, the environment, time effects, and again, possibly, inconsistencies in chip manufacture.

In contrast to *inter*-chip noise, *intra*-chip noise is often more subtle and not as easily discernible, due to the fact that there is usually no obvious reference situation for a comparison of a single microarray experiment outcome.

Next, we explore examples of noise in microarray data. Figure 3.13 below, courtesy of Todd Golub at the MIT/Whitehead Institute's Genome Center, displays the gene expression intensity–intensity plot from an experiment that has been repeated with the same sample hybridized onto two separate arrays of the same type. First, it should be pointed out that had the two arrays produced exactly the same data, a plot of the gene-by-gene expression level of chip 1 versus chip 2 would have shown the data points aligning themselves exactly along the diagonal line $y = x$. Instead, here we see a typical, and, in fact a rather qualitatively good, result for intermicroarray variation. This variation is a very typical manifestation of *inter*-chip noise. Note especially that the typical data point spread away from this $y = x$ line increases with decreasing expression intensity. As mentioned earlier, this is due to fact that at lower gene expression levels, the corresponding RNA amounts that are being measured are smaller and therefore the effect of noise in these measurements is relatively more significant. Thus, there is greater variation here between these two nominally identical probe assays in a repeated assay.

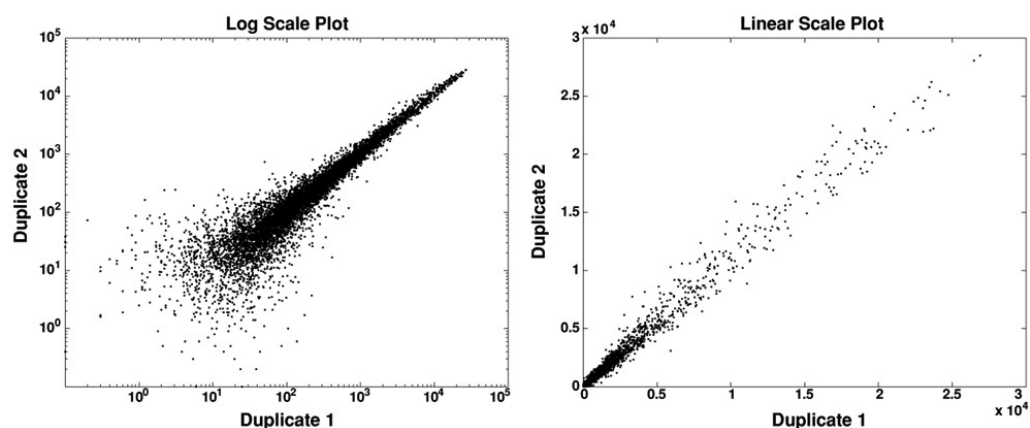


Figure 3.13: Apparent differences in source of variance due to log scaling. On the left is the apparently increased variation at lower expression levels using a plot of the logarithm of all the gene expression levels measured in one microarray versus the logarithm of all the of the genes measured on another microarray. When the expression values are plotted on a linear scale as on the right,

then it appears that the higher expression levels have higher variance. The former scale emphasizes the decreased reproducibility at lower expression levels and the latter emphasizes that there are fewer measurements or genes at higher expression levels. Note that for well-definedness, the logarithmic scale plot excludes measurements that had a negative value.

Golub *et al.* [78] also plotted two lines, $y = 2x$ and $y = \frac{1}{2}x$, which mark the locations of a 2.0-fold increase and a 2.0-fold decrease in gene expression, respectively, relative to the $y = x$ line of an idealized 1.0-fold increase (or decrease), *i.e.*, zero variation. Observe that at lower expression intensities, there is a sizeable number of genes whose expression varies outside of this 2.0-fold envelope despite the fact that these intensities came from essentially identical samples. A similar-looking pattern is seen in the following two figures. The second figure shows data from repeat hybridization of a sample that was hybridized to one chip, washed off from the first chip, and then rehybridized onto a second chip. In the third figure, RNA was extracted twice from a common sample and hybridized onto two different chips from the same lot simultaneously.

In all three scenarios, there is a similar and more prominent spread of data points about the $y = x$ line at lower intensity levels about the idealized line of zero variation $y = x$. At this point, we bring to attention a common *trompe l'oeil* that derives from the logarithmic scaling commonly employed to illustrate the concordance (or lack thereof) between two microarray hybridization data sets. As shown in figure 3.13, by scaling the data sets to a logarithmic scale, the apparent variance at the higher expression levels is reduced. However, as we will explore further in our analysis of noise in microarrays, it remains that a larger proportion of the poorly reproduced results with microarrays are obtained at the lower expression levels. The reason for the apparent wider scatter at higher expression levels in the non-logarithm-plotted graph in figure 3.13 is that there are relatively fewer genes with high measured expression levels.

In figure 3.8, we illustrate another example of the significance of noise in oligonucleotide microarrays, the Affymetrix GeneChip. Here, we show measured expression data or Avg Diffs plotted from an experiment where the same RNA sample was hybridized onto separate oligochips for human skeletal muscle. All gene expressions were plotted on one of the three graphs based upon their Affymetrix Abs Calls (Absent, Marginal, Present). As we noted in section 3.1.2, Affymetrix Abs Calls are determined from a decision matrix which is a function of the specificity of probe set expression for each gene. Note that the decision rules for this matrix are not currently available to the public. The lack of robustness or reproducible stability in both the Abs Calls and the Avg Diffs is evident from these plots. That is, for the same RNA sample both the expression level and the Abs Call will vary. Apropos the Abs Call instability, note that the total number of genes across the three experiments is well under the total number probes for each GeneChip. While such observed variances are not unexpected, it should warn the investigator against depending too heavily upon a particular microarray-reported value, statistic, or call in his or her research program.

The noise example above is not singularly unexpected or unusual within the gamut of microarray noise examples, nor is it particular to the Affymetrix technology. Similar manifestations of noise, though from possibly different sources, are just as consequential in other microarray technologies. For instance, consider data from robotically spotted Incyte GEM microarrays in figure 3.14, which are three rows of expression data from an Incyte file. Of relevance to our discussion are the location index of a probe on the physical microarray and its corresponding difference in expression. Incyte GEM microarrays are used in a technique similar to the two-dye fluorescence technique for cDNA chips pioneered at Stanford University, so that each microarray is hybridized to two differently labeled sample probes.

GEMID	Location	DiffExpr	BalancedDiffExpr	P1Signal	P1S/B	P1Area%
022PAQWV	2196	1.2	1.2	4330	60.3	90
022PAQWV	783	-1.1	-1.2	4325	43.8	94

P2BalancedSignal	P2Signal	P2S/B	P2Area%	Probe1	P1Description	Probe2
3724	3616	70.5	90	123Z3996	cocsxplusminus.si	12363997
5001	4856	64.1	94	123Z3996	cocsxplusminus.si	12363997

P2Description	GenelD	PlateRow	PlateCol	PlateID	ClonelD
csx	-137302	C	11	0210A00E	524442
csx	-131764	E	5	021VA0MR	463651

CloneSource	AccessionNum	Locus	IncyteClonelD	PCRStatus	GeneName
IMAGEConsortium	AI325648.1	02:15.5	mm45a10	Passed	RibosomalproteinL7[IMAGE:524442]
IMAGEConsortium	AA027730.1		mi15h10	Passed	PublicdomainEST[IMAGE:463651]

Figure 3.14: Incyte data file snippet. Heads and the two rows of values from an Incyte data file. Incyte microarrays are constructed using a robotic spotting process. The spotting process involves two proximal spots for the two dyes—Cy3 and Cy5—one of which is the targeted clone and the other a control spot.

Since cDNA microarrays, unlike oligochips, allow for two sample probes corresponding to different experiments to be hybridized simultaneously on the chip, the *inter*-experiment noise here may also *intra*-chip rather than exclusively *inter*-chip as we had seen with oligonucleotide microarrays. The two different colors or labels correspond to signals P1S and P2S respectively. These signals are corrected for background intensity and its variation, B, by the image analysis software used with the Incyte microarrays. Background corrected values are reported as P1S/B and P2S/B which correspond to the background-corrected intensities for the two probes. The P1S and the P2S area percentages correspond to the areas of potential hybridization that are recorded by the image analysis software to have undergone actual hybridization. In theory, if the appropriate control probe is chosen, say P2S, then this should correct for changes in hybridization conditions across different areas of the microarray that is caused by noise, such as localized temperature differences on the microarray surface.

The first thing to note from figure 3.15 is that we see an almost similar distribution of intensities for an identical RNA sample (the "hybridization cocktail") that is hybridized onto two separate Incyte arrays, P and P'. One way to view the noise profile of a robotically spotted microarray is as a plot of the ratio of signal P2S versus P2S/P2S' as in figure 3.16. In the ideal situation, the latter ratio ought to be uniformly equal to 1.0. That is, data points should only be distributed along the horizontal line $y = 1$ for any signal intensity P2S. In fact, at lower signal strengths of P2S, there are a number of genes whose P2S/P2S' ratios are significantly above 2.0-fold, even though on average the number of these outliers decreases with increasing P2S, and most genes tend to have a fold of 1.0. This demonstrates once again that the expression variance at lower expression values is more pronounced.

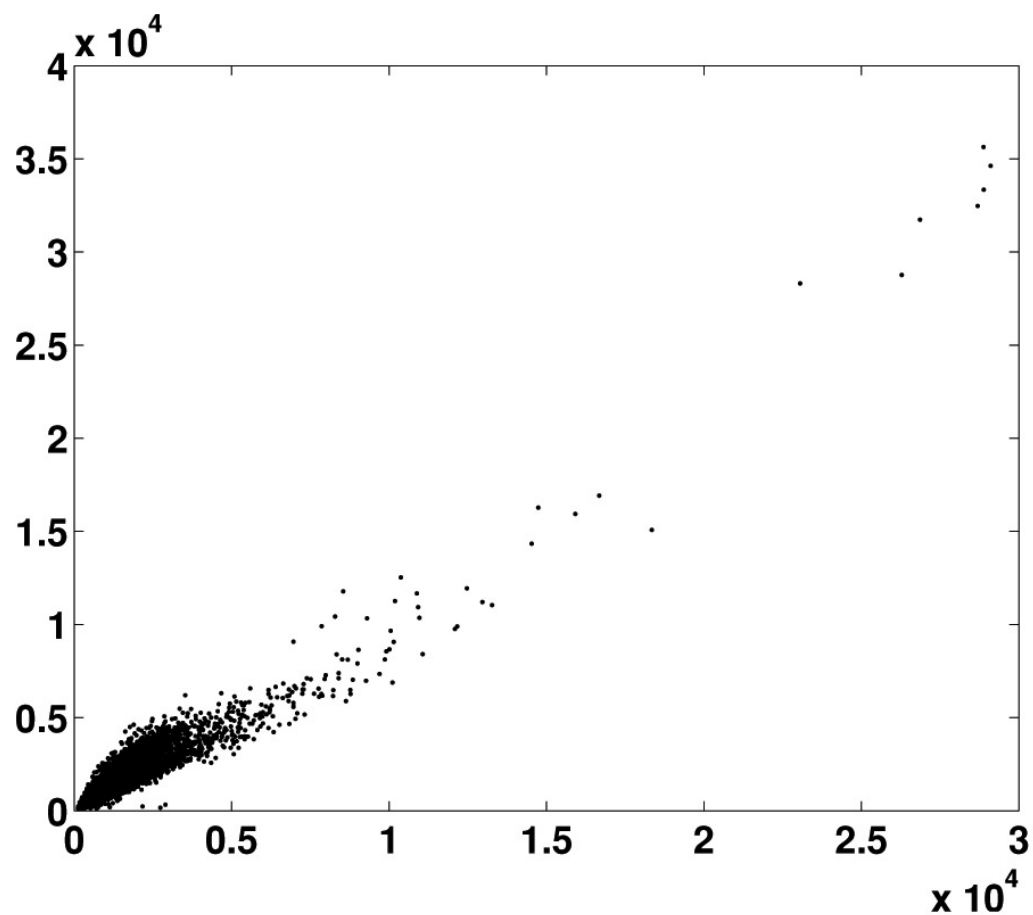


Figure 3.15: Duplicate cDNA assays of a common extract. The same RNA extract or hybridization "cocktail" measured on two Incyte spotted arrays. The P2S signals (the second of two dyes) from each microarray is plotted for each gene.

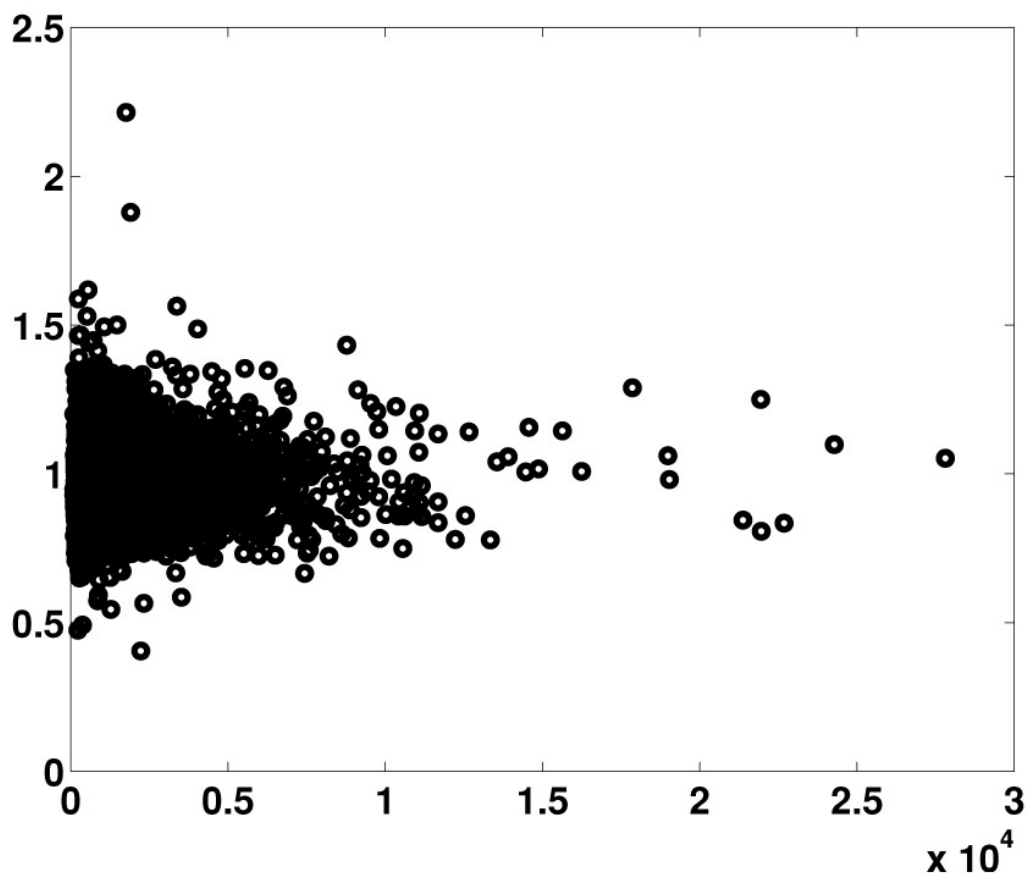


Figure 3.16: Ratio of probe intensities as a function of expression level. The ratio of the P2S/P2S' signals (y-axis) from two robotically spotted microarrays plotted against the P2S signal. The same RNA extract was used in both microarrays. Although the ratio is close to 1.0 for many genes, it does deviate sporadically and widely in a few instances. Also, as the expression level decreases, the distribution of the ratio spreads increasingly away from 1.0.

Microarray-reported data may also vary by the spatial location of the probe on the physical array. In order to see this, we map every probe and its associated reported expression amount from the two dimensional physical chip onto a one-dimensional array or line, and plot the expression level of each cell on the y-axis versus its location index on the x-axis. Figure 3.17 displays such a plot of experiments involving Incyte microarrays. A visually prominent feature in these graphs are four spikes marking dramatic increases in the expression level of the four probe cell locations on every chip. Furthermore, there appears to be a periodic pattern in reported expression level with respect to probe location on these microarrays. A careful review of chip manufacturer's specifications reveals that there have indeed been four control probe sectors (spiked controls of reference RNA transcripts) on each chip which would account for these spikes. The first preanalysis task then is to remove these spiked controls' data from their respective data sets and to renormalize these data sets for background effects, thus producing the next plots. In figure 3.17 the fold differences appear to be much less variable than previously and less obviously influenced by the probe location on the microarray.

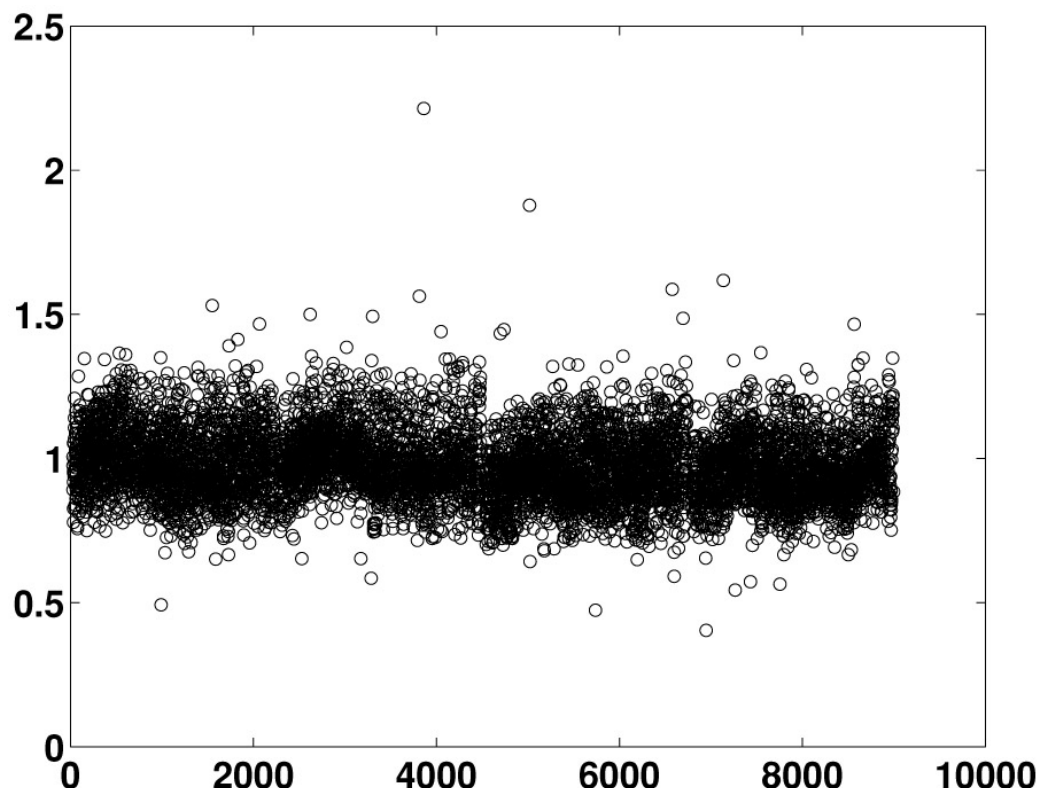


Figure 3.17: Ratio of expression level as a function of position. The ratio of the two *different* dye intensities per gene for a single microarray plotted against the position in the array (defined as a single number that is computed from the x and y coordinates of the probe on the array). The "control islands" that contain "spiked" controls on the array were not included in this plot. Also, all expression values were corrected for background noise. No particular pattern is obvious on inspection.

An eyeball inspection across thousands of data points as we have done above with the Incyte expression data set is often not sufficient for detecting subtler and less visually prominent patterns in the data representation. By applying standard signal-processing techniques such as the discrete Fourier transform, we find several major frequency peaks in the expression level across the chip (see figure 3.18). The highest peak here occurs at a frequency of 4 per array surface. This kind of periodic pattern in the Fourier transform expression data appears to be location dependent, and should lead the analyst to suspect that something systematically amiss occurred during the generation of this data set which could have happened during the manufacturing, hybridization, or the scanning stages.

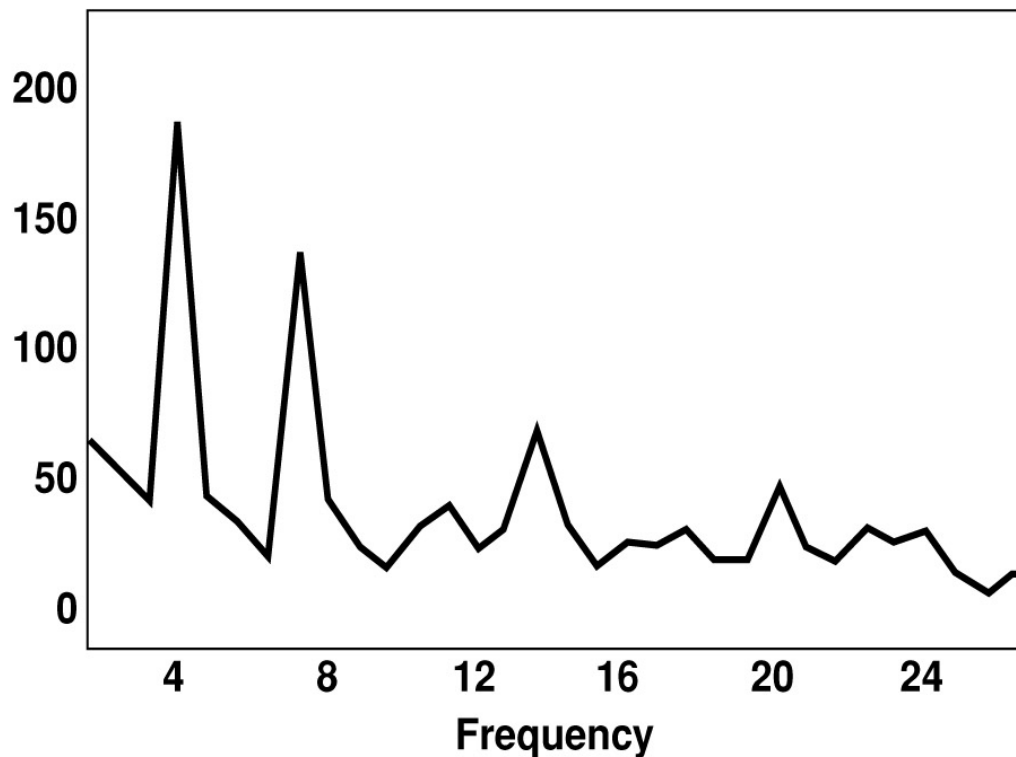


Figure 3.18: Fourier analysis of spatial series on spotted microarray. The same data illustrated in figure 3.17 is subjected to a Fourier analysis to extract systematic periodic signals. As shown, the frequency with the largest power is 4. That is, there appears to be a periodicity of 4 per microarray based on the position of the probe on the microarray. The magnitude of this systematic variation dominates fold changes in gene expression of the magnitude reported elsewhere as noteworthy. Consider the following image of the Incyte GEM microarray at the time this analysis was done (figure 3.19). The pattern seems to suggest that the periodicity in the reported data originated from four separate but similarly behaved sources. It turns out in this example that during the array manufacturing process, four separate pins had indeed been used to spot each microarray quadrant. It should not be surprising, then, that small but systematic physical and chemical variations in the amount of probe cDNAs that were picked up and deposited by each of the four pins are reflected and recorded in the overall expression pattern one sees here. Other possible sources of the lower-level systematic variations seen in figure 3.18 may include the frequency with which the pin assembly units are replaced during the microarray spotting procedure.



Figure 3.19: Incyte GEM Microarray *circa* 1999. In this photograph of a GEM microarray, note that there are four quadrants which are each spotted with separate pins. Again, although these noise examples were taken from Incyte gem microarrays and are of the *intra*-chip variety, it demonstrates that data from any microarray assay should be subject to at least some scrutiny or a quality control step as we have done here to screen out inherent and systematic biases in the measurement system. Numerous manifestations and sources of data acquisition error can be immediately caught by a simple visual inspection of the scanned images of a microarray. The images of the hybridizations illustrated for three different kinds of microarrays in figure 3.20 should be highly motivating in this regard.

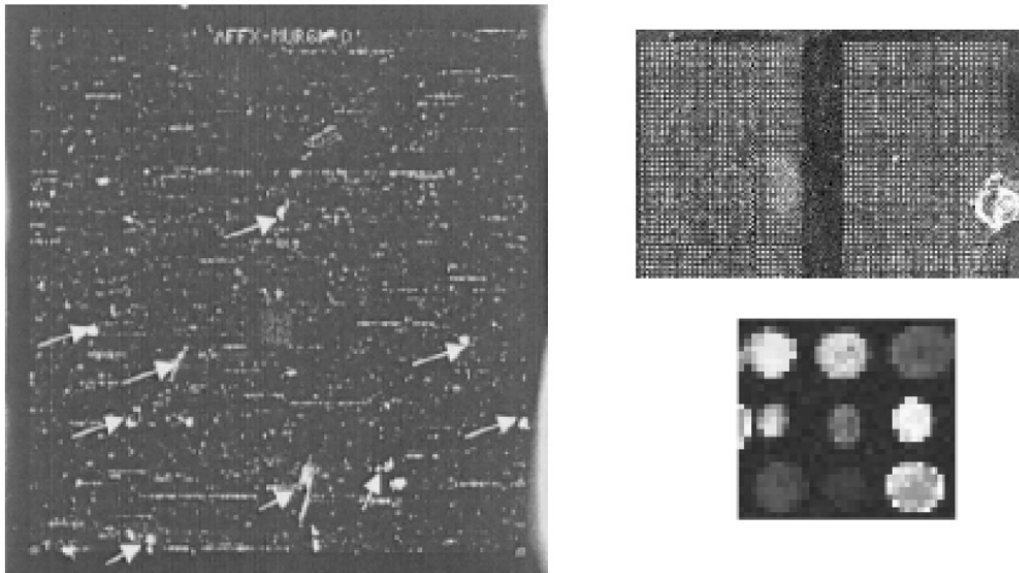


Figure 3.20: Problems in images acquired from microarrays. On the left: a contaminated D array from the Murine 6500 Affymetrix GeneChip set. Several particles are highlighted by arrows and are thought to be torn pieces of the chip cartridge septum, potentially resulting from repeatedly pipetting the target into the array [156]. On the right top: local changes in intensity due to contaminants and scratches. (Derived from <http://www.mediacy.com/arraypro.htm>.) Right bottom: high magnification of a scanned image of a spotted microarray. Note the different sizes and shapes of the spots [199].

3.2.7 Biological variation as noise: The Human Genome Project and irreproducibility of expression measurements

As of 2001, preliminary results of the Human Genome Project have provided further explanations and reasons for microarray measurements having poor reproducibility in addition to the ones iterated in section 3.2.5. We now know that there exists a relatively high frequency of *alternative splicing* products from each gene—see figure 3.21—and are more aware of the heterogeneity of common gene polymorphisms across individuals and populations.

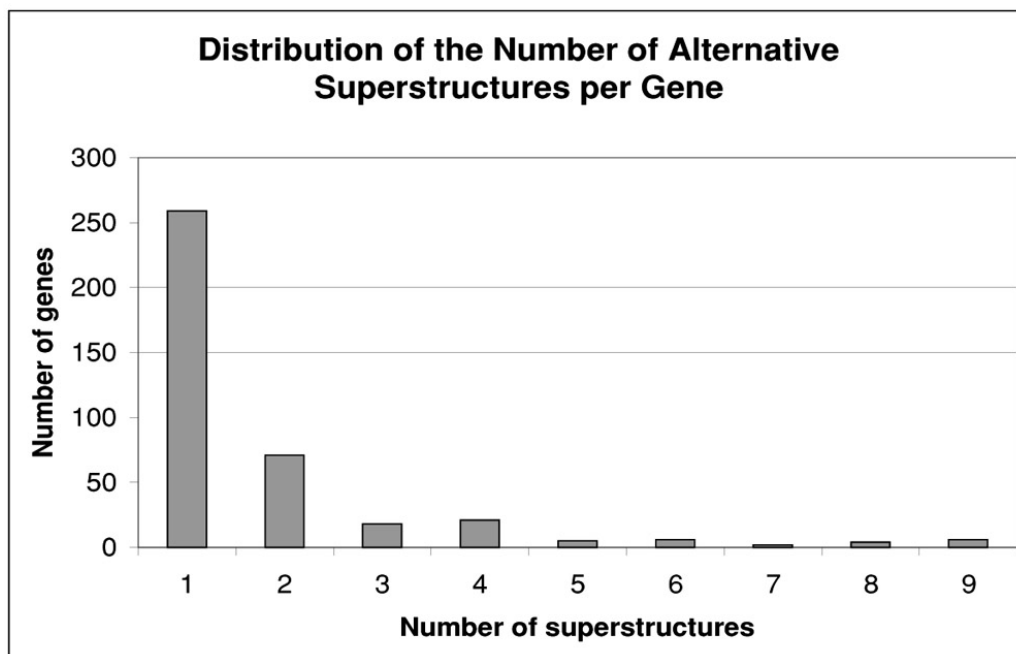


Figure 3.21: Estimate of the incidence of alternative splicing in genes. While most genes have only one spliced product, many have more than one, and this can affect the the gene product detection

efficacy and rate by microarrays. (Derived from Mironov *et al.* [132].)

Genes, which are composed of exons and introns, have the remarkably versatile property that their exons can be selectively added or deleted during transcription, giving rise to different proteins from the same gene. This form of *alternative splicing* appears to play a crucial role in the cellular physiology of higher organisms. For example, in the development of *Drosophila melanogaster*, a single gene determines the eventual sex (male or female) of the organism, depending upon how the gene is spliced [91]. In contrast to the most simple eukaryotes such as yeast, it appears that higher organisms such as humans have a much larger number of alternative splicing products per gene. The average number of distinct transcripts per gene averages 3.0 for humans as compared to 1.1 for yeast. Current oligonucleotide microarrays only target a set of subsequences for a particular individual transcript, and full-length cDNA microarrays only target a particular individual transcript. Consequently, both these technologies will show variation in signal intensity even if the same gene is being expressed at the same level but an alternatively spliced transcript is being transcribed rather than the splicing for which the microarray was engineered. Future microarrays may need to be constructed in a manner that is exon- and intron-specific, instead of gene-specific.

Apropos the heterogeneity of gene polymorphisms, there is, on average, one single nucleotide polymorphism (SNP) in every 1000 human nucleotide bases. Polymorphic variations of the same gene can be quite common, as noted by Chakravarti [43], and as a result, a cDNA or oligonucleotide probe that is engineered for a particular variant might hybridize considerably less effectively with another variant. This could be true even if the polymorphism does not change the final protein product of a gene. Although the genetic code is somewhat redundant in the sense that 64 codons code for 20 amino acids,^[4] two codons that code for the same amino acid may bind significantly differently to their complementary DNA molecule and these chemical differences might be detectable by the microarray.

3.2.8 Managing noise

Now that we have established that the current microarray technologies are bedeviled with *noise*, we will begin to address the data that are generated using such systems. The most direct route would be to have a direct measurement of the variation per gene of all microarray measurements. This is done by replicating the microarray assay(s). In practice, only duplicate or triplicate hybridizations are performed because of cost. The goal here is to obtain a subset of genes from these experiments whose reported expression level is robust across replicates, despite the noise. As we have alluded to in section 2.9 and as illustrated in figure 2.9, any of these techniques will inevitably have to balance between their significant false-negative or false-positive rate. That is, in order to avoid having to follow thousands of red herrings, as would be the case with a high false-positive rate, one has to be willing to discard many genes that might be genuinely involved in the biological process of interest on the grounds that their reported expression is inconsistent or irreproducible across replicates in the current assaying technology. We describe in this section, as an example, one technique that was developed to identify sets of genes that have been up- or downregulated, that had a low false-positive rate for the specific data set in consideration. When we consider the broader question of what constitutes a significant change in expression level for a gene, then the question of appropriate (dis)similarity measures is raised. This question is addressed in sections 3.3 and 3.6.

Modeling the expression-level-dependent noise envelope If, as has been mentioned in preceding chapters on reproducibility and replicates, the variance in reported gene expression level is a function of the expression level itself, then clearly single-value thresholds (*e.g.*, a 2.0- or 3.0-fold increase) applied to an entire expression data set, as is commonplace in the literature, are not sufficient for deciding which genes are significantly changed between different samples or conditions. At lower expression levels, these uniform thresholds will be associated with an

increased false–positive rate, and at high expression levels with an increased false–negative rate. One way to address this problem would be to develop fold thresholds that are themselves a function of the reported expression level, rather than a fixed number. This is simply done by obtaining at least a duplicate hybridization from the same hybridization cocktail hybridized onto two microarrays. As much as is possible, these are identical experiments in which the operating conditions, cell lines, culture media, incubation time, and so forth are controlled to be the same.

As the expression values should be identical, all variance is theoretically attributable to nonbiological sources. Then, for each set of duplicates, an identity mask (ID mask) is calculated in Tsien *et al.* [181] wherein are fold changes which are insignificant or attributable to noise alone. Two parameters are used for creating each ID mask: expression value range (or a sliding window of expression intensities), and either the scale value or the number of standard deviations. These are used to calculate the ID mask borders and can be adjusted for different trade–offs in sensitivity and specificity, depending on one's utility model as reflected in figure 2.9.

We illustrate the application of ID masks to six experiments (A, B, C, D, E, F) that were run in duplicate. Total RNA was isolated from the cell lines (MCF–7 human breast cancer cells and MG–63 human osteosarcoma) and hybridized onto Atlas Human cDNA Expression Arrays from Clontech (Clontech Laboratories, Inc., Palo Alto, CA). Each of these Atlas Arrays (Human 1.2 I, Human Cancer) is a nylon membrane on which approximately 1200 human cDNAs have been immobilized. Although this example uses a relatively low–density array technology, the methodology applies similarly to higher–density microarrays.

Two methods are then explored for creating ID masks. Method 1 relies on segmental calculation of standard deviations. A "data point" refers to an (x, y) pairing, in which x is an expression value of a gene g from the first hybridization, and y is the corresponding fold difference value (*i.e.*, the ratio of the expression of the same gene g in the second hybridization to x). Using all data points in a given sliding window of expression values (*e.g.*, from intensities 1001 to 2000), the standard deviation of the fold values is calculated. The average of expression values within that intensity window is then paired with the average fold value within the same window plus the number of standard deviations specified by the experimenter. This new pair becomes a candidate "upper mask border" point. Similarly, a candidate "lower mask border" point is created by pairing the average expression value of that window with the average fold value minus the number of standard deviations as specified by the user. Each successive group of data points in each sliding window of expression values (*e.g.*, all points from intensities 2001 to 3000, then all points from intensities 3001 to 4000, *etc.*) likewise give rise to candidate mask border points. A line is then fitted via least squares linear regression on the set of (expression value, fold value) pairs comprising the candidate upper mask border points. This line defines the upper mask border. Similarly, one computes the lower mask border from the set of calculated candidate lower mask border points. If one of the derived mask borders fits poorly—based upon its relationship to the original data points—the "reciprocal reflection" of the other (good–fitting) mask border can serve in its place. This simply means that each (x, y) point on the good–fitting (linear) border gives rise to a point $(x, 1/y)$ to create the reciprocal reflection border. Figures 3.22 and 3.23 show ID masks delimited by one linear regression border; the other border was derived by taking the reciprocal values of that linear regression border. The region between these borders represents the "identity" region of insignificant fold differences (*i.e.*, fold changes resulting from noise alone).

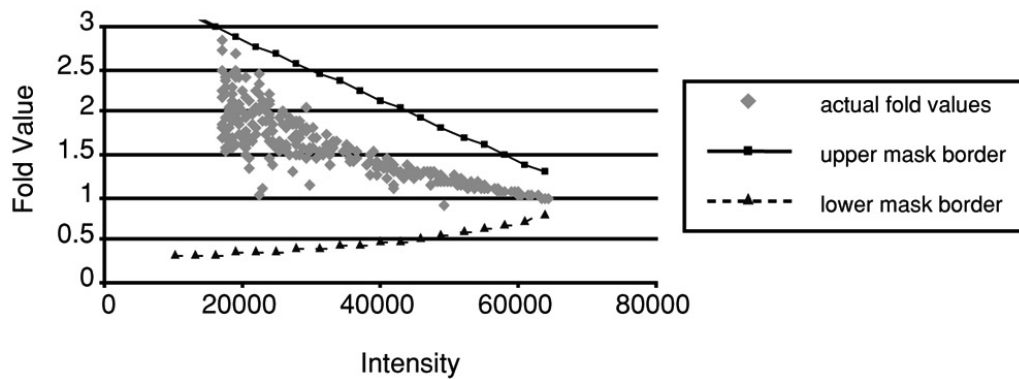


Figure 3.22: Identity mask for experiment A. Method 2 with parameters 9000 for expression value sliding window size and scaling factor 0.975 resulted in the lowest percentage of original data points lying outside of the mask region (0.7%).

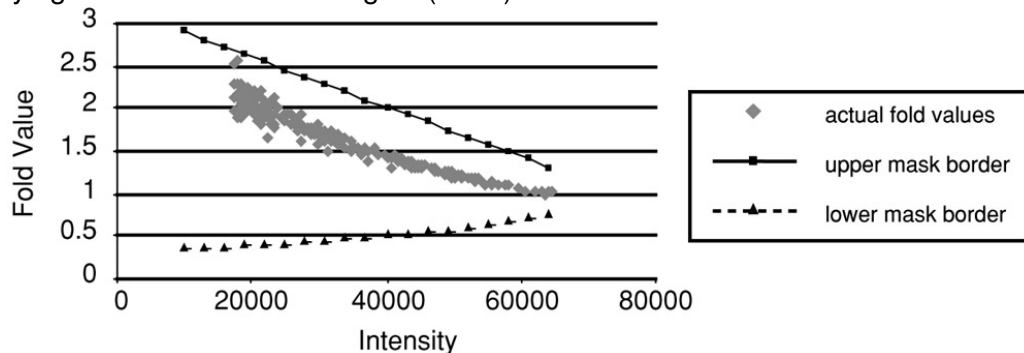


Figure 3.23: Identity mask for experiment E. Method 1 with parameters 5000 for intensity window size and 3 SD resulted in the lowest percentage of original data points lying outside of the mask region (0.9%).

The second method, method 2, for creating an ID mask uses candidate mask border points derived from maximal points in each sliding expression intensity window rather than from standard deviation calculations as in method 1. Specifically, among all data points in a given intensity window, the point with the greatest fold value is chosen. This is repeated for each successive window of expression values. These fold values can also be scaled before use in a linear regression to find the upper mask border. The lower mask border is analogously derived from the minimal fold values. Once the ID mask has been derived, all original data points are checked for inclusion or exclusion in the ID mask region. The percentage of data points lying outside of the ID mask region is recorded. We then automatically searched a large number of masks for those which provided the best performance. For both methods 1 and 2 of ID mask creation, sliding windows of ranges 1000, 5000, and 9000 on the expression value axis were chosen for experimentation. Only when calculations were not possible with one of these window sizes (*e.g.*, due to division by zero) was an alternative window size chosen. For method 1, the number of standard deviations (for calculation of candidate mask border points) was chosen to be 2.5 and 3.0. For method 2, the scaling factor was chosen to be 0.975 and 1.0. Twelve candidate ID masks were created for each pair of experiments (2 methods, times 3 intensity window sizes times 2 scale or SD factors). For each pair of experiments the mask with the lowest percentage of original data points lying outside of the mask region was selected. The results are shown in table 3.2.8.

The particular values that were obtained for these experiments are only shown for illustrative purposes. For any microarray technology adopted, even a simple analysis of this sort on duplicate experiments will result in much greater accuracy. That is, these masks will provide much greater sensitivity and specificity control than a single arbitrary fold-ratio threshold. For even greater accuracy, this kind of noise modeling must be done on a per-gene basis as described in section 3.3.

^[4]Twenty-one amino acids if one includes selenocysteine.

3.3 Prototypical Objectives and Questions in Microarray Analyses

We will now discuss common analytic approaches and issues surrounding the basic problems of finding similarities and differences in reported gene expression levels between the following (refer to table 3.3):

- **S1.** Genes in a microarray experiment, *e.g.*, *intra*-array comparisons of the expression levels in experiment E_m of genes G_k versus G_r in table 3.3, or between genes G_k and G_r in experiments E_m through E_y .
- **S2.** Experiments for a gene, *e.g.*, *inter*-array comparisons of the expression level of G_k in experiments E_m versus E_y , or between experiments E_m and E_y across all genes.

Table 3.3: Performance of the 12 identity masks. Each pair of identical experiments gave rise to 12 candidate ID masks. Six of these 12 were derived by method 1 (three with 3 SD and three with 2.5 SD). The other six were derived by method 2 (three with scale 1.00 and three with scale 0.975). Shown here are the percentages of original data points lying outside of the mask region for each of the 12 candidate ID masks derived for experiments A, B, C. $\tilde{A}$ = standard deviation; intensity range or window size of 2000 instead of 1000 is used in experiments B and C for the method 1 trials. (Derived from Tsien *et al.* [181].)

	A	Exp A	A	B	Exp B	B	C	Exp C	C
Range Size	1000	5000	9000	2000; 1000	5000	9000	2000; 1000	5000	9000
$\tilde{A} = 3$	3.1	2.7	2.2	93.2	80.5	1.7	99.6	100.0	2.0
$\tilde{A} = 2.5$	11.1	3.8	3.3	97.3	97.6	2.7	100.0	100.0	2.4
$\tilde{A} = 1.0$	19.2	6.2	0.7	100.0	99.0	2.4	100.0	100.0	2.4
$\tilde{A} = 0.975$	19.2	6.2	0.7	100.0	99.3	2.4	100.0	100.0	2.9

Note that for expositional clarity, we are assuming a one-to-one correspondence between an experiment and a microarray assay.

As has been noted previously, the two most prominent characteristics of microarrays are that they enable the parallel assay of numerous RNA species levels, and the irreproducibility of, and *noise* in, their reported expression levels. In view of these characteristics, the principal objective of microarray data analysis is to extract or discover *knowledge* about a biological system from the wealth of gene expression *information* that is obtained in the wake of each set of chip experiments. Specifically, *information* here primarily consists of noise-ridden gene expression data which may be pangenomic and which may be supplemented by *a priori* known biological facts. *Knowledge*, on the other hand, is certain aspects of the system under investigation that can be elucidated by gene expression data and which includes correlations between genes or conditions, genes whose expression levels distinguish dissimilar biological states, or, at a finer level, regulatory mechanisms and directions of causality between genes, or between conditions and genes. At this point in time, the question of which biological aspect can be unraveled by expression data has neither been entirely explored nor clearly defined.

Table 3.4: A prototypal microarrays experiment data set.

	...	Experiment E_m	...	Experiment E_y	...
$\hat{i}$					

Gene G_k		x_m^k		x_y^k	
$\hat{i}$					
Gene G_r		x_m^r		x_y^r	
$\hat{i}$					

In the posthybridization analysis, one will inevitably perform the basic *intra*– and *inter*–array comparisons **S1** and **S2**. We encounter situation **S2**, for instance, in studies that seek to find gene markers for biological or physiological states A versus B , where experiments E_1 through E_{m_A} are m_A replicate expression measurements in state A , and E_{m_A+1} through $E_{m_A+m_B}$ are m_B replicate measurements in state B . A time course study where experiments E_1 through E_m are the expression levels sampled at m time points $t_1, t_2, \dots, t_m$, is a typical scenario where **S1** is used. Furthermore, both **S1** and **S2** could apply in a study where experiments E_1 through E_{2m} are m different temperatures sampled twice per temperature condition. As mentioned in section 3.2, with the irreproducible nature of microarray reported measurements in mind, any attempt to reach the above objective in a consistent and rigorous manner will first require well–defined and mathematically workable notions of reproducibility and similarity of gene expression levels as reported by microarrays.

These definitions should capture or abstract, but not contradict, the biological experience, an immediate corollary of which would be an empirical approach to determine the *significance* of a reported gene expression level change. Since the full details of the thermodynamical or physical states of a chip experimental system cannot yet be practically obtained, nor can the sum total of all possible noise sources be characterized, any definition of reproducibility would typically have to be predicated upon preanalysis assumptions about the behavior of noise in that system.

3.3.1 Two examples: *Inter*–array and *intra*–array.

For heuristic purposes, we will describe inter–microarray and intra–microarray analysis situations which capture the most essential related features of **S1** and **S2** from the introduction of this chapter. While these examples only involve Affymetrix oligonucleotide microarrays, the analytic approach and problems are more general in scope and the methodologies can generally be extended to many other gene expression assaying technologies following easy domain–specific modifications.

Study S1: Yeast cell cycle A mycologist wishes to find groups of yeast genes whose expression profiles are similar over a 24–hour period. For this, she or he obtains gene expression measurements using Affymetrix S98 yeast genome microarrays for a synchronized sample of yeast cells over a 24–hour period by sampling the total RNA from this population at 30–minute intervals. A total of 48 separate time points were sampled twice (for duplicate measurements) at each time point: $T_1, \dots, T_{48}$ (and replicates $T2_1, \dots, T2_{48}$).

Study S2: A drug in tervention study A pharmacologist wishes to characterize the effect(s) of drug X 3 hours after it has been introduced into normal adult wild–type mice by the expression levels of liver cell genes that can be probed using Affymetrix Mu11K microarrays. The gene expression profiles of normal adult mice liver cells that are not treated with drug X are used as the control state. Call the preintervention or control state A , and the postintervention state B . For replicate measurements, liver samples were obtained without drug X application from M_A adult mice and another M_B adult mice liver samples were obtained after drug X was applied.

Detailed specifications of the Affymetrix technology are available in the Affymetrix GeneChip analysis manual [10]. A technical outline of this technology, namely on the interpretation of `.cel` and `.chp` files, is found in section 3.1.2. Affymetrix Mu11K murine chips, e.g., have total probe sets for $N = 13,179$ unique transcripts, including "housekeeping genes," whereas the yeast genome S98

array has N_H 6400.

The study properly begins at the experimental design stage: Mice liver explants in study **S2** are harvested and prepared under identical conditions—these most obviously being similar time of day, environmental conditions, and precautionary procedures taken to minimize the effects of non- X -related artifacts and noise in the gene expression, and *mutatis mutandis* for the yeast cells in study **S1**. Controlling for extraneous factors, which are potentially microarray-detectable in *inter*-array study **S2**, is clearly more difficult and some other factors that should be considered include the age, physical condition, and sex of the mice candidates. Despite the most stringent replicate protocol, it is clear that there always exists some biological or biochemical variation between any two mice, any two cells taken from a common host, or even between any two synchronized yeast cells. In practice, one can only try one's best to exclude explants from mice whose observed or known physiological state is believed to be associated with a different response pattern to X and which might skew the experimental data significantly. With the obvious modifications, whole yeast cells are harvested and duplicates are collected similarly at each of the 48 time points. Following a well-designed analysis scheme, one would hope to be able to classify such negligible variations as noise. To this end, it is important that replicate experiments be run for any chip study. Replicate measurements allow one to quantify the extent to which variations were detectable by microarrays, and to decide whether this noise will confound the reported measurements and attendant analyses. Study **S2** has M_A and M_B replicate experiments for each state. Chapter 2 presents a comprehensive treatment of an experimental design in the presence of noise.

For the remainder of this chapter, we will primarily be concentrating on *inter*-array study **S2**, and developing the mathematical tools, *e.g.*, normalization techniques and measures of (dis)similarity which will be used in chapter 4 to address analyses which are essentially *intra*-array such as study **S1**. We note, however, that most of the questions and methodologies in *inter*-array analyses are equally relevant to the *intra*-array instance, with the obvious contextual modifications. In analyses involving Affymetrix oligochip data, the `.chp` file is the usual starting point for a typical microarray analysis, (see section 3.1.2). From this file, we will be considering the Probe Set Name, Avg Diff and Abs Call columns. Consolidating the columns of the M_A , M_B `.chp` files from study **S2** into a table, we have table 3.3.1:

Let A_j and B_j denote the Avg Diff value corresponding to a microarray experiment before and after drug X intervention, respectively, with duplicates $j = 1, 2$. Note that the replicate hybridizations A_1 and A_2 might have been scanned into image files using the different user-defined settings or under different ambient conditions beyond the control of the experimenter. Therefore, the most natural question that one asks in any *inter*-chip study is whether one can validly compare expression levels, represented by the Avg Diff, for any one gene across `.chp` files that correspond to separate microarray experiments. For instance, for Gene 9785 in table 3.3.1 we may ask:

- **Q1.** Do order relations (*e.g.*, $<$, e , $=$) of reported Avg Diff values for Gene 9785 in different experimental conditions imply comparisons between the true expression level of the measured gene under these conditions? In particular, while one can convincingly argue that the expression level of Gene 9785 is less intense than the level for Gene 9784 because the *intra*-chip Avg Diff of Gene 9785 (250.5) is smaller than that of Gene 9784 (1211.1) in experiment B_1 —assuming that all probes are equally effective—it is not immediately clear whether it is reasonable to claim that the true expression level of Gene 9785 in condition B_2 is greater than in condition B_1 even though the *inter*-chip numerical Avg Diff of Gene 9785 in B_2 (141.2) is greater than the Avg Diff in B_1 (110.7).
- **Q2.** How would one characterize or distinguish the expression (*e.g.*, fold) change from pre- to postintervention states for Gene 9785 that is *most likely* due to a drug X intervention from

an expression change that is caused by measurement variations or, more broadly, noise?

Table 3.5: Avg Diff (Abs Call) data for *inter*-array study **S2** where $M_A = M_B = 2$

Probe Set Name	A ₁	A ₂	B ₁	B ₂
Gene 1	64.3 (P)	248.2 (P)	128.6 (P)	124.1 (P)
$\hat{\mu}$				
Gene 9784	1211.1 (P)	1250.0 (P)	18.6 (A)	18.3 (A)
Gene 9785	250.5 (P)	193.3 (P)	110.7 (P)	141.2 (P)
Gene 9786	54.3 (A)	-3.1 (A)	1.1 (A)	2.0 (A)
$\hat{\mu}$				
Gene 13,179	0.9 (P)	07.7 (A)	459.7 (A)	198.8 (A)

Note that in the ideal system, where every chip assays every transcript equally well and the experiments have been designed so that only expression changes which are a direct consequence of drug X intervention are recorded by the chips with all other attendant thermodynamical or physical variations being negligible and not microarray-detectable, these questions possess trivial answers. Namely, we expect that being replicates, the Avg Diffs $B_1 = B_2$ and $A_1 = A_2$ for each gene, and assuming that A_j, B_j are positive-valued, B_j/A_j quantifies fold change for each gene after intervention. Such utopian scenarios do not appear to be achievable at any time in the near future. In the following sections, we describe several ways to address the two preceding questions under different assumptions and conditions that are commonly used in the bioinformatics literature.

3.4 Preprocessing: Filters and Normalization

We will be referring to the data from study **S2** as tabulated in table 3.3.1. Since the number of replicate experiments in a typical microarray study does not normally exceed three, it is not easy to derive meaningful statistics for any one gene "latitudinally" across two or three repeat measurements, *e.g.*, the mean average reported intensity for Gene 9785 in state *B* is $(250.5 + 193.3)/2 = 221.9$. Descriptive statistics such as means or variances are generally informative of the underlying stochastic process in the system being investigated only when one has a large relevant data set to work on. Even though each microarray assay yields one noisy reading for each of the $\sim 10^4$ individual genes, the reported measurement for gene *k* will not inform us about the robustness of the measurement for gene *j* ($j \neq k$), in the absence of at least one other whole replicate assay.

On the other hand, even prior to any transformation(s) that will render the distinct replicate data sets comparable to each other, we can obtain informative statistics for each chip experiment data set "longitudinally" or *intra*-array across all genes 1 through *N* such as the following:

- **P1.** The mean, standard deviation, minimum, and maximum of Avg Diff in each microarray experiment. For example, for chip A_2 of study **S2** we could find that the mean is 232.4, standard deviation 888.1, minimum -2000.7, and maximum 19792.8.
- **P2.** The mean and standard deviation of Avg Diff differences for all duplicate pairs.

For example, the average and standard deviation of $(B_2 - B_1)$ for all 13,179 genes in study **S2** may be -35.4 and 200.3 , respectively. This tells us that, on average, a gene registers a lower expression level in B_1 in contrast to experiment B_2 . The standard deviation may be informally regarded as an inverse indicator of reproducibility of the replicate readings. This is not strictly an *intra*-array calculation.

- **P3.** The distribution range of Avg Diff values on each chip.
- **P4.** The distribution of genes which have J ($J \leq 4$) many Present (P) Abs Calls. For instance in study **S2**, 2547 genes have a P call in all experiments, 4112 genes have Absent (A) calls throughout all experiments, and 3792 genes have exactly three A calls in all four experiments.

Gross overall data statistics like the above can quickly be used to detect systematic *inter*-chip differences. For instance, if the genomicist in study **S2** finds that for **P2**, the average replicate difference $(B_1 - B_2)$ was 11714.4 , then he or she might have good reason to suspect that chip B_1 was scanned at a much higher ambient or background setting than was B_2 . Such statistics have been used as preliminary and very crude filters to reduce the number of potentially interesting candidate genes in some microarray studies. For instance, the genomicist in study **S2** could decide that she would only consider (following **P4**) the 2547 genes which have P Abs Calls throughout all 4 experiments for more refined data analysis. Alternatively, the genomicist could accept only genes whose Avg Diffs in some or all of the four experiments are above a predetermined cut-off value. These kinds of filters have their weaknesses. In table 3.3.1, we clearly see that Gene 9784 would be missing from the list of potentially interesting candidates for study **S2** following the all-P-Abs-Call filter, and yet from a cursory inspection of the raw data, and in the absence of further analysis, it seems that drug X triggers a significant reported expression decrease in Gene 9784 from its control levels.

In general, any gross preliminary filtering of the sort above leads to a loss of information. Loss of information is inevitable in typical microarray analyses and is not always a negative feature. After all, microarray data analyses are essentially reductionistic by nature due to the problems of irreproducibility as well as cost limitations in confirming candidate genes resulting from these analyses. The relevance of the loss is context-specific. It depends upon the sort of information that is lost and the scientific question being investigated. Gross statistics such as **P1** and **P3** have also been used to "scale" or *normalize* one chip data set against another so that transformed numerical indicators of expression intensity (Avg Diffs) are *comparable* in some probabilistic sense. This is the gist of question **Q1**.

3.4.1 Normalization

We shall use the term *normalization* to, informally, refer to the transformations that are applied onto data sets from chip experiments to render them *comparable* to one another in a probabilistic or statistical sense, or to take into quantitative account assumptions about how these data were generated. Normalization of a collection of different data sets is typically carried out with respect to a reference data set. The reference set could either be data from one particular experiment, or a postulated distribution of expression intensities. The following are three normalization methods in common practice:

- **N1.** Linear regression
- **N2.** Probabilistic distributions: first, second or k -th moments
- **N3.** Housekeeping genes, and spiked targets

The choice of transformations will depend upon assumptions that are made about the properties of individual data sets such as intensity distribution and the behavior of noise. This will subsequently

influence the characterization of expression or fold change significance at later stages in the analysis.

Linear regression Our discussion will first focus upon the assumptions correctable by linear transformations, **N1**. Referring to study **S2**, let x_i and y_i denote the reported expression intensity (Avg Diff) of Gene i in the duplicate experiments B_1 and B_2 respectively with $i = 1, \dots, N$, where $N = 13,179$ is the total number of unique probes on the Mu11K chip set. Let B_1 be the reference data set. For notational clarity, let $\vec{x}$ denote the vector of gene-ordered intensities in a chip experiment, $(x_1, \dots, x_N)$ in $\mathbb{R}^N$, and let $\bar{x} \doteq \frac{1}{n} \sum_{i=1}^N x_i$ be the mean of the gene expression intensities x_i in the chip set B_1 .

Now consider for the moment a hypothetical situation where chip B_2 has a systematic error in relation to its (reference) duplicate B_1 such that the Avg Diff for every Gene i , x_i in B_1 is remeasured or reproduced in the B_2 experiment as $y_i = \pm_1 x_i + \pm_0$ for some real constants $\pm_0$ and $\pm_1$. A global linear shift of this kind could arise, for instance, if chip B_2 were scanned at a different uniform ambient brightness from chip B_1 . Physically, $\pm_1$ is the magnification or *dilation* factor and $\pm_0$ is the *translation* factor. A scatter plot of x_i versus y_i would then look like a discretized line of slope $\pm_1$ with its vertical or y -intercept at $\pm_0$, provided that not all the x_i are equal to a constant. Clearly, if chips B_1 and B_2 were ideal duplicates, then the points (x_i, y_i) would be aligned on a line of slope one through the origin.

This graphical representation of our supposedly replicated raw data suggests an intuitive way to resolve the hypothetical problem. Knowing the values $\pm_1$ and $\pm_0$, the transformation $y_i \mapsto \hat{y}_i \doteq (y_i - \alpha_0)/\alpha_1$ will correct this systematic difference in B_2 with respect to B_1 so that a plot of x_i versus $\hat{y}_i$ is a line of slope one through the origin $(0, 0)$.

In the general situation, one usually has no *a priori* knowledge of the slope $\pm_1$ and y -intercept $\pm_0$ of the x_i versus y_i scatterplot. Instead, the main task in this case is to determine the values $\pm_0$ and $\pm_1$. Normally, one supposes that the data set B_2 is most likely to be systematically different from B_1 by a linear transformation just described, and assume that with B_1 and B_2 being duplicates that their scatterplot should ideally approximate, in the sense of *least squares*, a line of slope one through the $(0, 0)$. Using standard linear regression techniques, one computes $\pm_0$ and $\pm_1$ for the equation of the line which passes through the data (x_i, y_i) and which minimizes the sum of squared vertical or y -errors between this line and the observed data points (x_i, y_i) . In other words this line $f(x) = \pm_1 x - \pm_0$, say, should minimize $\sum_{i=1}^N (y_i - f(x_i))^2$. Solving for $\pm_1$ and $\pm_0$ using simple calculus, linear algebra gives us the dilation and translation factors,

$$\alpha_1 = D^{-1} \begin{vmatrix} \sum x_i y_i & \sum x_i \\ \sum y_i & n \end{vmatrix}, \quad \alpha_0 = D^{-1} \begin{vmatrix} \sum x_i^2 & \sum x_i y_i \\ \sum x_i & \sum y_i \end{vmatrix}, \quad (3.4.1)$$

$$D = \begin{vmatrix} \sum x_i^2 & \sum x_i \\ \sum y_i & n \end{vmatrix},$$

where $|M|$ indicates the determinant of the square matrix M . The normalization transformation then is $y_i \mapsto \hat{y}_i = (y_i - \alpha_0)/\alpha_1$. As an additional bonus, note that the averages of the normalized set B_2 and the reference set B_1 are equalized, i.e., $\frac{1}{N} \sum_{i=1}^N \hat{y}_i = \frac{1}{N} \sum_{i=1}^N x_i$. Verify this. Question: Are their respective standard deviations the same?

Variations of **N1** that have appeared in the literature include using a subset of genes for the regression calculation as, for instance, using only genes which have present Abs Calls in both the B_1 and B_2 sets, or excluding data points (x_i, y_i) when either x_i or y_i lie below a threshold.

Furthermore, if one supposes that the microarrays assay a large enough number of different genes and that the subset of these genes whose expression levels change significantly between conditions A to B is small compared to the total number of genes being measured, then one may also also normalize pairs of experiments A_j, B_j which are not true biological replicates. A useful exercise: Let us say that we have normalized data A_1, A_2 via **N1** with respect to B_1 to produce $\hat{A}_1, \hat{A}_2$ separately. Is the linear regression of data $\hat{A}_1$ versus $\hat{A}_2$ a line of slope one through (0; 0)? What happens when $\hat{A}_1$ and $(\hat{A}_1 - \hat{A}_2)$, regarded as ordered vectors in $\mathbb{R}^N$ are orthogonal to one another?

Probabilistic distributions The second category of normalization schemes **N2** operates under the premise that the distribution of expression levels or Avg Diffs in each chip experiment, or more stringently for duplicate sets of experiments, should be identical. Consider again the data sets $B_1 = \{x_i\}_{i=1}^N$ from table 3.3.1. Several studies in the bioinformatics literature have applied the standard central limit theorem–type transformation, $x_i \mapsto \hat{x}_i \doteq (x_i - \bar{x})/\sigma_x$, on each experimental data set. Here, $\tilde{A}_x$ denotes the standard deviation for the B_1 data set. so that the transformed data B_1 will now have mean 0 and variance 1. Exercise: Consider another data set $B_1 = \{y_i\}_{i=1}^N$ that has been normalized similarly $y_i \mapsto \hat{y}_i \doteq (y_i - \bar{y})/\sigma_y$. Show that the linear regression of $\hat{x}_i$ versus $\hat{y}_i$ is a line of slope $\sum_{i=1}^N \hat{x}_i \hat{y}_i / \sum_{i=1}^N \hat{y}_i^2$ ($= N^2 \text{Cov}(\hat{x}, \hat{y}) / N^2 \text{Var}(\hat{y})$) through the (0, 0). Here $\text{Cov}(\cdot, \cdot)$ is the symmetrical covariance function.

Note that this approach of setting the first two moments of each data set to constants does not guarantee that all the transformed sets of intensities have the same distribution, unless the original data sets were gaussian–distributed to begin with, in which case they would be completely characterized by their first (mean) and second (variance) moments. As an aside, it may be useful to check whether an arbitrary data set is indeed gaussian–distributed. We do this qualitatively with a quartile–quartile plot between the test data set and another data set of the same cardinality that is known beforehand to be gaussian–distributed. A gaussian–distributed data set can be generated using the pseudo–random number utility on most computing machines [144]. These two data sets are first individually ordered by magnitude and then plotted pairwise; for example, if these ranked data sets of cardinality n looked like $\{s_i\}_{i=1}^n, \{t_i\}_{i=1}^n$, then we plot (s_i, t_i) . If B_1 is gaussian–distributed, then its quartile–quartile plot against a gaussian–distributed reference set will be linear.

On equating the statistical moments of any two data sets, the reader should also note that even if two distributions share the same k th moments for all integral k , it does not imply that the two distributions are the same (or in the probabilistic parlance, almost surely equal). As a counterexample from Casella and Berger [42], the distributions with probability density functions $f_1(x) = \frac{1}{\sqrt{2\pi x}} \exp[-(\log x)^2/2]$ and $f_2(x) = f_1(x)[1 + \sin(2\pi \log x)]$ have k moments that agree; we clearly see from their graphs that these are different distributions. Furthermore, Press *et al.* [144] have observed that it is not uncommon to encounter real–life data sets that have finite means but arbitrarily large second moments. The reader ought to be aware of the existence of such pathological cases. These cases are not central to our discussion and the interested reader is referred to [144] for details. Less stringent variations of this normalization scheme are also in common use; *e.g.*, methods which equalize only the first moments (means) of intensity data sets as, for instance, $y_i \mapsto \frac{\bar{x}}{\bar{y}} y_i$, or $y_i \mapsto y_i - (\bar{y} - \bar{x})$, where the mean of the data set $\{y_i\}_{i=1}^N$ is transformed to coincide with the mean of a reference set $\{x_i\}_{i=1}^N$, by a dilation or a translation.

Housekeeping genes and spiked targets The **N3** category of normalization devices works on the assumption that certain probes or targets have a known constant hybridization behavior throughout all experimental conditions. For example, exact amounts of particular mRNAs are incorporated or spiked into the target that have a previously known and deterministic effect on specific probes on

the microarray under any condition. These special genes could function as housekeeping devices in any chip experiment. In this context, the normalization technique typically amounts to transformations on whole data sets, which will narrow or eliminate the statistical divergence in expression intensity values or Avg Diffs of these housekeeping genes across experiments.

General assumptions and principles of normalization The normalization methods that are outlined above postulate a linear and systematic nature in microarray measurement errors, and assume that every target–probe complex behaves in a similar manner, *i.e.*, hybridization rates are equal and independent of the transcript sequence. More general normalization techniques which are nonlinear or nonparametric have been described [156, 182]. Alternative normalization techniques for microarray data sets in the literature have included the use of eigenvectors [5], a scatterplot smoother [61], normalizing by both sample and gene [81], and mapping the expression data into a real interval between 0 and 1 [193].

Again, we emphasize that the choice of normalization procedure depends entirely upon the postulates, particularly in the noise–related assumptions that one makes for a set of microarray experiments. Recall that we had originally visited the topic of normalization with the aim of resolving **Q1**, *i.e.*, whether order relations (*i.e.*, d , $>$, $=$) between *inter*–chip Avg Diffs for a gene *as is* imply a comparison of the true *inter*–chip expression level of a gene across different chip experimental conditions. While normalization techniques like **N1** attempt to correct global and *deterministic* measurement errors which have the form of a linear transformation, it does not resolve the *stochastic* component of the error, in other words, error or variations due purely to *chance*. Chance–type errors are often subsumed under the term *noise* and are almost always modeled as a stochastic process. Graphically, stochastic effects are reflected by the scatter pattern of data points about some line of regression, as one sees in figure 3.15. In accounting for noise, the genomicist would often make assumptions regarding the stochastic model. For instance, she or he could postulate that the repeat measurements are random variables of intensity–dependent gaussian distributions whose means lie along the regression line and possess a uniform standard deviation. Methods like **N2** implicitly assume that each microarray data set is a set of samples drawn from a probabilistic distribution which is strongly characterized by its first k –moments. This and other modeling assumptions of the noise effect can naturally be used to determine the significance of a change in the expression level of a gene which we will discuss in an upcoming section on fold significance. The reader is advised that even after applying any one of the normalization techniques above, it might not make sense to extrapolate order relations of the true expression levels across experiments or conditions from order relations on the numerical microarray data.

In view of this, one might begin to ask whether there is a weaker notion of *inter*–chip comparison. That is, while we cannot definitely claim that the true expression level for Gene 9785 under condition B_2 is greater than in B_1 because the Avg Diff value for Gene 9785 in B_2 (141.2) is greater than in B_1 (110.7), can we derive meaningful biological comparisons of expression levels from the data pair (110.7, 141.2) at all, or at least within some statistical or probabilistic framework?

Consider the scatterplot of the replicate data sets B_1 and B_2 with points (x_i, y_i) where $x_i \in B_1$ and $y_i \in B_2$. Assume that the replicate measurement for any gene is a random variable which is characterized by gaussian distributions whose means are expression intensity dependent and have a uniform standard deviation along the line of regression. In this situation, it is reasonable to expect the data points to be mainly clustered within a 1–standard deviation envelope of the regression line. The *outliers* or data points which lie outside or away from this 1–SD envelope may be interpreted as genes whose expression intensity report or Avg Diff reproduces poorly across replicate experimental conditions. So, even though an order relation in the Avg Diff values does not imply a same–order relation between the corresponding true expression levels, the coordinates of the data (x_i, y_i) can be an indicator of reproducibility of a microarray assay depending upon the postulated

underlying noise distribution in the system. Next, we consider the scatterplot for the data set pair (B_1, A_1) , assume as previously that measurement errors are gaussian distributed about a regression line, and similar to the above we define the outliers as data points that lie outside the 1-SD envelope away from the regression line. Then a naïve interpretation of the outliers in the (B_1, A_1) plot which are not outliers in the (B_1, B_2) plot is that these are genes that have a expression change in going between states A and B that is less likely to be due to measurement error or noise. We can also naturally assign a magnitude to this expression change by considering the number of standard deviations the outlier in the (B_1, A_1) plot lies away from the regression line. This subject rightly belongs under the topic of gene expression change or fold significance which is explored in the next section.

3.5 Background on Fold

In this section, we describe several approaches to resolving the second question **Q2** posed in subsection 3.3.1. That is, how would one characterize an expression level or a fold change between different experimental states, for instance, preversus postintervention, in our ongoing example study **S2**, that is most likely or significantly due to changes in the biological state of a target rather than to measurement error or noise.

Most methods that have been presented in the bioinformatics literature assume that, with all probes being equally effective, a gene which has a statistically larger change in expression level is more likely to have that change due to a difference in biological state rather than to noise. It should be noted, however, that unless we have enough replicate measurements to be able to determine the robustness of a gene expression reading, we cannot entirely rule out the possibility that noise may cause large changes in reported expression levels. Conversely, not all small changes in expression level are produced by noise alone and, in fact, gene expression changes which are below a preset threshold of noise might very well trigger significant physiological effects in a system. Typically, biological states are macroscopic in scale, and transitions from one state to another are, arguably, continuous phenomena, whereas noise occurs at microscopic scales and is discontinuous.

For an informative contrast and guide as to how one might proceed in microarray fold analysis, let us briefly review how fold change for a particular mRNA transcript is determined using the traditional Northern blot. Typically in a blotting run, one starts by isolating total RNA on the order of 10 μ g per gel lane from the system of interest. Each gel lane corresponds to an experimental or replicate condition. Depending on the gel lane to be loaded, the sample load may be spiked with standard controls such as glyceraldehyde-3-phosphate dehydrogenase (GAPDH) for background normalization or loading control calculations. The gel loading step is usually performed manually in approximately uniform quantities. An electric current is passed through the gel complex and by electrophoresis, RNA fragments will move at different speeds in the complex according to their size. Afterward, separated RNA is transferred from the gel onto nitrocellulose membrane to produce a blot. Labeled probes are constructed with radioactivity-labeled nucleotides and are designed to be specific to each RNA. The control and test RNA samples are incubated with the probes. The blot is then washed to remove probes that have not specifically hybridized, exposed to an X-ray film, its image scanned, and the intensity **fold** change quantified by a molecular or phosphor imager.

In each gel lane, the intensity of the mRNA of interest is first normalized with respect to its loading control probe intensity. Rather than an absolute and precise quantification, it would be more appropriate to describe the task of the phosphor imager as fold estimation. In the scientific literature, it is common to quote integer-valued fold changes such as 2 or 100. It is rare to find reports of 1.01- or 10.307-fold in the literature. For the average biological scientist, these numerical fold values have a qualitative rather than a precise quantitative meaning. At each stage of the blotting

protocol, many occasions exist for error and noise to enter into the analysis and subsequently into the final scanned image and quantification, in ways that are not entirely understood or avoidable, that would render a 10^3 precision of folds biologically redundant or meaningless unless one possesses the error statistic for the blotting and imaging protocol. Another factor governing fold precision is the machine sensitivity of the imager. Recall the setup for a typical microarray analysis. As observed previously, a chip experiment may be regarded as the *reverse* of performing $\sim 10^4$ different Northern blots in parallel on a common two-dimensional substrate of area $\sim 10^3 \text{ m}^2$ of probes upon which one hybridizes the labeled target. Each microarray probe set would correspond to its associated mRNA sample in one gel lane without the loading control and, in the case of Affymetrix oligochips, with an additional set of mismatched probe controls for nonspecific hybridization. Each whole chip has its set of control or housekeeping probes corresponding to the loading control in a gel lane of the Northern blot.

In view of their similarities, one may reasonably expect any noise and reproducibility issue affecting a Northern blot to manifest itself as prominently in a microarray experiment. Due to the microscopic dimensions of each probe feature on a microarray, small irregularities during scanning or fabrication could lead to discontinuous and possibly contradictory outcomes that might not be immediately or practically detectable from among 10^4 other distinct probes. It is equally instructive to reexamine the starting points for the representative fold calculations in both the Northern blot and microarray technologies.

For microarrays, fold analysis starts off from the level of a text file (e.g., a `.chp` file in the case of the Affymetrix technology) which contains information such as a probe identifier and corresponding indicators of the level of sample probe or transcript that was detected. Theoretically, this text file is a numerical representation of the scanned chip image, specifically of its detected levels of different RNAs, and is generated by the chip manufacturer's software program which implements (often) proprietary statistical image-processing algorithms that are typically opaque to the user. There is always some loss of information in going from the true image to its machine image file, and to the numerical representation of the image file. The relevance of this loss is, of course, context-specific. Normally, in chip data analysis, the bioinformatician will perform statistical tests, classification, or clustering algorithms based on these preprocessed numerical representations of RNA levels alone. Thus the generic chip data analysis conclusions are at least twice removed from the source or microarray image of RNA levels.

In contrast, the end product of a Northern blot analysis is an X-ray film or an image of the blot, and an estimated quantification of fold change for the RNA of interest after the image is processed by a phosphor imager whose working principles might appear to be more transparent and seem closer to the physical phenomenon under investigation. With microarrays, it is possible to begin fold calculations from their image file. First, this approach may require the bioinformatician to develop his or her own image-processing program which might mean that he or she would have to acquire a working knowledge of several specialized disciplines, including image analysis and software engineering, which is a daunting task in itself. Second, the bioinformatician will need to obtain specific information from chip manufacturers about probe identities and their respective coordinates on the array for every different make of chip that he or she will use. Creating this dictionary location-probe feature lookup may be time-intensive, as in the case with some Affymetrix oligonucleotide microarrays where each probe set of 16 to 20 for a transcript is split up and dispersed throughout the physical array.

It should be noted that while the use of the fold or ratio of intensities is a natural way to quantify relative changes in expression level within the framework of Northern blots, it might not be an appropriate relative measure nor might it retain the same meaning when ported over *as is* into microarray data analysis. In calculating the fold change for a gene between conditions *A* and *B* as

assayed by microarrays A_1 and B_1 , one would very frequently and simply take the arithmetic ratio numerical representation of reported gene expression in the A_1 and B_1 chips. There are problems in doing this. For instance, Affymetrix represents the intensity of a transcript by the Avg Diff which may be a nonpositive number in which case the ratio of nonpositive quantities is physically meaningless. Taking the data from study **S2** for example, how does one quantify the fold of Gene 9784 from duplicate states B_1 (−4.6) to B_2 (18.3)? Furthermore, the ratio is not defined when the denominator is zero. For clarity, we shall mainly concentrate on examples of fold analyses on the .chp text files in the Affymetrix technology of study **S2**.

3.5.1 Fold calculation and significance

The discussions in this section refer to study **S2**. A prototypal question concerns whether the expression level, as represented by Avg Diffs, of Gene i is *significantly* different between states A and B . By *significance* one almost always means *statistical significance*—specifically, a difference that is unlikely to be due to measurement errors or noise, modeled as a random process that is characterized by a postulated probability distribution which is in turn derived either from the replicate data or some *null hypothesis*. The delicate point here lies in deciding on a consistent and well-defined criterion for statistical significance, and more fundamentally on the choice of a null hypothesis.

Similarly to the normalization procedure, diverse approaches exist to account for different assumptions and postulates about the initial condition of the biological system, the behavior of the target–probe complex, and the constitution and properties of noise. As detailed in [144] for a more general applicational framework, these approaches typically start off by computing a set of statistical parameters or statistics from the empirical data such as means, variances, and k th moments which define a test distribution. This is followed by a choice of a suitable null distribution and some calculations to decide where the parameters fall within this null distribution, *i.e.*, comparing between the test and null distributions. If the test statistic falls in a probabilistically unlikely region on the null distribution, then one may conclude that the null hypothesis is false for the data set, which in this context means that the data or test distribution is statistically different from the postulated noise or null distribution. As emphasized in [144], the reader should bear in mind that one can only disprove a null hypothesis, never prove it. That is, the fact that a data statistic falls within a probabilistically likely region of the null distribution does not imply that the fold is equivalent to the postulated noise effect. When the number of samples used for calculating a statistic is reasonably large, a difference of means that is less than the standard deviation may be significant. Whereas when samples are sparse, a difference of means that is much larger than the standard deviation may not be significant. This fact is relevant considering that the generic microarray study typically uses a small number of expression data per gene, corresponding to different experimental or replicate conditions, for calculating the statistic of the gene.

From study **S2** suppose that the data sets $\{A_j\}_{j=1}^{M_A}$ and $\{B_j\}_{j=1}^{M_B}$ have been suitably normalized following any of the methods outlined in the last section. In order to avoid the clutter of notation, we will use the same symbols A_j , B_j to denote the un-normalized and normalized data sets; the assignments should be clear from the context. Let A_j^i denote the Avg Diff for Gene i as assayed by the microarray A_j . Define B_j^i likewise. First, there are several ways to compute the fold statistic, *e.g.*, the mean, of the Gene i ($i = 1, 2, \dots, N$):

- **F1.** Average of A_j over the average of B_j : $\mu^{\text{Gene } i} = \frac{\frac{1}{M_B} \sum_{j=1}^{M_B} B_j^i}{\frac{1}{M_A} \sum_{j=1}^{M_A} A_j^i}$
- **F2.** If $M_A = M_B = M$: $\mu^{\text{Gene } i} = \frac{1}{M} \sum_{j=1}^M \log \left(\frac{B_j^i}{A_j^i} \right)$
- **F3.** More generally in **F2**: $\mu^{\text{Gene } i} = \frac{1}{M} \sum_{j=1}^M \log \left(\frac{B_{\tilde{A}(j)}^i}{A_j^i} \right)$ where $\tilde{A}$ is any permutation on the

set $\{1, 2, \dots, M\}$

Some studies have alternatively chosen to compute the fold of Gene i from states A to B as $\frac{B^i - A^i}{A^i}$ instead of $\frac{B^i}{A^i}$ as above. In order to resolve the problem of non-positive-valued Avg Diffs, it is commonplace in the literature, before taking arithmetic ratios, to threshold the Avg Diff values in every chip data set to an arbitrarily chosen minimum positive number, as e.g., in setting all Avg Diffs of less than 10.0 to 10.0. Alternatively, some studies translate all the intensity data in a chip data set so that the minimal translated Avg Diff in each set is positive. For example, if the minimum element in the A_2 data set is -2000.7 , then when 2001.7 is added to every Avg Diff reading in A_2 , all the translated Avg Diffs will be positive-valued. Note that such solutions will skew the intensity statistic of the microarray data which is not always a desirable thing. For symmetry reasons, the logarithm of folds rather than just the folds alone are averaged. Consider the data for Gene 1 un-normalized: From A_1 to B_1 , the intensity of Gene i changed 2.0 $\left(\frac{128.6}{64.3}\right)$ -fold, whereas from A_2 to B_2 , it changed 0.5 $\left(\frac{124.1}{248.2}\right)$ -fold so that on average the intensity of Gene i should intuitively be unchanged, i.e., have a fold change of 1.0. However, it is obvious that the arithmetic average of 2.0 and 0.5 does not equal 1.0. A logarithmic transformation of the individual folds solves this problem, as does the use of the geometric rather than the arithmetic mean of the folds.

As an exercise, the reader should verify that the order of taking logarithmic ratios in **F2** and **F3** does not change the resulting fold, $\frac{1}{4}^{\text{Gene } i}$. After computing the fold of all genes in the data set in any one of the preceding ways, one typically wants to know whether a fold average or a fold distribution is different from, or *statistically significant* in relation to, a postulated *null distribution* which represents the effects of measurement errors or noise. As we have already noted, it is essentially the choice of the *null hypothesis* which distinguishes the different methods for determining fold significance. Some studies associate the null hypothesis with an interval of nonsignificance informally called a noise envelope. The distribution for the null hypothesis has also been calculated from the microarray data set permuted, especially when the number of replicate data points is small. In general, replication improves the estimation of the null distribution. Below, we outline several common null distributions:

- **G1.** A null distribution for each Gene i whose mean is obtained by averaging latitudinally across duplicates, $\mu^{\text{Noise } i} = \frac{1}{2} \left(\left| \log \left(\frac{A_1^i}{A_2^i} \right) \right| + \left| \log \left(\frac{B_1^i}{B_2^i} \right) \right| \right)$
- **G2.** A null distribution for all genes whose mean is obtained by averaging longitudinally across all N genes, $\mu_A = \frac{1}{N} \sum_{i=1}^N \left| \log \left(\frac{A_1^i}{A_2^i} \right) \right|$ or $\mu_B = \frac{1}{N} \sum_{i=1}^N \left| \log \left(\frac{B_1^i}{B_2^i} \right) \right|$
- **G3.** For each gene row, permutation of the intensity data, for instance, exchange data between conditions A_1 " B_2 , etc., and then recalculate **G1**.

A coarse, qualitative method may combine **F1** and **G2** to decide that a Gene i with $\frac{1}{4}^{\text{Gene } i} > \max(\frac{1}{4}_A; \frac{1}{4}_B)$ or $\frac{1}{4}^{\text{Gene } i} < -\max(\frac{1}{4}_A; \frac{1}{4}_B)$ is significantly different, foldwise, from the average fold statistic resulting from noise as calculated from duplicate conditions. Another approach could be to rank the N genes by their $\frac{1}{4}^{\text{Gene } i}$ value from **F2** and to decide that the top and bottom 5% of these ranked genes are significantly changed.

There exist equally diverse non fold-centered methods for determining the significance of a change in expression. The intuitive idea behind all these approaches is to call a change statistically significant if the expression change *inter*-states A and B is maximal, and the expression change *intra*-state within the replicate conditions $\{A_j\}_{j=1}^{M^A}$ and $\{B_j\}_{j=1}^{M^B}$, respectively, is minimal. Again, the choice of a null distribution representing the postulated noise distinguishes these methods. Being non fold-based, traditional parametric and nonparametric statistical tools for analyzing means or variances between data groups such as the χ^2 , Student t , Mann-Whitney, and F tests and ANOVA

have been applied in the literature. These methods are reviewed comprehensively in [144] and the reader should be aware of the implicit assumptions and underlying null hypotheses in these standard tests prior to application. A drawback to these traditional approaches is that their conclusions are "asymptotic", *i.e.*, they are statistically valid only when one has a large number of replicate data.

At the end of the normalization section 3.4, we had briefly described a way to graphically visualize the determination of significant difference in expression data between A_1 and B_1 . To reiterate, on the scatterplot of the un-normalized data sets A_1 versus B_1 , we compute and draw the linear regression line. If chip assay reproducibility is reasonably robust and if we assume that the majority of genes do not undergo a dramatic change in expression level in going from state A_1 to B_1 , then the data points should cluster close to this regression line which should have slope one and should pass through (0, 0). The envelope of a standard error or deviation spread away from the regression line is our object of interest. This 1-SD envelope may be defined by a pair of lines that lie along a horizontal distance $\left(\frac{1}{N} \sum_{i=1}^N (B_1^i - f(A_1^i))^2\right)^{\frac{1}{2}}$ to the left and right of the regression, $f(x) = \pm_1 x + \pm_0$.

The intuitive reasoning here is that a data point which is outside this envelope is a statistically *insignificant* event relative to a postulated stochastic distribution, usually gaussian, of the data around the regression line due purely to noise (chance). Such an outlying data point represents, in the reverse context, a gene which has undergone a statistically *significant* expression change from A_1 to B_1 .

3.5.2 Fold change may not mean the same thing in different expression measurement technologies

In one common experimental design, DNA microarrays are built by printing or spotting an array of cDNA on a glass slide, which provides gene-specific probes for hybridization to targets. Two different samples of mRNA are labeled with different fluorescent dyes and simultaneously hybridized onto each probe. The data for each probe (gene) consists of two fluorescence intensity measurements, typically red and green, representing the expression levels of the same gene in the red-labeled (with the dye Cy5) and the green-labeled (with Cy3) mRNA samples. Early analyses of the data obtained through these microarrays will identify differentially expressed genes by taking the ratio of red intensity to green intensity, *i.e.*, the *intensity ratio*, and choosing an arbitrary cut-off above or below which the genes will be interpreted as being differentially expressed in the two samples. The crucial point is that one typically does not use the absolute measure of intensity, but uses, instead, the ratio or degree of difference between samples.

In one of the earliest uses of this design, Schena *et al.* [157] called a gene differentially expressed if the expression level in the two mRNA samples differed by more than a factor of 5. DeRisi *et al.* [56] called genes differentially expressed if the logarithm of their ratio differed by more than a factor of 3, after standardizing the intensity measurements using a set of "housekeeping" genes. More recently, sophisticated statistical techniques have been developed to provide more reliable calculations of the intensity ratios by taking into account the distribution of the intensities across the whole microarray. In the first statistical analysis of these data, Chen *et al.* [45] proposed a method to identify statistically significant changes between the two mRNA samples, under several assumptions on the distribution of the intensities, including normality, null intercept, and constant variation coefficient. Newton *et al.* [135] offered a bayesian approach to the problem, using a hierarchical model for intensities and identified differentially expressed genes on the basis of the posterior odds of change. It is worthwhile to mention that Arfin *et al.* [11], in a study involving cDNA arrays expression profiling of *E. coli*, found little correlation between fold difference in the reported expression levels of a gene and its accuracy.

Using synthetic oligonucleotide microarrays, on the other hand, one identifies differentially expressed genes by comparing expression measurements made on two separate microarrays. On these microarrays, there is no simple measure of intensity: instead, 16 to 20 probe pairs are used to interrogate each gene. Each probe pair has a PM and MM intensity signal. In an ideal setting, if a gene is truly present, each of the PM probes would have a high intensity and each of the MM probes would have a near-zero intensity. The ratio PM/MM closely resembles the intensity ratio used with spotted microarrays; however, here we are using this ratio to determine an absolute expression level for this gene. Since each gene is represented by many probe pairs, the average $\log(\text{PM/MM})$ is called the *Log Average Ratio*. A probe pair would be considered positive if $\text{PM} - \text{MM}$ is greater than or equal to a threshold, and if PM/MM is greater than or equal to another threshold (similarly, $\text{MM} - \text{PM}$ and PM/MM are calculated in considering a probe pair negative). The average of the $\text{PM} - \text{MM}$ differences for all probe pairs in a probe set is called the *Average Difference*, and this value is used as the absolute amount of expression for a gene.

In order to determine whether a gene expression is significantly different, current experimental protocols require the (possibly repeated) pairwise comparison of two microarrays, each hybridized to each of the different samples being compared. The intensity ratio used in the spotted arrays is thus replaced by the indirectly calculated ratio between the the average difference of each gene in the two different arrays. Typically, a cut-off threshold is arbitrarily selected between 2 and 3 (see, e.g., [97, 198]). That is, the expression level of a gene will be considered differentially expressed in an experiment if the (indirectly calculated) average difference is two or three times higher in one microarray than in the other. However, this threshold has been set as low as 1.7 [117].

The major manufacturer of these microarrays [10] provides software implementing a different measure of differential expression. Let x represent the average difference of a gene on one microarray and y its average difference on the second microarray. For each gene, the level of change across the two microarrays is computed by a weighted ratio:

$$W = \frac{y - x}{\max(\min(x, y), Q_c \cdot Q_m)}$$

If $y \leq x$, then $+1$ is added to W , otherwise $W - 1$. Q_c is the maximum noise level of the two microarrays, or the difference in pixel intensities seen in the lowest intensity probes, and Q_m is a constant depending on the resolution of the microarray—2.1 for 50 $\frac{1}{4}\text{m}$ feature microarrays and 2.8 for 24 $\frac{1}{4}\text{m}$ feature microarrays. However, the decision of whether a gene is *significantly* differentially expressed in the two microarrays is based on a complicated interpretation of the change of each of its 14 to 20 probe pairs. Let $(\text{PM} - \text{MM})_x$ be the difference between PM and MM for a probe pair in the first array and $(\text{PM} - \text{MM})_y$ be the same difference for the same probe pair on the second array. A probe pair shows a significant change (increased or decreased) across two microarrays if:

1. the difference $(\text{PM} - \text{MM})_y - (\text{PM} - \text{MM})_x$ is higher than an arbitrary threshold called the *Change Threshold*, and
2. the ratio, $\frac{(\text{PM} - \text{MM})_y - (\text{PM} - \text{MM})_x}{\max(Q, \min((\text{PM} - \text{MM})_x, (\text{PM} - \text{MM})_y))}$ exceeds another arbitrary threshold called the *Percent Change Threshold*, where Q represents the noise on the chip.

Given this definition of a significantly increased or decreased probe pair, a gene is considered significantly differentially expressed based on four aggregate measures:

1. The number of increased or decreased probe pairs, which ever is higher.
2. The ratio of the number of increased probe pairs to the number of decreased probe pairs.
3. The difference or subtraction of the log average ratios of the gene in the two microarrays.
4. The difference in the number of positive probe pairs minus the difference in the number of negative probe pairs, divided by the number of probe pairs.

The decision of whether a gene is significantly differentially expressed between two microarrays is decided by a user-modifiable matrix indexed by the above four results. The user can set thresholds to define the boundaries where each metric changes the decision. Default values for these thresholds have been set by the manufacturer through in-house empirical testing.^[5] Since systematic variations are expected to occur between any two microarrays—due to biological noise, changed environmental conditions, or other factors affecting equally all genes on an array—various normalization techniques are used to render comparable the expression levels across the two microarrays. Section 3.4.1 discusses the various techniques used.

It should be clear that there is a great deal of difference in the interpretation of intensity measurements, intensity ratios, and gene expression levels across the two systems. Fundamentally, spotted arrays are currently used to measure a *difference* in transcription levels between two samples, whereas oligonucleotide arrays are used to measure *absolute* transcription levels in a single sample. Even if one discounts the differences in the quantitative and differencing algorithms, it should be obvious to the reader that the "results" of an experiment involving spotted arrays are going to be significantly influenced by the choice of the reference or control sample, already implying that the resulting values are not directly comparable.

^[5]As complicated as this methodology sounds in our description here, we are ignoring the fact that in the actual algorithm, not all available probe pairs are used in these calculations. Probe pairs with an intensity difference significantly beyond the mean intensity difference are considered outliers and are ignored in these calculations.

3.6 Dissimilarity and Similarity Measures

In time course studies such as study **S1**, it is natural to talk about how two genes G_k and G_j are (dis)similar to one another with respect to their expression levels sampled in time. In the bioinformatics literature, a gene and its corresponding expression levels are often represented as a data point or a vector in $\mathbb{R}^k$, where k is a positive integer (denoted by $\mathbb{N}$). Each vector component or coordinate slot denotes the expression level of that gene under a different experimental condition with k conditions altogether. A measure of similarity or dissimilarity such as a metric is then defined on $\mathbb{R}^k$ which maps pairs of vectors, representing genes and their corresponding expression information, to real numbers. When the number of genes and the expression data set are very large, as in most micro array experiments, one often asks which genes or groups of genes behave *similarly* under these k different or replicate conditions. In an attempt to answer this question, one resorts to classification and clustering methods, and of particular importance prior to the analysis, one has to define what one means by the term *similar*. Similarity is characterized by the choice of a (dis)similarity measure, such as a metric.

As noted by Jain and Dubes [101], a *cluster method* takes as its input a finite data set $\mathcal{X}$ of at least p ($p > 2$) elements together with a list of finitely many attributes of these data, or, alternatively, it may consist of a measure of similarity or dissimilarity between pairs of objects in $\mathcal{X}$. The output is an internal structure of $\mathcal{X}$, which is determined entirely by the input and *not* dependent on any external criterion. Following [103], even though there is no universal agreement in the field with regard to terminology, when the rules for determining the output are externally defined, the algorithm will be called a *classification* rather than a cluster method. For example, suppose that we have for our set $\mathcal{X}$ microarray data which are represented as points in $\mathbb{R}^3$, *i.e.*, equipped with the standard Euclidean distance, a metric. Then, a cluster method would ideally yield groups of genes which are physically close (in Euclidean distance) to each other, and hence behave similarly over three conditions, forming very tangible visual clusters when plotted. On the other hand, questions such as which genes appear in the first octant of $\mathbb{R}^3$ would fall under the classification category. The distinction between the terms *clustering* and *classification* is not purely of academic interest. Clustering algorithms generally have a higher order of complexity, and are therefore more

computationally intensive, than classification methods. We begin the discussion of (dis)similarity measures with several formal and standard definitions. Let $\mathcal{X}$ be any nonempty set.

Definition 1 A function $\rho: \mathcal{X} \times \mathcal{X} \rightarrow [0, \infty)$ is a metric if for all $x, y \in \mathcal{X}$:

- **M1.** $\rho(x, y) = 0$ if and only if $x = y$ (definite).
- **M2.** $\rho(x, y) = \rho(y, x)$ (symmetrical).
- **M3.** $\rho(x, y) \leq \rho(x, z) + \rho(z, y)$, for all $z \in \mathcal{X}$ (triangle inequality).

Some examples of common metrics are:

- Let $\mathcal{X} = \mathbb{R}^n, n \in \mathbb{N}$. The classical Euclidean distance, $\rho(x, y) = (\sum_{i=1}^n |x_i - y_i|^2)^{\frac{1}{2}}$ for any $x, y \in \mathcal{X}$, where $x = (x_1, \dots, x_n), y = (y_1, \dots, y_n)$.
- Let $\mathcal{X} = \mathbb{R}^n, n \in \mathbb{N}$. $\mathcal{L}^p$ -type distances with $p \in \mathbb{N} \setminus \{0\}$, $\rho(x, y) = (\sum_{i=1}^n |x_i - y_i|^p)^{\frac{1}{p}}$ for any $x, y \in \mathcal{X}$. When $p = 2$, we have the preceding Euclidean distance.
- Suppose that $\mathcal{X}$ is the set of real-valued continuous functions on the closed real interval $[0, 1]$. Let $f, g \in \mathcal{X}$, and $\rho(f, g) = \max_{x \in [0, 1]} |f(x) - g(x)|$. Exercise: Confirm this.
- The distance between two random variables X and Y , $\rho(X, Y) = H(X, Y) - H(X; Y)$ where $H(X, Y)$ is the joint entropy of X, Y , and $H(X; Y) = H(X) - H(X|Y)$ is the mutual information between X and Y , a symmetrical function. An overview of entropy and information-theoretic concepts is presented in subsection 3.6.2. We refer the interested reader to Cover and Thomas [54], for a detailed review of information theory.

The reader should be warned that there is a widespread misuse of the terminology *metric* in the current bioinformatics literature to include functions such as the linear (Pearson's) correlation coefficient and mutual information. It is a simple exercise to show that these functions do not satisfy the axioms of a metric. Next we define a weaker class of functions than the metric, a *dissimilarity coefficient*.

Definition 2 A dissimilarity coefficient is any function $d: \mathcal{X} \times \mathcal{X} \rightarrow [0, \infty)$ which satisfies two axioms:

- **D1.** $d(x, x) = 0$ for all $x \in \mathcal{X}$.
- **D2.** $d(x, y) = d(y, x)$ for all $x, y \in \mathcal{X}$ (symmetric).

A dissimilarity coefficient d is *definite* if it satisfies a third condition:

- **D3.** $d(x, y) = 0$ if and only if $x = y, x, y \in \mathcal{X}$

Any metric is a definite dissimilarity coefficient, and note that $1 - r$, where r denotes the linear (Pearson's) correlation coefficient is a definite dissimilarity coefficient. Exercise: Confirm these.

3.6.1 Linear correlation

Given data points (x_i, y_i) for $i = 1, \dots, n$, it is common to ask whether a *linear* relationship exists between the quantities x_i and y_i , and to what degree are the data linear, at least statistically. The graphical equivalent would be to ask whether the scatter plot of x_i versus y_i looks like (discretizations of) a line. The linear or Pearson's correlation coefficient r is the mathematical device that is most often used to do this,

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (3.6.2)$$

where $\bar{x}, \bar{y}$ denote the means of x_i, y_i respectively. As an aside, observe that,

$$\begin{aligned} r^2 &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (y_i - \bar{y})^2} \\ &= \alpha_1^{(x,y)} \alpha_1^{(y,x)}. \end{aligned} \quad (3.6.3)$$

In other words, r is the geometric mean of the slope of the linear regression of x versus y and the slope of the linear regression of y versus x , denoted $\pm^{(x,y)}_1$ and $\pm^{(y,x)}_1$ respectively. By the Cauchy–Schwarz inequality,

$$\langle x, y \rangle = \|x\| \|y\| \cos \theta, \quad x, y \in \mathbb{R}^n, \quad (3.6.4)$$

where $\langle x, y \rangle \doteq \sum_{i=1}^n x_i y_i$ is the standard dot product on $\mathbb{R}^n$, $\|x\| \doteq \langle x, x \rangle^{\frac{1}{2}}$ and θ is the angle between vectors x and y , it can be shown that,

$$-1 \leq r \leq 1,$$

$r = 1$, if and only if, $x = y$; x, y are positively correlated,

$r = -1$, if and only if, $x = -y$; x, y are negatively correlated.

Suppose that one regards x, y as random variables with samples x_i, y_i indexed by $i = 1, \dots, n$. If x and y are stochastically independent, then $r = 0$. The converse is not generally true, as, *e.g.*, when we take $t_i = i \cdot 2\pi/n$, $x_i = \cos t_i$ and $y_i = \sin t_i$ for any integer n . Exercise: Verify this. When r is near 0, x and y are said to be uncorrelated. A review of the concept of stochastic independence may be found in Shiryaev [163]. Note that r has a nonrigorous transitive property such that if x and y , and y and z , are strongly correlated with correlation strength $r_{x,y}$ and $r_{y,z}$, x and z have correlation strength r . As noted by Press *et al.* [144], while r provides a summary of the strength of the association if the correlation is found to be significant, r is not generally useful for deciding whether an observed correlation is statistically significant. This weakness arises from the fact that the computation (3.6.4) ignores the distributions of x and y . One solution to this problem is to use nonparametric or rank correlation, which details are also found in [144]. It can also be shown that r is more sensitive to small perturbations of data points that are farther away from the centroid of the system than to perturbations near the centroid. This may result in a bias. Linear correlation has been used to find pairwise associated genes across different experimental conditions, as in [63].

3.6.2 Entropy and mutual information

Suppose that we have the expression information for a pair of genes G^x, G^y across n experimental conditions. In the most extreme cases, the expression of G^x and G^y may be completely dependent (if this dependence is linear, this implies that $r = \pm 1$), or independent (which implies that $r = 0$) across these n conditions. Question: Can we measure or quantify the dependence of G^x and G^y levels in this case? Equivalently, how much would the expression data for G^x inform us about the corresponding expression of G^y , and *vice versa*? In this regard, we may appeal to relevant information-theoretic and conceptual devices that were formulated to solve the fundamental problem in communications of reproducing at the receiver endpoint, either exactly or approximately, a message from the transmitter end point. The messages transmitted and received will correspond

to the expression data of G^x and G^y in the biological scenario. Suppose that X is a random variable with the set of events or outcomes $\mathcal{F}_X = \{x_i\}_{i=1}^n$, and corresponding probabilities $\mathbb{P}(X = x_i) = p_{x_i}$ $\in [0, 1]$. By definition, $\sum_{x \in \mathcal{F}_X} \mathbb{P}(X = x) = 1$. Following Shannon [160], we want to find a measure $H(p_{x_1}, \dots, p_{x_n})$ of how much "choice" is involved in the selection of events, or of how uncertain we are of the outcome. Intuitively and reasonably, we will require that H have the following three properties:

- **E1.** H is continuous in p_{x_i} .
- **E2.** If all p_{x_i} 's are equal, i.e., $p_{x_i} = \frac{1}{n}$, then H should be a monotonically increasing function of n . With equally probable events, there is more choice or uncertainty in the outcome when there are more possible events x_n .
- **E3.** If a choice is broken down into two successive choices, the original H should be the weighted sum of the individual values of the later H 's. For instance, let $n = 3$ and $\{p_{x_1}, p_{x_2}, p_{x_3}\} = \{\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\}$, and we consider three possible decompositions of a choice of the three outcomes: Then $H(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}) = H(\frac{1}{2}, \frac{1}{2}) + \frac{1}{2}H(\frac{2}{3}, \frac{1}{3}) = H(\frac{5}{6}, \frac{1}{6}) + \frac{5}{6}H(\frac{3}{5}, \frac{2}{5})$.

It can be shown that the only explicit form of H satisfying **E1** to **E3**,

$$H(p_{x_1}, \dots, p_{x_n}) = -K \sum_{i=1}^n p_{x_i} \log p_{x_i},$$

where K is positive constant. We let $K = 1$. We shall call $H(X)$ the entropy of random variable X . Consider the case of a Bernoulli random variable X , i.e., $n = 2$, with $p_{x_1} = p$ and $p_{x_2} = 1 - p$ where $p > 0$. Then, when X is one-dimensional,

$$H(X) = -p \log p - (1 - p) \log(1 - p).$$

Using a base-2 logarithm, H plotted as a function of p is concave, non-negative, symmetrical and has a maximum of 1 at $p = 1/2$ (see figure 3.24).

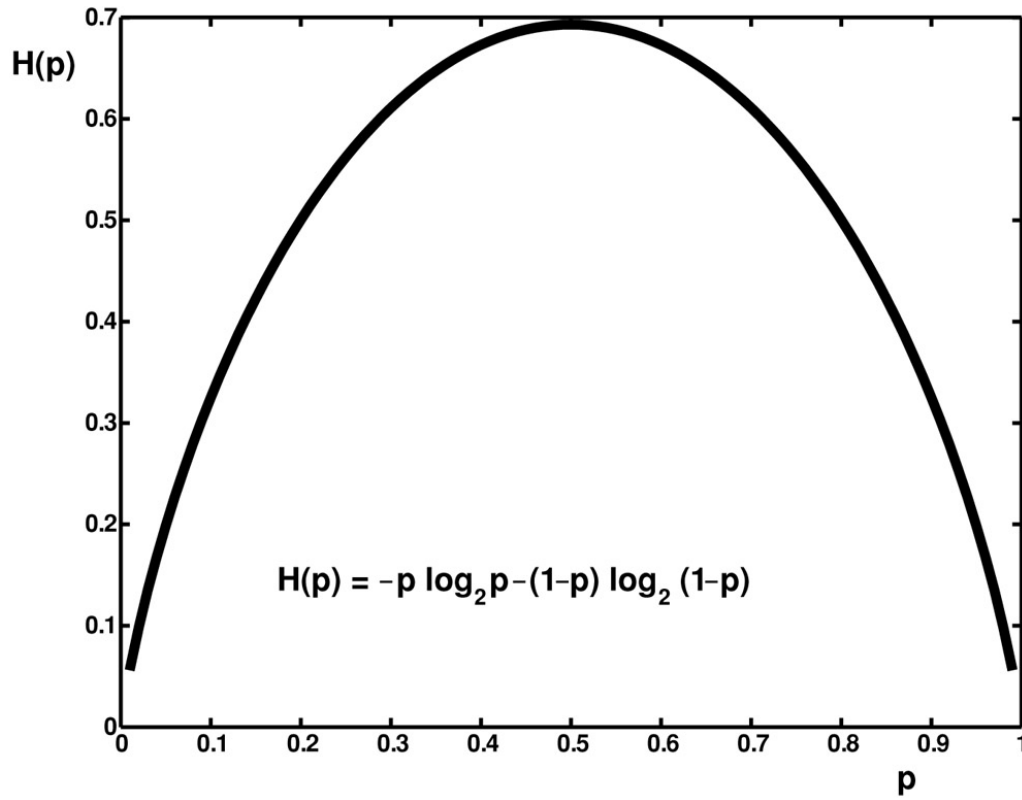


Figure 3.24: Entropy plot

As noted in [160], H has several interesting properties that substantiate it as a reasonable measure of choice or uncertainty.

H1. $H \in 0$ with equality if and only if all but one $p_{x_i} = 0$ and the non-zero event has probability 1. Therefore, when we are entirely certain about the outcome H the measure of uncertainty vanishes.

H2. For any number of possible outcomes n , H is a maximum ($\log n$) when all the p_{x_i} 's are equal, i.e., $p_{x_i} = 1/n$. Intuitively, this is the most uncertain scenario.

H3. Suppose that Y is a random variable with the set of outcomes $\mathcal{F}_Y = \{y_i\}_{i=1}^m$ and corresponding probabilities $P(Y = y_i) = p_{y_i} \geq 0$ and $\sum_{y \in \mathcal{F}_Y} P(Y = y) = 1$. Let p_{x_i, y_j} be the probability for the joint occurrence of $X = x_i$ and $Y = y_j$. Define the joint entropy of X and Y to be

$$H(X, Y) \doteq - \sum_{x \in \mathcal{F}_X, y \in \mathcal{F}_Y} p_{x, y} \log p_{x, y} = - \sum_{i=1}^n \sum_{j=1}^m p_{x_i, y_j} \log p_{x_i, y_j}$$

and by basic properties of marginals of joint distributions,

$$\begin{aligned} H(X) &= - \sum_{i,j=1,1}^{n,m} p_{x_i, y_j} \sum_{j=1}^m \log p_{x_i, y_j} = - \sum_{i=1}^n p_{x_i} \log p_{x_i} \\ H(Y) &= - \sum_{i,j=1,1}^{n,m} p_{x_i, y_j} \sum_{i=1}^n \log p_{x_i, y_j} = - \sum_{j=1}^m p_{y_j} \log p_{y_j} \end{aligned}$$

Exercise: Show that $H(X, Y) \leq H(X) + H(Y)$ with equality if and only if X and Y are independent, i.e., $p_{x_i, y_j} = p_{x_i} p_{y_j}$. The uncertainty of a joint event is less than or equal to the sum of individual uncertainties. Note that $H(X, X) = H(X)$.

H4. Any change toward equalizing the p_{x_i} 's increases H . Computationally, any averaging operation, $p_{x_i} \mapsto p'_i = \sum_{k=1}^n a_{ik} p_{x_k}$, where $\sum_i a_{ik} = \sum_k a_{ik} = 1$ with $a_{ik} \in [0, 1]$ increases H —except in the special case of a permutation where for every i all but one $a_{ik} = 0$, the non-zero constant being 1, in which case H is invariant.

H5. Recall the definition of conditional probability, $P(X = x|Y = y) = \frac{P(X=x, Y=y)}{P(Y=y)}$. if $P(Y = y) \neq 0$, i.e., $p_{x|y} = \frac{p_{x,y}}{p_y}$. An easy consequence of this definition is Bayes' theorem,

$$p_{y|x} = \frac{p_{x|y} p_y}{p_x} = \frac{p_{x|y} p_y}{\sum_{y' \in \mathcal{F}_Y} p_{x|y'} p_{y'}}$$

The conditional entropy of X given $Y = y$ is defined as

$$H(X|Y = y) \doteq - \sum_{x \in \mathcal{F}_X} p_{x|y} \log p_{x|y}$$

where as the conditional entropy of X given Y is defined by averaging the $H(X|Y = y)$ over $y \in \mathcal{F}_Y$,

$$\begin{aligned} H(X|Y) &= \sum_{y \in \mathcal{F}_Y} p_y H(X|Y = y) \\ &= - \sum_{x \in \mathcal{F}_X, y \in \mathcal{F}_Y} p_y p_{x|y} \log p_{x|y} \\ &= - \sum_{x \in \mathcal{F}_X, y \in \mathcal{F}_Y} p_{x,y} \log p_{x|y} \end{aligned}$$

This quantity measures how uncertain we are of Y on average when we know X . The chain rule for entropy can then be shown. Exercise:

$$H(X, Y) = H(X|Y) + H(Y) = H(Y|X) + H(X)$$

Now following the exercise in **H3**, we can conclude that,

$$\begin{aligned} H(X) + H(Y) &\geq H(X, Y) = H(X|Y) + H(Y) \\ H(X) &\geq H(X|Y) \end{aligned}$$

which says that *on average* our knowledge of Y does not increase our uncertainty of X . The reader is encouraged to come up with a pair of X, Y such that $H(X|Y = y) > H(X)$ but $H(X|Y) \leq H(X)$. When X and Y are independent, our uncertainty of X is unchanged; otherwise it decreases.

H6. The mutual information between X and Y is defined as

$$M(X, Y) \doteq H(X) - H(X|Y)$$

Exercise: Show that $M(X, Y) \geq 0$ and $M(X, Y) = M(Y, X)$. Mutual information measures the average reduction in uncertainty about X given that we have knowledge of Y , and *vice versa* by the exercise. When X and Y are independent, $M(X, Y) = 0$, i.e., there is no reduction in uncertainty. At the other extreme, when $X = Y$, $M(X, Y) = H(X)$ so that we have reduced all uncertainty— $H(X)$ —about the

system. Here we also see that M is *not* a metric since $M(X, X) \neq 0$. The distance between random variables X and Y is defined as

$$D(X, Y) \doteq H(X, Y) - M(X, Y)$$

Exercise: Show that D satisfies the three axioms of a metric in an earlier section.

H7. The cross or Kullback–Liebler entropy between the probability distributions p_x and q_x over a common space of possible outcomes $\mathcal{F}_X$ is defined as,

$$D_{\text{KL}}(p, q) \doteq \sum_{x \in \mathcal{F}_X} p_x \log \frac{p_x}{q_x}$$

Note that in general, $D_{\text{KL}}(p, q) \neq D_{\text{KL}}(q, p)$.

In order to apply the preceding tools in the context of microarray data analysis, suppose that we have the gene expression data for genes G^x, G^y across m different experimental conditions. Divide the ranges of the G^x, G^y expression data into uniformly spaced n intervals or "bins." Let p_{x_i} be the probability that the expression for G^x falls into the i th interval, and similarly for G^y . Suppose that three out of the m different experiment intensities for G^x fell within the k th bin, then we let $p_{x_k} = 3/n$. A graphical example of this is shown in figure 3.25.

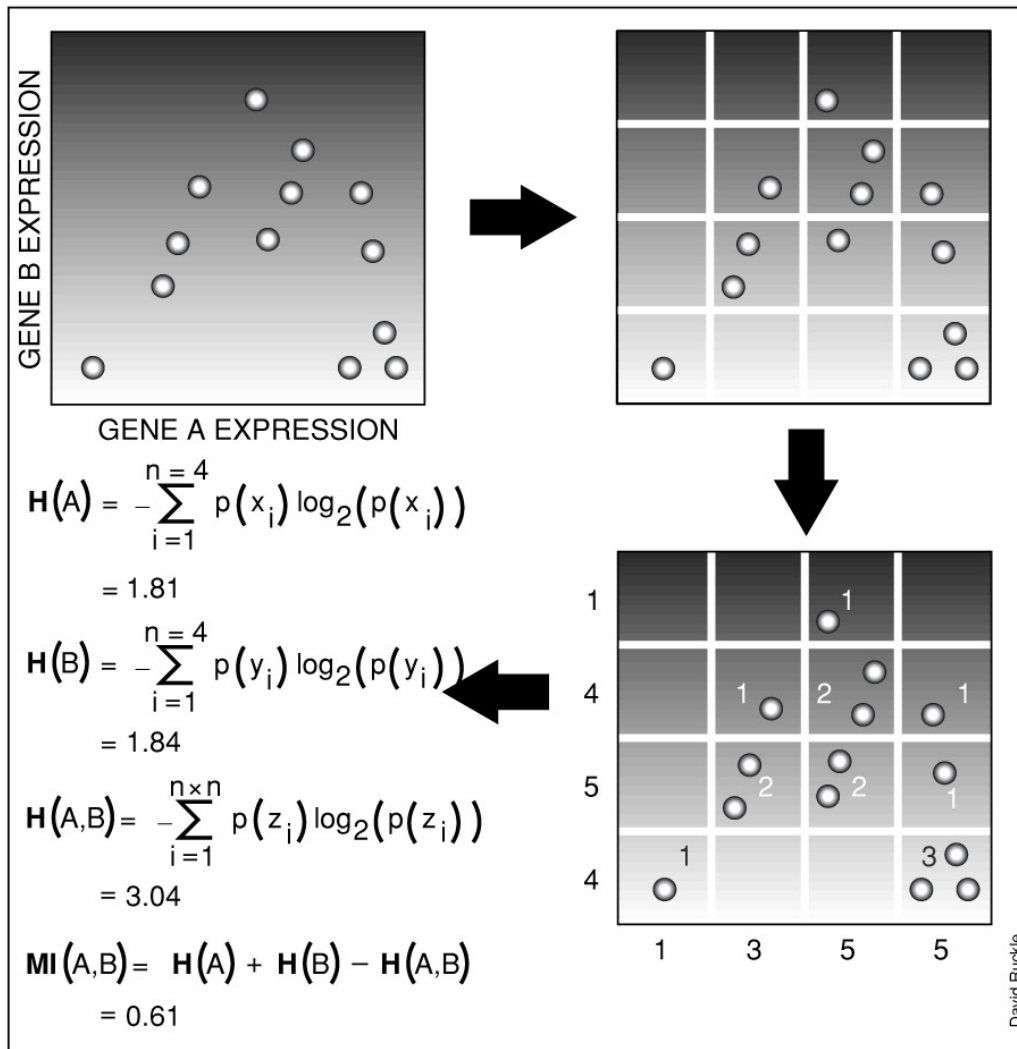


Figure 3.25: Graphical example of mutual information calculation. First, a scatterplot of the expression measurements of the two genes is created, and a grid is imposed. In this example, each expression measurement is quantized into four bins (one can think of these as "low," "low-medium," "high-medium," and "high," though any number and positioning of bins can be considered). The entropy for each gene is then calculated using the row and column sums, and the joint entropy is calculated from the grid.

3.6.3 Dynamics

Dynamics provides a powerful example of how the choice of the appropriate similarity measure can affect the analysis of a genomic system. We use the term *dynamics* to refer to the rate of change of genetic expression over time, calculated as the first-order difference of the genetic expression levels ($E_{t2} - E_{t1}, E_{t3} - E_{t2}$). This is different from the simple temporal pattern of genetic expression (E_{t1}, E_{t2}, E_{t3}) that we refer to as statics. The primary motivation for studying gene expression dynamics is that existing static techniques may not identify all the important relationships. Some genes may have associated dynamic behaviors but may not have associated static expression behaviors. A hypothetical example is shown in figure 3.26: Gene A codes for an enhancer protein that regulates the expression of Gene B—a high level of Gene A causes an upregulation of expression in Gene B. Since Gene B can be at many possible expression levels before being affected by Gene A, the enhancer-type relationship between the two genes cannot be noticed by simply examining the correlation of static expression patterns. Instead, one needs to examine the dynamics of gene expression—the way in which the expression level of Gene A leads to a change in Gene B—to detect the underlying dynamic relationship. We therefore hypothesized that using dynamics

measures as our fundamental similarity measure has the potential to discover relationships between genes that are not detectable using static similarity measures.

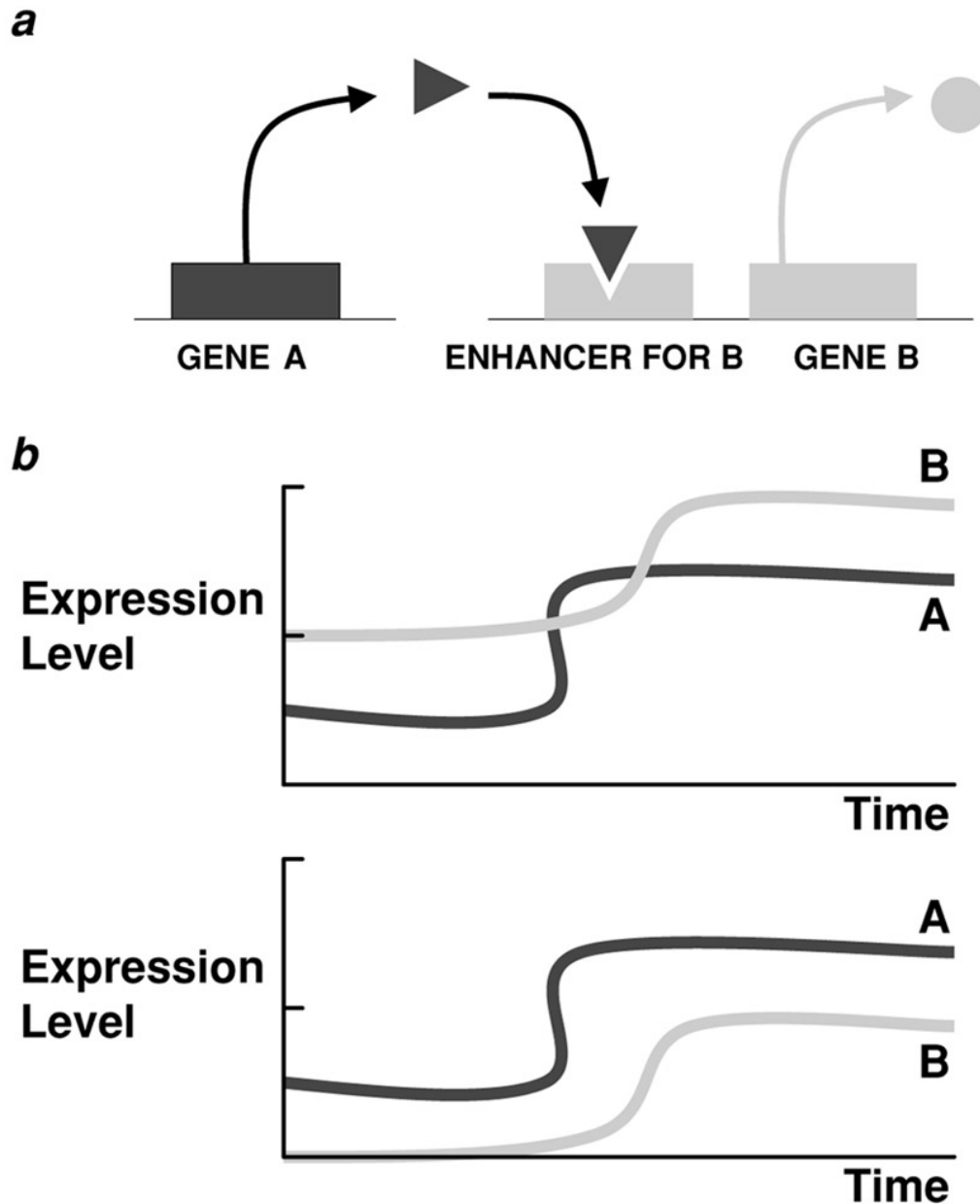


Figure 3.26: Dynamic relationships between genes. (a) The expressed product of Gene A binds an enhancer region that increases transcription of Gene B. (b) Gene B's initial expression level before being affected by Gene A can vary throughout the experiment. As a result, measuring the correlation between the absolute levels of Genes A and B will not reveal the underlying enhancement relationship between the two. Instead, this can only be done by analyzing the expression dynamics—the change in expression level of Gene B in relation to the expression level of Gene A. (Derived from Reis *et al.* [150].)

Reis *et al.* [150] investigated the *Saccharomyces cerevisiae* mRNA expression data aggregated from several experiments reported by Eisen *et al.* [63] in which the response of the yeast cells to several different stimuli is recorded. The data contain 79 data points in 10 time series measured under different experimental conditions. Of over 6000 genes in the yeast genome, Eisen *et al.* included only 2467 genes that had functional annotations. We analyzed the same subset of genes as described. Dynamics similarity measures were calculated as slopes of the change of expression over time. Slopes were calculated between each adjacent pair of expression data points, E_{t_n} and

$E_{t_{n+1}}: \text{Slope}_{(n,n+1)} = \frac{\text{Expression_Level}_{n+1} - \text{Expression_Level}_n}{\text{Time}_{n+1} - \text{Time}_n}$ As slopes are only calculated between data points within the same time series, the 79 data points in 10 time series are reduced to only 69 slope measurements. The units of the slope measurements are in normalized expression level units per minute. The authors found those pairs of genes that exceed a correlation coefficient of .78 based on a permutation analysis (see section 4.12.1). We compared the pairs of genes found to be correlated at that threshold or higher using both the standard static expression levels and the slope similarity measures. We found that 133 genes appeared in both the static and dynamic analyses, leaving 215 genes that were exclusive to the dynamics analysis. However, only about half of the 133 shared genes appear linked to the same genes in both analyses—most appear linked to other genes.

Of many examples, figure 3.27 shows the distribution of slopes between RAD6 and MET18 which are highly correlated by the slope similarity measure but not by the static gene expression levels. RAD6 is a ubiquitin–conjugating enzyme concentrated in the nucleus that is essential to mediating the degradation of amino end rule–dependent protein substrates(21). MET18, also known as MMS19, is a protein concentrated in the nucleus that affects RNA polymerase II transcription[150]). These are inversely related in their dynamics, with an R^2 of .791. It is not surprising that a gene responsible for protein degradation might have an inverse relationship to a gene responsible for RNA transcription leading to protein synthesis. Again, the significance here is that the discovery of this relationship depended upon picking the right similarity measure. Besides this one, there is undoubtedly a large unexplored universe of potentially useful measures.

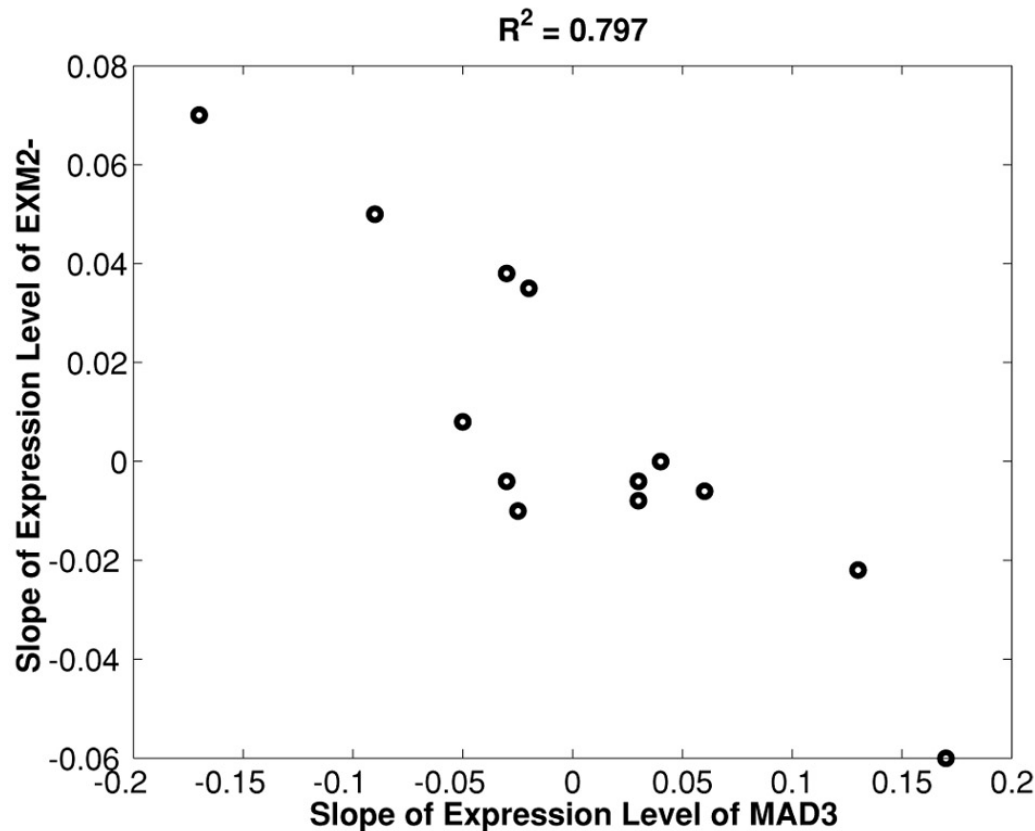


Figure 3.27: Negative dynamic correlation. The distribution of slopes of MAD3 and EXM2 plotted one against another.

Chapter 4: Genomic Data–Mining Techniques

4.1 Introduction

This chapter starts with an analysis of the components in molecular biology for which analysis can be performed. We then move to defining genomic data mining and what it actually means biologically to apply these clustering techniques in this domain. Although many analytic techniques are currently available, one should clearly not apply these techniques blindly without some hypothesis, so we discuss potential broad hypotheses one can start with. We then cover techniques for data reduction and filtering. Using examples written in pseudo–code, we cover the most commonly used techniques, discussing the advantages and disadvantages of each technique. We will then address the postanalysis process, including determining significance through permutation testing. Finally, we end with some thoughts on the automatic determination of genetic networks, including bayesian networks.

4.2 What Can Be Clustered in Functional Genomics?

Before launching into a discussion of the various clustering algorithms available to bioinformaticians, one needs to carefully consider the motivations behind data mining. *Data mining* can be loosely defined as determining the relationships of the elements in a data set. A variety of objects in biology can be captured in functional genomics data sets, a few of which should be mentioned here.

- *Genes*: Certainly, the majority of functional genomics bioinformatics analysis involves determining the relationships among genes. Genes can be clustered not only by gene expression but also by gene sequence, nucleotide composition, linkage, and chromosomal position. Genes can obviously be clustered using measurements from normal physiological states, as well as abnormal or pathological states.
- *Alternative splicing products*: An increasing number of genes are now predicted to have alternative splicing products, or a varying combination of exons. Various alternatively spliced products could be clustered to assess their pattern of expression compared to each other.
- *Tissues*: Normal tissues can be clustered to assess their relative degrees of similarity or dissimilarity. Unknown tissues, such as pathological specimens, could then be compared with clusters of known tissues.
- *Diseases*: Tissues from single or multiple diseases can be clustered. What is currently thought to be one disease may in fact be distinguishable into two or more subtypes.
- *Phenotype*: Cell lines, organisms, or patients can be clustered based on a series of measurements on them, of which gene expression may be only one component.
- *Patients*: In medical informatics, patient clusters are distinguishable in many ways: geographical location or home, presence or absence of a disease, the quantitation of specific laboratory measurements, or details of a longitudinal course of medical care.
- *Promoters*: Transcription factors are known to bind these specific regions of DNA to cause increased or decreased expression of the controlled gene. These could be clustered based on sequence similarity or association with functional genomics measurements.
- *Environment*: Environmental toxins and factors can be clustered by the pattern of gene expression, or the change in expression, seen after exposure.

As diagrammed in figure 2.11, all of these data types may end up in a large heterogeneous data table that can be used to advantage in the automated datamining techniques described below. It is precisely the lack of *a priori* limitations or assumptions of which data sources and data types will be

of clinical or biological interest that is one of the bulwarks of the genomic approach to discovery.

4.3 What Does it Mean to Cluster?

There are certainly laboratories that criticize data mining or clustering as a "fishing expedition," believing that the analytic technique starts with absolutely no hypothesis. There are also laboratories that apply every existing clustering method to their microarray data sets, hoping to find some "significant genes or clusters." The ideal analytic method lies between these two approaches. Certainly, different clustering methods are useful for different purposes. One should not blindly apply every clustering algorithm without knowing which method is best for answering particular kinds of questions. The analyst should start the discovery process with a hypothesis; this hypothesis may not be limited to a particular gene or set of genes but should ask a specific question of the data. At this point, it may be useful to list some potential questions one could start with:

- What uncategorized genes have an expression pattern similar to these genes that are well characterized?
- How different is the pattern of expression of gene X from other genes?
- What genes closely share a pattern of expression with gene X?
- What category of function might gene X belong to?
- What are all the pairs of genes that closely share patterns of expression?
- Are there subtypes of disease X discernible by tissue gene expression?
- What tissue is this sample tissue closest to?
- Which are the different patterns of gene expression?
- Which genes have a pattern that may have been a result of the influence of gene X?
- Which genes have a pattern that caused the expression pattern of gene X?
- What are all the gene–gene interactions present among these tissue samples?
- Which genes best differentiate these two group of tissues?
- Which gene–gene interactions best differentiate these two groups of tissue samples?

As we shall see as we progress through this chapter, different algorithms are more particularly suited to answer some of these hypotheses, compared with others.

4.4 Hierarchy of Bioinformatics Algorithms Available in Functional Genomics

Although there are only a few commonly used analytic techniques in functional genomics, it is important to remember that these techniques exist in a hierarchy of methods. One can model this taxonomy of techniques many different ways, as outlined in section 2.2; in this section, we chose to distinguish these methods as *supervised* or *unsupervised* [73]. An example of supervised learning is simply calculating the fold increase or decrease in expression of each gene in two microarray measurements. In this type of analysis, the particular context of each measurement is known, and the end result is a list of individual genes behaving differently in each of the different contexts. Another example of supervised learning is when it is algorithmically determined which genes are best able to determine an external labeling,^[1] such as distinguishing one type of leukemia from another [78]. This type of approach is used in neural network construction [155], and decision tree construction [146], which can then be easily translated into diagnostic algorithms [33,98].

In unsupervised learning, the patterns inherent in the data itself are determined, without *a priori* labels or contexts. Relevance networks, dendrograms, and self–organizing maps analyze every